

**TEMPLATE**

# KEY PROJECT INFORMATION & PROJECT DESIGN DOCUMENT (PDD)

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VERSION **v. 1.2**

RELATED SUPPORT

**- TEMPLATE GUIDE Key Project Information & Project Design Document v.1.2**

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Key Project Information

[SECTION A](#) – Description of project

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## KEY PROJECT INFORMATION

|                                                   |                                                                                                                                                                                                                                           |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GS ID of Project                                  | GS11466                                                                                                                                                                                                                                   |
| Title of Project                                  | Anakot Thmei Safe Water Project                                                                                                                                                                                                           |
| Time of First Submission Date                     | 21/12/2021                                                                                                                                                                                                                                |
| Date of Design Certification                      | 06/06/2023                                                                                                                                                                                                                                |
| Version number of the PDD                         | 7.4                                                                                                                                                                                                                                       |
| Completion date of version                        | 01/02/2024                                                                                                                                                                                                                                |
| Project Developer                                 | EcoAct SAS                                                                                                                                                                                                                                |
| Project Representative                            | EcoAct SAS                                                                                                                                                                                                                                |
| Project Participants and any communities involved | 1001fontaines                                                                                                                                                                                                                             |
| Host Country (ies)                                | Cambodia                                                                                                                                                                                                                                  |
| Activity Requirements applied                     | <input checked="" type="checkbox"/> Community Services Activities<br><input type="checkbox"/> Renewable Energy Activities<br><input type="checkbox"/> Land Use and Forestry Activities/Risks & Capacities<br><input type="checkbox"/> N/A |
| Scale of the project activity                     | <input type="checkbox"/> Micro scale<br><input checked="" type="checkbox"/> Small Scale<br><input type="checkbox"/> Large Scale                                                                                                           |
| Other Requirements applied                        | GS4GG Principles & Requirements Version 1.2, October 2019                                                                                                                                                                                 |
| Methodology (ies) applied and version number      | Gold Standard Methodology for emission reductions from safe drinking water supply, Version 1.0 (03/05/2021)                                                                                                                               |
| Product Requirements applied                      | <input checked="" type="checkbox"/> GHG Emissions Reduction & Sequestration<br><input type="checkbox"/> Renewable Energy Label<br><input type="checkbox"/> N/A                                                                            |
| Project Cycle:                                    | <input type="checkbox"/> Regular<br><input checked="" type="checkbox"/> Retroactive                                                                                                                                                       |

**Table 1 – Estimated Sustainable Development Contributions**

| Sustainable Development Goals Targeted | SDG Impact (defined in B.6.)                                        | Estimated Annual Average | Units or Products       |
|----------------------------------------|---------------------------------------------------------------------|--------------------------|-------------------------|
| 13 Climate Action (mandatory)          | Average Emission Reductions                                         | 6,225                    | tCO <sub>2e</sub> /year |
| 6 Clean water and sanitation           | Average Quantity of safe water distributed to project beneficiaries | 70,626,040               | Litres/year             |
| 8 Decent Work and Economic Growth      | Average Number of jobs created                                      | 91                       | No./year                |

## SECTION A. DESCRIPTION OF PROJECT

### A.1 Purpose and general description of project

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Anakot Thmei Safe Water Project has the objective to provide access to safe and affordable water to rural and peri-urban communities across Cambodia.

Inadequate access to safe drinking water is a continuously health threatening issue to 2.2 billion people<sup>1</sup>, primarily in developing countries. In Cambodia, one on three inhabitants lack access to safe drinking water<sup>2</sup> due to water infrastructures being not available or not reliable enough for local households and communities. Considering more than 60% of Cambodians live in rural areas, decentralized solutions appear as a relevant solution to provide households and communities with such basic needs.

In this context, the “Anakot Thmei Safe Water Project” project consists in establishing community-based water kiosks, which are small-scale water purification and distribution units. The project is thus primarily designed for the long-term improvement of the living conditions of the

<sup>1</sup> <https://www.who.int/news-room/fact-sheets/detail/drinking-water>

<sup>2</sup> <https://www.unicef.org/cambodia/water-sanitation-and-hygiene>

local people of Cambodia. Teuk Saat & 1001f community-based water kiosks are a long-term, sustainable solution that will give vulnerable populations the possibility to fulfill their most essential need: access to safe water. Indeed, many households do not have access to safe water and must boil water to purify it before drinking it. In some cases, inhabitants even drink unsafe water because of lack of knowledge on water quality, lack of financial support either to buy clean water or fuel to boil it or lack of time to do so. Thus, the implementation of the project will provide a safe, efficient, and cost-effective solution to the communities and overcome the challenges they currently face.

Through a system of franchise and creation of local microenterprises, the project will likely create an economic activity that will allow a family to escape extreme poverty (GDP per capita in 2020: 1,513 USD<sup>3</sup>). The social entrepreneurial model will bring social and economic benefits: 4 direct jobs (1 Entrepreneur + 3 Operators) will be created on average for each kiosk, bringing new source of revenue for the households and recruited entrepreneurs who will receive continuous training/coaching and support during the project from 1001fontaines & Teuk-Saat. Also, the durability of the kiosk operation is guaranteed as the operator has a personal interest in keeping the kiosk running. 1001fontaines and TeukSaat will provide technical support, water quality controls as well as coaching to local operators.

Each kiosk has 1 or 2 delivery vehicles or can be more to distribute on daily average 95 bottles to 120 bottles of water (1 water bottle = 20 L). Water is distributed to households by motorbikes equipped with trailers. Delivery vehicles follow pathways in nearby 10-15 villages where they proceed to the water distribution in plastic bottles or jugs of 20L. There is a possibility that some of the households can also visit the kiosk to purchase water.

1001fontaines and its franchise network ensures the water delivery and its quality through continuous monthly tests, following the methodology guidelines WHO. Two samples from the water kiosks are tested on a monthly basis, within one of 1001fontaines laboratories. The water quality test is done on a sample which is directly taken at the outlet of the treatment system. A second sample is taken from a sealed

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<sup>3</sup> <https://data.worldbank.org/indicator/NY.GDP.PCAP.CD>

bottle to ensure that the hygiene procedures guaranteeing water quality until the points of use are respected. At present 1001fontaines and TeukSaat have 4 water testing laboratories across 4 different locations in Cambodia. 1001fontaines and TeukSaat also have 5 local offices in Cambodia.

Moreover, the project has a social goal as it aims at providing safe water at an affordable price for local communities. Price has been chosen to be around 3% of monthly household income which is sufficient to cover water production expenses, general maintenance, and salaries. The price is not too high so that potential beneficiaries chose this solution rather than drinking unsafe water or using more expensive/time-consuming boiling solution.

The technology used is presented in section A.3. It consists of a pretreatment, filtering and purifying water from a local source before conditioning safe water in jugs. In terms of environmental benefits, the use of a local and decentralized system to purify water reduces demand of non-renewable biomass that would have been used to boil the water as a mean of purification in the absence of the water kiosks.

Anakot Thmei Safe Water Project aims at covering all regions across the country through the progressive construction of ~150 water kiosks in the 5 years starting from March 2020, focusing on rural and peri-urban communes where access to safe water is difficult for local inhabitants, because either no installation of piped water systems and sanitation facilities are planned nor current piped water systems matching water quality WHO standards.

## About:

**EcoAct:** EcoAct assists large groups, institutions, associations, territories and governmental bodies in the implementation of carbon strategies and climate actions. EcoAct designs, deploys and pilots creative solutions that transform their societal and environmental challenges into performance levers (change management, feasibility studies, carbon offset programs, etc.). Recognized as one of the best specialists in carbon neutrality, EcoAct daily advises pioneers in the climate change economy to increase their competitiveness, within the framework of measuring, reducing and offsetting their GHG emissions. EcoAct is also developing carbon offset projects on the

field, through its own teams and also thanks to the support of field partners. The project is a voluntary action launched by EcoAct SAS, "Project Developer" and Project Representative.

**1001fontaines:** 1001fontaines is a French NGO that aims at improving the health of vulnerable populations through access to safe drinking water. Present in Cambodia since 2005, 1001fontaines and its local partner Teuk Saat now reach out to more than 800,000 people in the country. 1001fontaines identifies, develops and implement or contribute to the development and implementation of low carbon projects for its account or for its clients and partners in Cambodia. 1001fontaines is Project Participant.

**TeukSaat:** For the last 17 years, TeukSaat 1001 has been pioneering and optimizing community-owned water kiosks, able sustainably to reach the last mile with safe drinking water. TeukSaat 1001 is already providing safe drinking water to hundred thousand of rural families and primary schools but there are still underserved communities for which water kiosk would offer a way toward a healthy thriving population and an improved sustainable local economy.

1001fontaines and its subsidiary TeukSaat oversees installing the water kiosks and managing and operating the water kiosks and the project at ground level and will be called as "Project Implementer". 1001fontaines and TeukSaat will provide technical support, water quality controls as well as coaching to local operators.

## **Roles & Responsibilities:**

### **EcoAct:**

- EcoAct SAS will communicates with Gold Standard on all matters related to the project and owns the GS VERs (Carbon Credits) generated from the project.
- EcoAct will manage the Gold Standard certification process to be in position to obtain the issuance of VERs in respect of GHG Reductions generated by the Project.
- EcoAct will be responsible for the Project registration and for taking any necessary measures to ensure the certification and issuance of VERs.
- EcoAct is entitled to all the VERs that will be generated by the project as part of the project's activities, throughout the life of the project
- EcoAct shall be responsible for carrying out the Gold Standard certification process which comprises (without exclusion) guiding 1001fontaines to conduct

the local stakeholder consultation and writing the related local stakeholder consultation report, drafting, and finalizing the GS Project Design Documentation,

- Contact and retain an independent accredited auditor (e.g., a Verifier or DOE) to review and validate the Project and shall do its best effort to obtain the verification of the Project.
- To ensure that the same approved baseline and monitoring methodology is applied in project.
- Training and capacity development of 1001f, TeukSaat staff

#### **1001fontaines:**

- 1001fontaines will be responsible for the purchase, commissioning of the kiosks, distribution, and monitoring of the safe water, hire of staff and management, implementation of the safe water Program in partnership with local communities and local partners
- General management and pre financing of the kiosk.

#### **TeukSaat:**

- To identify local partners, who'll supply the components as per specifications and materials as communicate by 1001f
- To execute agreement with the kiosk entrepreneur, for transfer of emission reductions in favor of themselves
- Disbursement of incentives/salaries to the Entrepreneur (according to their role and as per 1001f instructions, if any).
- Inspection of installed kiosk, Collection of kiosk data from the field and recording / archiving of collected data
- EX-post Monitoring of kiosks installed for performance, usage as per monitoring requirements set out in the registered monitoring plan
- Training of Entrepreneurs on installation, maintenance and after-sales services of kiosk and maintaining records
- Any other task and responsibilities assigned by CME to the CPA implementer, as and when required.

#### **A.1.1. Eligibility of the project under Gold Standard**

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The project is eligible as it applies the Principles and Requirements (version 1.2, 23<sup>rd</sup> October 2019) and Community Services Activity Requirements (version 1.2, 23<sup>rd</sup> October 2019). It follows an established Gold Standard approved methodology (Emission Reductions from Safe Drinking Water Supply Version 1.0 (03/05/2021)).

As per GS4GGG Principles & Requirements 4.1.3, the project is eligible for Gold Standard Certification as it is associated with Impact Quantification Methodology "Emission reductions from Safe Drinking Water Supply" (version 1.0, 03/05/2021).

Para 4.14 to 4.1.6 of GS4GGG Principles & Requirements is not applicable to this project as project is automatically additional in line with the para 4.1.3 of the GS4GGG Principles & Requirements 4.1.3 (apply Gold Standard approved Impact Quantification Methodology-Emission reductions from Safe Drinking Water Supply" (version 1.0, 03/05/2021).

As per GS4GG Principles & Requirements 4.1.7, the project does not support geo-engineering or energy generation from fossil fuel or nuclear, fossil fuel switch, nor supports, enhances, or prolongs such energy.

As per Para 4.1.49 (b) of Principles and Requirements (version 1.2, October 2019):

*Retroactive projects shall submit the required documents for preliminary review (time of first submission) within one year of the project start date. Retroactive Project submitted at a date later than one year from the project start date will not be eligible for Gold Standard certification.*

**The Project Developer (EcoAct SAS) has requested a deviation from Gold Standard to deviate from Para 4.1.49 (b) of Principles and Requirements (version 1.2, October 2019). The GS has granted deviation from para 4.1.49.(b) of the Principles and Requirements considering the COVID situations<sup>4</sup>.**

As per section 3.1.1 of GS4GG PRINCIPLES & REQUIREMENTS Version 1.2 dated October 2019, eligibility criteria are defined below:

| N° | Eligibility Criterion | Description | Mean of verification |
|----|-----------------------|-------------|----------------------|
|    |                       |             |                      |

<sup>4</sup> Refer DEV\_253 approved on 28/06/2022 from GS. The deviation request can be found here: <https://platform.sustain-cert.com/public-project/2667>



|         |                                          |                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                               |
|---------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.1.1.a | Types of project                         | Eligible projects shall include physical action/implementation on the ground                                                                                                                                                                                             | Project is eligible under Community Services Activity requirements section 3.1.1.(b) as it is providing safe drinking water through community level water treatment technologies (CWT) instead of boiling water using non-renewable biomass                                                                                                                                   |
| 3.1.1.b | Location of Project                      | Projects may be located in any part of the world                                                                                                                                                                                                                         | The location of project is Cambodia, as specified in section A.2                                                                                                                                                                                                                                                                                                              |
| 3.1.1.c | Project area, Project boundary and scale | The Project Area and Project Boundary shall be defined                                                                                                                                                                                                                   | Project area and boundary of project is specified in section B.3. Kiosks will be constructed all over the host country i.e. Cambodia. Project is small scale, with a maximum of 60,000 tCO <sub>2</sub> e ER/year (The ex-ante annual average ERs estimated as 6,225 tCO <sub>2</sub> e/year).                                                                                |
| 3.1.1.d | Host country requirements                | Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations                                                                                                                                                   | Project is in compliance with Cambodia requirements. In particular, the project is in line with the 2019-2023 Nation Action Plan on water access, coordinated by the Ministry of Rural Development. Regarding water quality standards, the project follows rigorously WHO requirements while the Cambodian Ministry of Rural Development is editing Water Quality Guidelines. |
| 3.1.1.e | Contacts details                         | Project Developer shall provide (i) name and (ii) contact details of 7all Project Participants; AND in case of an organisation (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing | Contact details can be found in appendix 2                                                                                                                                                                                                                                                                                                                                    |

|         |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.1.1.f | Legal ownership                                   | Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC)                                                                                                                                           | <p>Legal ownership of the Products generated by the Project is transferred from beneficiaries to EcoAct SAS (project developer).</p> <p>This is transparently explained in the following:</p> <ol style="list-style-type: none"> <li>1. Carbon Credit Transfer Agreements/MoU</li> <li>2. Carbon Credit display poster</li> <li>3. local stakeholder consultation conducted at kiosk level and other as project documentation.</li> </ol> <p>Refer section A.1.2 below for details.</p> |
| 3.1.1.g | Other rights                                      | As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to further project implementation in affected areas | Not applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3.1.1.h | Official Development Assistance (ODA) Declaration | All Project Developers applying for project activities located in a country named by the OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support                                                                                                                                                                                   | The public funding received for the construction of new kiosks do not involve any carbon unit counterpart. Credits generated by the project will not be transferred, neither directly nor indirectly, to the donor country providing support. See ODA declaration.                                                                                                                                                                                                                      |

|                  |                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                               |
|------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4. principle 1.b | Contribution to Climate Security & Sustainable Development: baseline scenario and project scenario | The Project shall define both the Baseline and Project Scenarios                                                                                                                                                                                                                                                      | Baseline scenario is described in B.4. Project scenario is described in section 1 and B.3, B.6 and B.7                                                                                                                                                                                                        |
| 4. principle 1.c | Contribution to climate security & Sustainable development                                         | All Projects shall demonstrate a clear, direct contribution to sustainable development, defined as making demonstrable, positive impacts on at least three Sustainable Development Goals (SDGs), one of which must be SDG 13 (defined herein as Emissions Reductions or Removals and/or Adaptation to climate change) | SDGs impacts are described in B.6                                                                                                                                                                                                                                                                             |
| 4. principle 2   | Safeguarding principles                                                                            | Projects shall conduct a Safeguarding Principles Assessment and conform to Gold Standard Safeguarding Principles and Requirements                                                                                                                                                                                     | Section D and appendix 1                                                                                                                                                                                                                                                                                      |
| 4. principle 3   | Stakeholder Inclusivity                                                                            | Projects shall identify and engage Relevant Stakeholders                                                                                                                                                                                                                                                              | Section E and Stakeholder Consultation Report<br><br>A physical stakeholder consultation meeting was held on 28/06/2022 at Teuk Saat 1001 office in Cambodia. 1001fontaines also conduct physical stakeholder consultation meeting before opening of each kiosk to receive the feedback from local community. |
| 4.1. principle 4 | Demonstration of real outcomes                                                                     | Development of monitoring and reporting plan                                                                                                                                                                                                                                                                          | Section B.7                                                                                                                                                                                                                                                                                                   |

|                 |                                                    |                         |                                                                                                                                                                                                      |
|-----------------|----------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.1 principle 5 | Financial Additionality and Ongoing Financial Need | Financial Additionality | No need to demonstrate the additionality of the project because it is in Cambodia, which is a LDC country <sup>5</sup> as per GS4GG Community Services Activity Requirements, section 4.1.9 (b) p 6. |
| 4.1 principle 5 | Financial Additionality and Ongoing Financial Need | Ongoing Financial Need  | No need to demonstrate the additionality of the project because it is in Cambodia, which is a LDC country as per GS4GG Community Services Activity Requirements, section 4.1.9 (b) p 6.              |

The project is not registered with any other voluntary or compliance scheme.

Also, Cambodia does not have an emission reduction cap enforced nor has the possibility to trade emissions that include the scope of this project.

#### Double counting

There are other safe water projects presents in Cambodia. However, 1001fontaines leads a strong investigation and analysis of the potential area before any rehabilitation or construction of a kiosk. New kiosks are constructed or rehabilitated only if the result of the investigation is that there is no more than 3 competitors within the targeted commune, all with higher prices. Regarding 1001fontaines own kiosks, two kiosks cannot be closer than 10 km one from each other. Teuk Saat & 1001f maintain a kiosk installation database. The kiosk database consists of kiosk location along with GPS coordinates.

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<sup>5</sup> <https://unctad.org/topic/least-developed-countries/list>

Further 1001f confirm the following:

- The “Anakot Thmei Safe Water Project” not included in any other voluntary or compliance standards programme
- The water kiosks that are part of the “Anakot Thmei Safe Water Project” are not included in any other voluntary or compliance project
- Each of the water kiosk included in the “Anakot Thmei Safe Water Project” have a unique kiosk code and its GPS coordinates.

A declaration in this regard is being submitted by 1001f.

## Eligibility under Gold Standard Community Services Activity (CSA) Requirements

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As per section 2 of GS4GG COMMUNITY SERVICES ACTIVITY REQUIREMENTS Version 1.2 dated October 2019, project type eligibility criteria as defined below:

| Eligibility Category                                                | Criteria | Eligibility criterion-- Required condition                                                                                                                                                                                                                                                                                                                                                | Justification                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Eligible Project Types                                           |          | All CSA Projects shall lead to climate change mitigation and/or adaptation by providing or improving access to services/resources at the household or community or institution level. Eligible services include electricity and energy, water and sanitation, waste management, housing, etc.                                                                                             | The goal of the project is to providing safe drinking water to household or community or institution through community water treatment (CWT) technology.                                                                                                                                                                                                                     |
| 2. GENERAL ELIGIBILITY CRITERIA-- Type of project                   |          | <b>(b) End-use energy efficiency:</b> Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end-user of the products and services are clearly identified and when the physical intervention is required at the user end. For example, efficient cooking, heating, lighting, etc. | The project involves providing safe drinking water through community water treatment (CWT) technology in project scenario instead of boiling water using non-renewable biomass in base line scenario. The availability of safe water by CWT avoids the energy requirements as compared to baseline scenario without affecting the level and quality of services or products. |
| 3. GENERAL ELIGIBILITY CRITERIA -- Project Area, Boundary and scale |          | Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements.                                                                                                                                                                                                                                                      | The physical boundary of the project is the households, communities, institutes (schools etc). that use the purified water where water kiosks are installed in Cambodia.<br><br>The project boundary is limited to the geographical boundary of Cambodia.<br><br>The project is small scale project.                                                                         |
| 4. GENERAL ELIGIBILITY CRITERIA --                                  |          | (a) Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting,                                                                                                                                                                                                                                                                     | a) The project beneficiaries transfer their rights on ownership of carbon credits to CME (EcoAct SAS) via                                                                                                                                                                                                                                                                    |

| Eligibility Category   | Criteria | Eligibility criterion-- Required condition                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Justification                                                                                                                                     |
|------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Legal Ownership</b> |          | <p>electricity generation, water treatment technology such as water filter, etc. shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with the FPIC requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided.</p> <p>(b) The transfer of Product ownership shall be discussed during local stakeholder consultations for projects.</p> | <p>agreements / display on the kiosks etc.</p> <p>b) The transfer of Carbon Credits is also discussed during local stakeholder consultations.</p> |

## Eligibility under Gold Standard GHG EMISSIONS REDUCTION & SEQUESTRATION PRODUCT REQUIREMENTS (Optional Requirement)

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As per section 2 of GS4GG GHG EMISSIONS REDUCTION & SEQUESTRATION PRODUCT REQUIREMENTS Version 2.1 dated 24/02/2022, general eligibility criteria as defined below:

| Eligibility Category                     | Criteria | Eligibility criterion-- Required condition                                                                                                                                                                                                                              | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. GENERAL ELIGIBILITY CRITERIA#1</b> |          | Unless otherwise stated elsewhere in the Principles & Requirements, Projects involving a mix of eligible and ineligible components can only claim credits for the Emission Reductions and/or sequestration associated with the eligible component of the project.       | <p>The project involves providing safe drinking water through community level water treatment (CWT) technology in project scenario.</p> <p>The project applies GS approved "Emission reductions from Safe Drinking Water Supply" impact quantification methodologies.</p> <p>Hence as per the GS4GG Principles and Requirements version 1.2 section 4.1.3 (a), the project becomes automatically eligible.</p> <p>The project does not involve any ineligible component.</p> |
| <b>2. GENERAL ELIGIBILITY CRITERIA#2</b> |          | Bundled Projects: Where Projects are submitted together for certification within a bundle, each Project within the bundle shall individually conform to all GS4GG Requirements. Eligibility criteria with regards to the scale of the Project shall apply to the bundle | <p>Not Applicable.</p> <p>The project is an individual project.</p>                                                                                                                                                                                                                                                                                                                                                                                                          |

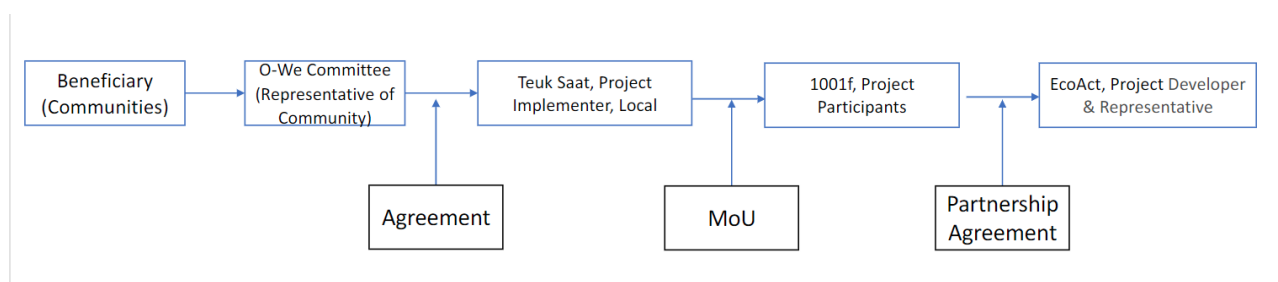
| Eligibility Category                     | Criteria | Eligibility criterion-- Required condition                                                                                                                                                                                                                                                                                                                                                                                                               | Justification  |
|------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
|                                          |          | as a whole and not to the individual Projects.                                                                                                                                                                                                                                                                                                                                                                                                           |                |
| <b>3. GENERAL ELIGIBILITY CRITERIA#3</b> |          | Programme of Activities (PoA): (LUF – N/A) Where a group of Projects are submitted together for Gold Standard Design Certification within a Programme of Activities, each of these Projects shall conform to all Requirements including the Programme of Activity Requirements. A microscale VPA can only be part of a Microscale PoA and shall conform to all requirements including those listed in Annex A of the Programme of Activity Requirements. | Not Applicable |

### A.1.2 Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

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Legal ownership of the Products (GS VERs) generated by the Project is transferred from beneficiaries to EcoAct SAS (project developer). This is transparently explained during the local stakeholder consultation, during the first week launch of the water kiosks, as well as in project documentation i.e. contract agreements. In addition, it is stated through specific posters on water kiosks.

Regarding rights related to water use, kiosks are constructed in places chosen together with local governments at subnational level and villages. More precisely, areas where it is technically feasible and with a lack of access to safe water are targeted. The water sources to be used are chosen altogether with Teuk Saat, local village representatives and commune council. The diagram below explains how the carbon credits transfer from beneficiaries to EcoAct SAS.



#### 1. Carbon Credit Transfer from Communities to O-We Committee:

- O-We committee represent the communities (project beneficiaries).

- b) O-We Committee, local authorities, who are elected, recommend people to Teuk Saat to sit at the O-We committee. It can be director of the school, etc. Anyone the local authorities fit to represent the local communities. There is formal agreement between community and the O-We committee.
- c) Key community leader meeting before start of operation of each kiosk and the transfer of Carbon credits discussed in the meeting
- d) There is one O-We committee per kiosk

## 2. **Carbon Credit Transfer Agreement from O-We community to Teuk Saat**

- a. Old agreement between O-We Committee & Teuk Saat Carbon waiver for kiosks installed from March 2020 to Feb 2022.
- b. New agreement between O-We Committee & Teuk Saat that includes the carbon credit ownership clause for kiosks implemented from March 2022 onwards.
- c. The old & new agreement have clause related to land ownership and for supply of water for kiosk (commune council own the land and water supply authority)

## **Carbon Credit Transfer from Teuk Saat to 1001fontaines**

- a. MoU between Teuk Saat & 1001fontaines.
4. Carbon Credit Transfer from 1001fontaines to EcoAct
- b. a. Partnership Agreement between EcoAct & 1001fontaines

## **A.2 Location of project**

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The Project aims at implementing the local water kiosk solution in all provinces in the Kingdom of Cambodia.

**Host Country:** Cambodia

**Region/State/Province:** All within host country (Cambodia)

**City/town/community:** All within host country (Cambodia)





**Image: Map of Cambodia (host country)**

Kiosks are constructed by Teuk Saat on public land (municipal land) in coordination with local authorities and local communities of the rural and peri-urban areas.

A list of kiosks installed, and their exact locations is available in the installation database<sup>6</sup>. The coordinates of the Teuk Saat office in Phnom Penh are: N11.537767, E104.913466<sup>7</sup> This list will be updated for every new kiosk commissioned during project operation for monitoring purposes. It is foreseen *ex-ante* that 150 kiosks estimated to be developed and the the project yielding the maximum cap of 60,000 tCO<sub>2e</sub> per year.

Estimation of Water Kiosks in first crediting period

| Year           | Period                | Number of kiosks built <sup>8</sup><br>(As on 30/03/2023) | Cumulative number of<br>kiosks built<br>(As on 30/03/2023) |
|----------------|-----------------------|-----------------------------------------------------------|------------------------------------------------------------|
| 0 <sup>9</sup> | 09/03/2020-05/06/2021 | 40                                                        | 40                                                         |
| 1              | 06/06/2021-08/06/2022 | 37                                                        | 77                                                         |
| 2              | 06/06/2022-08/06/2023 | 12                                                        | 89                                                         |

<sup>6</sup> Will be submit to the VVB.

<sup>7</sup> St 95 , #100 Sangkat Boeng Trabek, Khan Chamkamorn Phnom Penh, 12300, Cambodia

<sup>8</sup> The number of kiosks installation data given in row 1 to 3 is actual installation data while the number given from row 4 to 5 is estimated to be install (not installed yet).

<sup>9</sup> Year is 0 is the period between project start date and project crediting period start date. ER from this period are not claimable as they are not part of the crediting period. But kiosks installed during this period can be included within project scope as they are installed after project start date.

|   |                       |    |     |
|---|-----------------------|----|-----|
| 3 | 06/06/2023-08/06/2024 | 30 | 119 |
| 4 | 06/06/2024-05/06/2025 | 35 | 154 |
| 5 | 06/06/2025-05/06/2026 | 0  | 154 |

In terms of site selection, municipalities where kiosks are implemented are selected using the criteria as below:

- Families: at least 1000 families or households,
- Population: at least 4,500 people covered in 5 Km radius or more
- Density:
- Water sources: round year, usage, and tested it with minimum WHO Standard for water quality as set,
- Road access to the villages of the municipalities: accessibility for delivery,
- Location: land available for the kiosk is not nearby grave, funeral site, pagoda, close to the health center, not far from the village, practical road access to the kiosk enabling deliveries,
- Distance from existing kiosk > 10Km,
- Distance to access water source from potential kiosk (maxi. 250m),
- Access electric grid connection from potential kiosk (maxi. 200m),
- Land size (15mx15m) and soil consolidation/filling,
- Land for the kiosk belongs to the municipality.

After such a selection of sites is done, final site validation is based on data analysis of water testing and discussions with 1001fontaines' site selection committee: the latter is composed of community facilitator, technical team, water quality team, and the business team. Then 1001fontaines signs a cooperation agreement with the relevant municipality authorities.

### A.3 Technologies and/or measures

>>

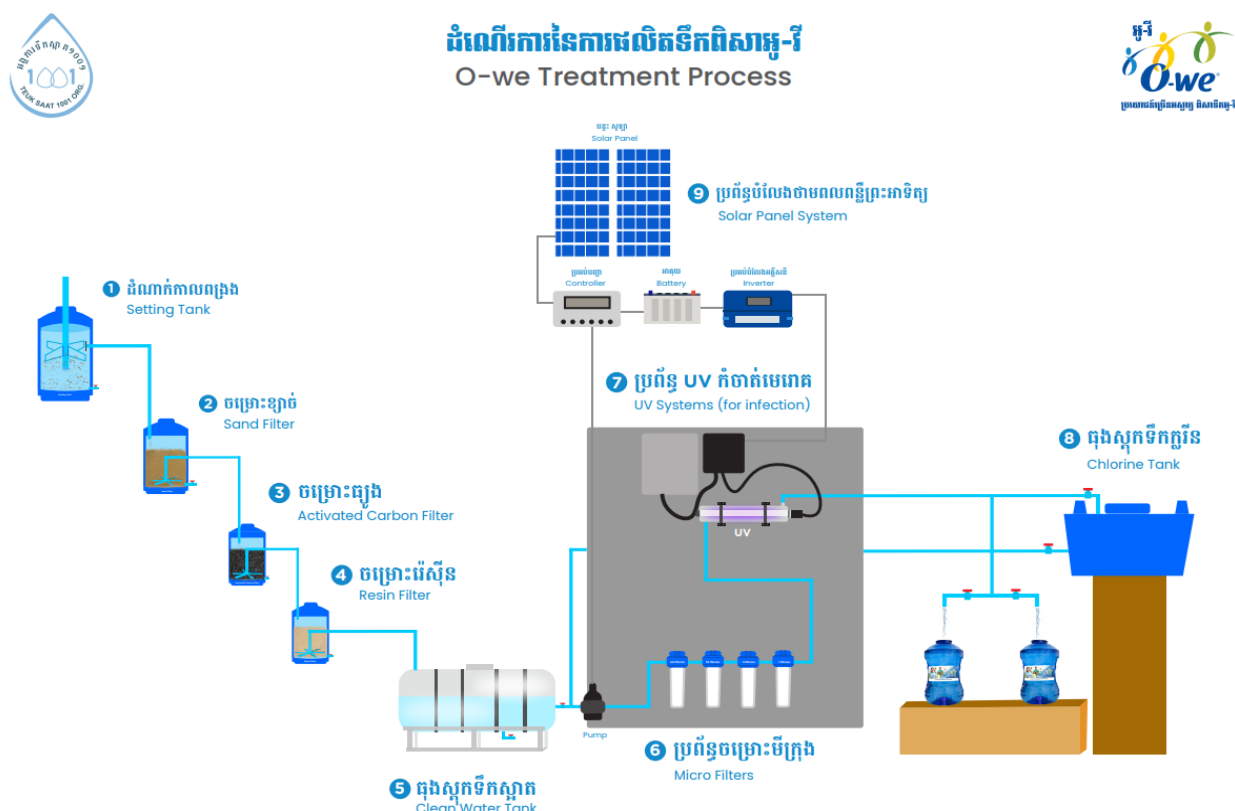
The project aims at implementing and operating water kiosks. It is a community level water treatment technologies (CWT) that is defined – as per the Emission reductions from Safe Drinking Water Supply v.1.0 – as a “Technologies that treat water at a central point to obtain safe water”. The safe water may be supplied directly to the premises of users, or users may retrieve water from the central point or another distribution point (motorbikes or trucks). Water to be treated comes from an existing water supply or from a community water supply technology. In this project's case, the purification of

water is made within the kiosk and water is distributed to local households at affordable price.

The water-treatment has been enabled by years of experimentation in Cambodia and follows the steps below:

- Locally available water is pumped from the nearby source (ponds, rivers, wells...)? Water source is not more than 250 m away from the kiosk.
- Water then goes through a multistep filtration process (see graphic below) and is purified with a UV-lamp to reach WHO standards defined in the SDWS v1.0 p.4 (World Health Organization. (2017). Guidelines for Drinking-water Quality: fourth edition incorporating the first addendum. Geneva: World Health Organization) for producing safe and clean drinking water.
- Purified water is conditioned in 20-liter sanitized and sealed jugs, providing 2 to 3 days of consumption for an average household (4-5 family members).
- Jugs are delivered directly to the homes of beneficiaries, or they can buy from kiosk.
- Empty jugs are collected, cleaned/disinfected, and reused.
- Water quality is monitored every month to comply with the World Health Organization's standards.

Figure 1 Water flow diagram from pumping to distribution in 1001 Water Kiosk.



A typical kiosk has the following components installed:

| Set up 2020-1           | Description/Items                          | Unit | Item description              | Description of use              | Qty per site                |
|-------------------------|--------------------------------------------|------|-------------------------------|---------------------------------|-----------------------------|
| Micro Filters Cartridge | Filter 1 µm ( មីក្រូ ហ្វីលធីរ ១មីក្រូន )   | Pc   | AFIMO,puomelt                 | microfiltration / purification  | 3                           |
|                         | Filter 5 µm ( មីក្រូ ហ្វីលធីរ ៥មីក្រូន )   | Pc   | AFIMO,puomelt                 | microfiltration / purification  | 3                           |
|                         | Filter 10 µm ( មីក្រូ ហ្វីលធីរ ១០មីក្រូន ) | Pc   | AFIMO,puomelt                 | microfiltration / purification  | 3                           |
|                         | Filter 50 µm ( មីក្រូ ហ្វីលធីរ ៥០មីក្រូន ) | Pc   | AFIMO,puomelt                 | microfiltration / purification  | 3                           |
| Resin                   | Resin                                      | sack | 25 kg per sack, Resinex KW-8  | material filtration tank 2      | 2                           |
| Activated Carbon        | Activated Carbon                           | sack | 25 kg per sack, Kobon KB 850  | material filtration tank 3      | 4                           |
| Sand                    | Sand (size 1)                              | sack | 20 kg per sack (size 12-18mm) | material filtration tank 1      | 8                           |
|                         | Sand (size 2)                              | sack | 20 kg per sack (size 8-12mm)  | material filtration tank 1      | 12                          |
|                         | Sand (size 3)                              | sack | 20 kg per sack (size 4-8mm)   | material filtration tank 1      | 20                          |
| Storage                 | Plastic tank 200L                          | Pc   | 200 litre                     | Tank for cleaning bottle, equip | 2                           |
|                         | Plastic Tank 300L                          | Pc   | 300 litre                     | Resin & activated charcoal      | 2                           |
|                         | Plastic Tank 700L                          | Pc   | 700 litre                     | Sand tank                       | 1                           |
|                         | Plastic Tank 1500L                         | Pc   | 1500 litre                    | Flocculation Tanks              | 2.00                        |
|                         | Stainless Steel tank 2000L                 | Pc   | 2000 litre                    | Clean water tank                | 1                           |
| UV                      | UVc Reactor & lamp & controller            | Pc   | Viqua VH410M                  | purification                    | 1                           |
| Ballast                 | Ballast controller                         | Pc   | Viqua BA-ICE-CM               | purification                    | 1 (included in UV lamp set) |
| Pipe and fitting        | PVC Pipe and fitting                       | Pc   | Material UPVC, brand SCG      | filtration/purification         | As needed                   |
| Solar                   | PV Solar Panel                             | Pc   | Monocrystalline 100 Wp,12V    | solar                           | 2                           |
|                         | Solar charge controller                    | Pc   | 20A PWM (12/24 Auto)          | solar                           | 1                           |
| Electric pump           | DC Pump (12v), Seaflo ( ម៉ូទ័រ Seaflo )    | Pc   | SFDP1-050-060-51              | microfiltration                 | 1                           |
| Battery                 | Gel battery                                | Pc   | 12v,150Ah, Narada 12NDT150S   | solar                           | 1                           |
| Regulator               | Inverter 500watt ( អាំងវឺតទ័រ )            | Pc   | Convertor from DC to AC       | For UV power supply             | 1                           |

Here is following the specifications for UV lamp, motor pump, solar panel, and designed capacity that are part of the kiosk's system.

Table 4 : Component's specifications of the kiosks

| Hardware specification |                                                             |
|------------------------|-------------------------------------------------------------|
| UV lamp                | Lamp power: 36W, 6A<br>Life time: about 8000 hours of using |
| Motor pump             | Maximum flow rate: 1500 l/min<br>Life time : 3 years        |
| Solar panel            | 12 VDC supply with regulator<br>Life time: 20 years         |
| Designed capacity      | 4,800 l/day <sup>2</sup>                                    |

Please note that the UV lamps use in water kiosks have an 8,000 hour of usage life expectancy and which can be replaced after the 8,000 hours of operation at multiple times. Further the water kiosks use sand filters and other kind of filters for water filtration that can be easily replaced and with such replacement, kiosk provide safe water quality and new life to the system. Hence the water kiosks have a life of more than 15 years as it renewed every time on maintenance and replacement.

For villagers to be able to pay for drinking water, it is necessary to reach the lowest production cost. This explains certain technical choices, such as using surface water (or limited depth well drillings) or resorting to a technology (i.e., UV disinfection) that can be supplied by solar energy. Above all, a key element is to create local production capacities devoted to local consumption only, which allows reducing distribution costs to almost nothing.

Water kiosks rely on 100% renewable energy provided by solar panels installed on the buildings. For the other usage, kiosks purchase grid electricity.

The social business model consists in the creation of a microenterprise for each kiosk: the operator runs the facility that purifies water, bottle it in 20L jugs and distribute/sell it to surrounding inhabitants. The sales should cover employee salaries as well as maintenance and operational costs.

### Contribution to identified SDGs

**SDG 6:** This project provides, thanks to the distribution following production with the kiosks, access to safe drinking water to surrounding inhabitants of each kiosk, positively contributing directly to #SDG6. The impact will be followed through the indicator "Quantity of safe drinking water distributed"

**SDG 8:** This technology and business model enable economic dynamism and creation of jobs for each kiosk through microenterprise creation, directly contributing to #SDG8. The sustainability of the business model ensures a long-term activity for the microentrepreneurs. The impact will be followed through the indicator “Number of jobs created”

**SDG 13:** In addition, the process for making water safe is much more efficient than water boiling with cookstoves households often use, as proved by the Emission Reduction calculations. The project thus directly contributes to #SDG13. The impact will be followed through the emission reduction calculations.

#### A.4 Scale of the project

>>

Small Scale, project type III

The project meets the project activity requirements for a small-scale project. According to the Community services activities 3.1.2, and precised by GHG Emissions Reduction & Sequestration Product Requirements (version 2.1, 24/02/2022) 9.1.2.c., small scale is defined by the 60,000 ton CO<sub>2e</sub> per year threshold<sup>10</sup>. Indeed, emission reduction of this project will not exceed 60,000 tCO<sub>2e</sub> in any year of the crediting period. The estimated number of VERs calculated ex-ante are described in part B.6.4 for the SDG 13.

As per the paragraph 3.4.2 of the methodology selected for water technologies, Methodology for Emission Reductions from Safe Drinking Water Supply, ver 1.0 states that:

*For household end-users currently drinking unsafe water, the principles of suppressed demand are applied, such that the general baseline scenario is assumed to be that users would have boiled water for drinking in the absence of the project activity<sup>11</sup>.*

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<sup>10</sup> <https://globalgoals.goldstandard.org/501-pr-ghg-emissions-reductions-sequestration/>

<sup>11</sup> As per GS4GG Principle and Requirements, application of suppressed demand baseline is limited to Micro & Small-scale activities. In Refer to GHG emission reduction & sequestration product requirements for definition. For the purpose of this methodology the small scale project applies Type III definition i.e, GHG emission reduction for small scale project shall not exceed more than 60,000 ton CO<sub>2e</sub> in any year of the crediting period.

Hence, as per the applied methodology the suppress demand baseline is allowed for Small-scale project activities. As of now the PD is only developing small scale project for the water technologies hence the suppress demand is allowed and no assessment is required.

### **A.5 Funding sources of project**

>>

There is no public or ODA funding for this project activity, all revenue for the project will be derived from the sales of VERs and the sales of water.

Incomes from the sale of water grossly cover operational costs (including microentrepreneur salaries). Incomes from the sale of VERs finance the construction of the new kiosks or rehabilitation of old kiosks, it covers high investments costs. Indeed, new kiosks could not be built without the revenues from sales of VERs.

## SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

### B.1. Reference of approved methodology (ies)

>>

**GHG Baseline & Monitoring Methodology:** This project applies for the Gold Standard "Methodology for emission reductions from Safe Drinking Water Supply v.1.0" published on the 03/05/21<sup>12</sup>.

**Standardized baseline:** N/A

**Mandatory GS Guideline:** N/A

### B.2. Applicability of methodology (ies)

>> The methodology is applicable under the following conditions. It is demonstrated for each applicability condition that the project meets these conditions. As per para 2.2.1 of "Methodology for emission reductions from Safe Drinking Water Supply v.1.0" published on the 03/05/21":

**Table 5**

| Applicability condition                                                                                                                                                                                                                                                          | Explanation                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <i>"Eligible household water treatment technologies (HWT), institutional water treatment technologies (IWT), and community level water treatment technologies (CWT) include bleach/chlorine, water filter (ceramic, sand, composite, membrane, etc.), UV disinfection, etc."</i> | As explained in the part A.3 of this PDD, the project technology used is the community level water treatment technologies (CWT). |
| <i>"Eligible community water supply technologies (CWS) include new installation of new borehole hand-pumps, borehole hand-pumps rehabilitation, solar powered"</i>                                                                                                               | Not applicable as the project is CWT.                                                                                            |

<sup>12</sup> <https://globalgoals.goldstandard.org/429-ee-sws-emission-reductions-from-safe-drinking-water-supply/>



|                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>drinking water pumps, etc. Water pumps powered by fossil-fuel engines are not eligible, with the exception of backup fossil-fuel engines that are used for no more than 10% of operating hours (parameter SWDS 33)."</i></p>                                                                                          |                                                                                                                                                                                                                                                                                                                                                          |
| <p><i>"All projects involving CWT and CWS technologies must also include ongoing maintenance and repair of the project technology".</i></p>                                                                                                                                                                                 | <p>Each kiosk will be maintained by a local entrepreneur who will be trained. This entrepreneur will assure the maintenance and repair of kiosk, following the methodology of monitoring explained in part B.7.</p>                                                                                                                                      |
| <p><i>"Where the project involves the rehabilitation of an existing technology, the project developer shall provide evidence that the existing technology is non-operational and that there is no planned maintenance or repair for at least 3 months after the date it became non-operational (parameter SWDS 2)."</i></p> | <p>Please note that all the kiosk installed till now (30/03/2023) are newly built. But in future 1001f have plans to rehabilitate some of the old kiosks they have installed earlier. In the rehabilitation of old kiosk, project developer/implementer will provide the evidence to prove the non-operationality of the old kiosk in current state.</p> |
| <p><i>"This methodology allows for project activities to include safe water treatment and/or supply technologies implemented for end-users in households, and/or commercial premises such as shops or institutional premises including half or full day/boarding schools, prisons, army camps &amp; refugee camps."</i></p> | <p>This methodology will serve individual households, community and could also be provide safe water to commercial premises such as shops or institutional premises including half or full day/boarding schools, prisons, army camps &amp; refugee camps.</p>                                                                                            |
| <p><i>"In cases where the safe water is retrieved at the CWT or CWS location, the water in its improved form shall be available within a distance of 1 km or less from the end-users, as demonstrated by satellite imaging or GPS</i></p>                                                                                   | <p><b>As per the CWT definition on page 3 of the methodology the distribution's point (distribution vehicle's) can be considered as a source of safe water.</b> Thus, the project end-users (households, schools, commercial establishments etc.) will be within</p>                                                                                     |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>coordinates<sup>13</sup> of each CWT or CWS location. Alternatively, as a proxy, a total collection time of 30 minutes or less for a round trip, including queuing, using the travel modes of walking or pedaling may be demonstrated (parameter SDWS 1)."</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>1 km or less from these distribution points (motorbike, trucks etc.).</p> <p>Water is distributed at the kiosk's location but also via an efficient distribution network (trucks, motorcycles etc.) to households. Each kiosk is indeed equipped with 1-2 vehicles or more that distribute water to households within a radius of at least 5 km. The number of households potentially served by the distribution system will be therefore calculated. There is a possibility that some of the households can also visit to the kiosk to purchase water.</p> |
| <p><i>"Project technology performance level (HWT and IWT): It shall be demonstrated based on report of laboratory testing<sup>14</sup> or official notification<sup>15</sup> that the project technology or equipment achieves either (i) the performance target classification 3-star or 2-star level, meaning "Comprehensive Protection," as per the WHO International Scheme to Evaluate Household Water Treatment Technologies<sup>16</sup> (World Health Organization, 2011) or (ii) compliance with the national standard or guideline<sup>17</sup> for household drinking water treatment technology; if no national guideline or standard is available, then the project technology shall comply with</i></p> | <p>Not concerned as the project is a Community Level Water Treatment Technology (CWT).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

<sup>13</sup> Acceptable formats for GPS coordinates include DMS (degrees, minutes and seconds), DMM (degrees and decimal minutes), and DD (decimal degrees).

<sup>14</sup> The testing should be undertaken under conditions that are representative of the operation conditions of the project site(s) including feedwater.

<sup>15</sup> For example notifications from the national authority on health.

<sup>16</sup> <https://www.who.int/home/cms-decommissioning>, WHO Refer to the list of products tested by WHO [https://www.who.int/water\\_sanitation\\_health/waterquality/household/hwts-products-evaluated/en/](https://www.who.int/water_sanitation_health/waterquality/household/hwts-products-evaluated/en/)

<sup>17</sup> The national standard or guideline shall be based on laboratory efficacy testing that, at a minimum, includes quantitative microbial measures of pre- and post-treatment challenge waters that are representative of potential drinking water sources, and that includes measured reductions based on at least one pathogen class (bacteria, viruses, protozoa). "Challenge water" is synonymous with "test water". This is the experimental water that has been spiked with microbes (a "microbial challenge") in order to demonstrate the potential for the technology to reduce microbes.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>the WHO International Scheme requirements as per (i) (parameter SDWS 2)."</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p>Project technology performance level (CWT and CWS): For each individual CWT or CWS, it shall be demonstrated at the start of each crediting period with water quality testing reports that the water directly supplied by the project water technology/source achieves both:</p> <ul style="list-style-type: none"> <li>i. microbial quality in line with either (i) national standards or guidelines for microbial quality of drinking water, or in the absence of such requirements, (ii) the guideline values for verification of microbial quality from the Guidelines for drinking-water quality (Table 7.10, WHO, 2017)<sup>18</sup>; and</li> <li>ii. compliance with (i) national standards or guidelines on priority chemical contamination and physical and aesthetic aspects, or in the absence of such requirements, (ii) international standards or guidelines on priority chemical contamination<sup>19</sup> and physical and aesthetic aspects. (parameter SWDS 3).</li> </ul> | <p>Project compliant with Cambodia water quality requirements. In particular, the project is in line with the 2019-2023 Nation Action Plan on water access, coordinated by the Ministry of Rural Development. Regarding water quality standards, the project follows rigorously WHO requirements while the Cambodian Ministry of Rural Development is editing Water Quality Guidelines. The monitoring part B.7 will describe the process of annual monitoring concerning water quality.</p> |
| <p><i>"The project must conduct annual water hygiene education campaigns for the end-users. (parameter SDWS 20)."</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>The monitoring part B.7 will describe the realization of annual water hygiene education campaigns.</p>                                                                                                                                                                                                                                                                                                                                                                                    |
| <p><i>"A project applying this methodology may make SDG claims if relevant monitoring parameter(s) is included in the monitoring plan to demonstrate and confirm the</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>The project will make SDG claims and same is included in the monitoring plan. Refer B.7 for details.</p>                                                                                                                                                                                                                                                                                                                                                                                  |

<sup>18</sup> World Health Organization. (2017). Guidelines for Drinking-water Quality: fourth edition incorporating the first addendum. Geneva: World Health Organization.

<sup>19</sup> At the global level, the priority chemical contaminants are arsenic and fluoride. In absence of relevant national standards, compliance with the WHO guideline values (maximum 10 µg/L and 1.5 mg/L, respectively) shall be demonstrated.

project’s contributions to SDGs<sup>20</sup>. See parameter SDWS  
 19”

<sup>20</sup> <https://www.sdg6monitoring.org/indicator-611/>

Compliance with the regulatory national framework for safe water supply Table 2 - regulation for safe drinking water supply

| Regulations or guidance for safe drinking water supply                                                                                                                                                                                                                                                                                                                                                                                                                     | Project compliance with the regulatory framework                                                                                                                                        | Link                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Law on Environmental Protection and Natural Resource Management 1996 (MoE)</p> <p>The Ministry of Environment shall collaborate with the concerned ministries to require the owners or responsible of the factories, pollution sources, industrial zones or those zones which have natural resource development activities to: - install or use of monitoring equipment, - provide samples; - prepare or keep files and submit records and reports for examination.</p> | <p><i>The project includes monitoring activities ensuring the provision of samples, files, records, and reports for examination.</i></p>                                                | <p><a href="https://cambodia-redd.org/wp-content/uploads/2013/05/LAW-1296-36-96-Environmental-Protection-Natural-Resources-Mgt-E.pdf">https://cambodia-redd.org/wp-content/uploads/2013/05/LAW-1296-36-96-Environmental-Protection-Natural-Resources-Mgt-E.pdf</a></p>                           |
| <p>Sub-decree #27 on Water Pollution Control 1999 Ministry of Water Resources and Meteorology (MoWRM)</p> <p>CHAPTER 5: Water Pollution - Monitoring in Public Water Areas<br/>Article 25: The water quality standard of public water areas for the purpose of the protection of public health is stipulated in the Annex 5 of this sub-decree. (see below the table)</p>                                                                                                  | <p><i>The water distributed by the project respects the water quality standards of public water areas for the purpose of the protection of the public health as shown in SDWS 3</i></p> | <p><a href="http://www.vertic.org/media/National%20Legislation/Cambodia/KH_Sub-decree%20No%2027%20on%20the%20Control%20of%20Water%20Pollution.pdf">http://www.vertic.org/media/National%20Legislation/Cambodia/KH_Sub-decree%20No%2027%20on%20the%20Control%20of%20Water%20Pollution.pdf</a></p> |
| <p>Law on Water Resource Management 2007 (MoWRM)</p>                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                         | <p><a href="https://portal.mrcmekong.org/assets/v1/documents/Cambodian">https://portal.mrcmekong.org/assets/v1/documents/Cambodian</a></p>                                                                                                                                                       |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The Royal Government may grant a reward or incentives to those who engage in research on, or the development of, new technologies, installations of modern equipment, that will contribute to the reduction of waste and improvement in water quality, and increase water use efficiency.</p>                                                                                                                                                                                                   | <p><i>The objectives of the project meet the expectations of developing installations of modern equipment, which will contribute to the improvement of water quality, and increase water use efficiency.</i></p>                                                  | <p><a href="#">-Law/Law-on-Water-Resources-Management-(2007).pdf</a></p>                                                                                                                                                                                                                                                                                                        |
| <p>National Strategy on Rural Water Supply, Sanitation and Hygiene 2011-2025 Ministry of Rural Development (MRD)</p> <p>Promote water quality safeguard:<br/>Promote the practice of safe handling of water from source to point of use by effectively disseminating the MRD's Hygiene Promotion guidelines.</p>                                                                                                                                                                                   | <p><i>The water distributed by the project respects the water quality standards of public water areas for the purpose of the protection of the public health as shown in SDWS 3.</i></p>                                                                          | <p><a href="https://data.opendevelopmentcambodia.net/laws_record/national-strategy-for-rural-water-supply-sanitation-and-hygiene-2011-2025">https://data.opendevelopmentcambodia.net/laws_record/national-strategy-for-rural-water-supply-sanitation-and-hygiene-2011-2025</a></p>                                                                                              |
| <p>Sub-decrees supporting the law of 2007 (MoWRM, 2015)<br/>Asian Development Bank (ADB)<br/>Sub-decree on water quality (drafting process).<br/>The drafting of sub-decrees for water licensing and water quality monitoring, submitted by MOWRAM to the Council of Ministers for approval by the Prime Minister, although not yet approved.</p> <p>September 2021: The Industry Ministry has finished the draft Law on Clean Water Management and will submit it to the Council of Ministers</p> | <p><i>The water distributed by the project respects the water quality standards of public water areas for the purpose of the protection of the public health as shown in SDWS 3 and will follow the new water quality standards that are being developed.</i></p> | <p><a href="https://www.adb.org/sites/default/files/project-documents/38558/38558-013-pcr-en.pdf">https://www.adb.org/sites/default/files/project-documents/38558/38558-013-pcr-en.pdf</a></p> <p><a href="https://www.khmertimeskh.com/50930397/draft-clean-water-law-ready-for-review/">https://www.khmertimeskh.com/50930397/draft-clean-water-law-ready-for-review/</a></p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| for review. The draft law has 15 Chapters and 91 Articles. The initiative of the draft law was started in 2016.                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                  |
| <p>DRINKING WATER QUALITY STANDARDS (DWS) for Cambodia (2004)</p> <p>The Royal Government of Cambodia has established a comprehensive policy on National Water Supply and Sanitation, covering both urban and rural water supplies. These standards are developed by an inter-ministerial process initiated by Ministry of Industry, Mines and Energy and concerned ministries with support from the World Health Organization (WHO).</p> | <p><i>As the DWS are applicable as the minimum requirement to all sources of drinking water in both urban and rural areas, public or private water supply regardless of its source, the water distributed under the project ensures that it meets all the requirements of the DWS (respect of microbial contamination, inorganic and organic constituents, microbiological, physical, chemical, aesthetic parameters, timing and frequency of sampling, sampling and analysis methods, sanitary inspection and survey, reporting).</i></p> | <p><a href="https://rdic.org/wp-content/uploads/2014/12/MIME-Drinking-Water-Quality-Standards-2004-en.pdf">https://rdic.org/wp-content/uploads/2014/12/MIME-Drinking-Water-Quality-Standards-2004-en.pdf</a></p> |

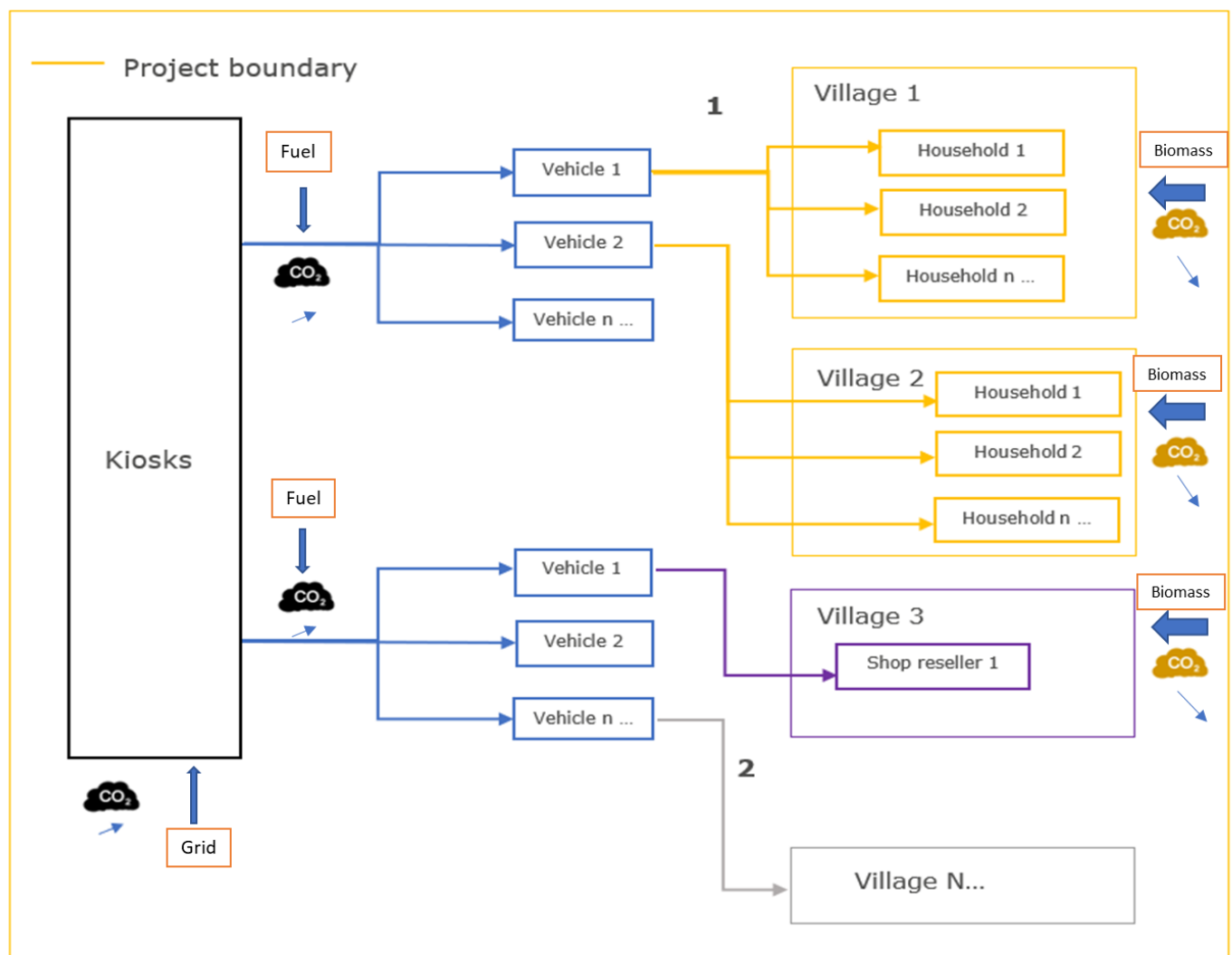
### B.3. Project boundary

>>

The physical boundary of the project is shown in section A.4 of this document, which is the rural and peri-urban zones in Cambodia. Each kiosk has GPS location as listed in kiosk installation database.

The scheme below provides an overview of the project boundary, whereby the emission related to the use of non-renewable firewood for making water drinkable are included in the baseline scenario and the emissions from transport are considered in the Baseline scenario.

Figure 2 - Project boundary from water kiosks to final distribution to end users





## Legend



CO<sub>2</sub>e Emissions released linked to the transport and electricity consumption from the grid



CO<sub>2</sub>e emissions reduced due to the project

Energy source (Electricity from the grid, Fuel, Biomass)



1. Transport of safe water by containers of 20L from kiosks to beneficiaries' households, and recuperation of the empty containers for reuse after cleaning it.
2. Some shops are intermediaries to distribute safe water from kiosks to end-user (households, schools, commercial establishments etc.). In this case, conditioned water is transported to the shops where beneficiaries can come and buy it.

Moreover, the technologies that are part of the project are not included in another voluntary market or CDM project activity (i.e no double-counting takes place) within the project boundary.

Table 6

| Source            |                                                                                    | GHGs                        | Included?       | Justification/Explanation      |
|-------------------|------------------------------------------------------------------------------------|-----------------------------|-----------------|--------------------------------|
| Baseline scenario | Emissions from wood                                                                | CO <sub>2</sub>             | Yes             | Major source of emissions      |
|                   | fuels utilized for obtaining safe drinking water displaced due to project activity | CH <sub>4</sub>             | Yes             | Negligible source of emissions |
|                   |                                                                                    | N <sub>2</sub> O            | Yes             | Negligible source of emissions |
|                   |                                                                                    | Emissions from fossil fuels | CO <sub>2</sub> | Yes                            |
|                   | utilized for obtaining safe drinking water displaced due to project activity       | CH <sub>4</sub>             | No              | Negligible source of emissions |
|                   |                                                                                    | N <sub>2</sub> O            | No              | Negligible source of emissions |
| Project scenario  | Emissions from electricity for                                                     | CO <sub>2</sub>             | Yes             | Minor source of emissions      |
|                   |                                                                                    | CH <sub>4</sub>             | No              | Negligible source of emissions |

|  |                                                                                     |                  |     |                                                                                  |
|--|-------------------------------------------------------------------------------------|------------------|-----|----------------------------------------------------------------------------------|
|  | Operating project water supply/treatment technology                                 | N <sub>2</sub> O | No  | Negligible source of emissions                                                   |
|  | Emissions from fossil fuels for operating project water supply/treatment technology | CO <sub>2</sub>  | Yes | Minor source of emissions: Limited fuel energy maybe required for transportation |
|  |                                                                                     | CH <sub>4</sub>  | No  | Negligible source of emissions                                                   |
|  |                                                                                     | N <sub>2</sub> O | No  | Negligible source of emissions                                                   |

#### B.4. Establishment and description of baseline scenario

As per the section 3.4 of applied methodology "Emission reduction from safe Drinking Water supply"" version 1.0 published on 03<sup>rd</sup> May 2021:

- ***the general baseline scenario is that users would have boiled water for drinking in the absence of the project activity.***

Hence, this "target population" that "would have boiled water for drinking in the absence of the project activity" is a representative baseline for the project.

In addition of this target population, the principles of suppressed demand<sup>21</sup> applies for "household end-users currently drinking unsafe water, such that the general baseline scenario is assumed to be that users would have boiled water for drinking in the absence of the project activity<sup>22</sup>" as described in the same methodology.

#### Justification of the Baseline scenario

As mentioned in the methodology, to establish the baseline scenario, the project proponent is required to carry out the following studies, whose implementation and results are presented after:

##### *1) Baseline non-renewable biomass (NRB) assessment*

<sup>21</sup> <https://www.goldstandard.org/project-developers/standard-documents>

<sup>22</sup> As per GS4GG Principle and Requirements, application of suppressed demand baseline is limited to Micro & Small scale activities. In Refer to GHG emission reduction & sequestration product requirements for definition. For the purpose of this methodology the small scale project applies Type III definition i.e, GHG emission reduction for small scale project shall not exceed more than 60,000 ton CO<sub>2</sub>e in any year of the crediting period

As the project activity is employing a non-renewable biomass (NRB) baseline, the fractional non-renewability of biomass ( $f_{NRB}$ ) must be identified.

The fraction of non-renewable biomass ( $f_{NRB}$ ) for Cambodia has been calculated as **52.03%**. At the time of writing the PDD, the UNFCCC default values for Cambodia published in EB 67 Annex 22<sup>23</sup> had expired. Hence, the PP has calculated the  $f_{NRB}$  value in line with the CDM Tool 30: Calculation of the fraction of non-renewable biomass version 3.0<sup>24</sup>. The  $f_{NRB}$  calculation procedure is given below.

**Step 1: Estimation of renewable biomass (RB) in the country of Cambodia as per equation (4) of the  $f_{NRB}$  tool:**

The quantity of renewable biomass available in the applicable area (RB) is estimated using the following equation:

$$RB = \sum (MAI_{forest,i} \times (F_{forest,i} - P_{forest,i})) + \sum (MAI_{other,i} \times (F_{other,i} - P_{other,i})) \quad \text{Equation 1}$$

**Where:**

- $MAI_{forest,i}$  = Mean Annual Increment of woody biomass growth per hectare in subcategory i of forest areas in the relevant period (tonnes/ha/yr)
- $MAI_{other,i}$  = Mean Annual Increment of woody biomass growth per hectare in subcategory i of other land areas in the relevant period (tonnes/ha/yr)
- $F_{forest,i}$  = Extent of forest in sub-category i in the relevant period (ha)
- $F_{other,i}$  = Extent of other land in sub-category i in the relevant period (ha)
- $P_{forest,i}$  = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (in subcategory i) in the relevant period (ha)
- $P_{other,i}$  = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within other land areas (in subcategory i) in the relevant period (ha)

<sup>23</sup> [https://cdm.unfccc.int/Reference/Notes/meth/meth\\_note12.pdf](https://cdm.unfccc.int/Reference/Notes/meth/meth_note12.pdf)

<sup>24</sup> [https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v1.pdf/history\\_view](https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v1.pdf/history_view)

- $i$  = Sub-category  $i$  of forest areas and other land areas

#### Forest & Other Wooded Land Area:

| Description                   | Value (ha) | Source                                                                      |
|-------------------------------|------------|-----------------------------------------------------------------------------|
| Forest Area (2020)            | 8,068,370  | Page 25, Global Forest Resources Assessment 2020 Report Cambodia, FAO, 2020 |
| Other Wooded Land Area (2020) | 543,720    | Page 25, Global Forest Resources Assessment 2020 Report Cambodia, FAO, 2020 |

#### Protected Forest Area:

| Description          | Value (ha) | Source                                                                                                         |
|----------------------|------------|----------------------------------------------------------------------------------------------------------------|
| P Forest             | 7,500,000  | KINGDOM OF CAMBODIA, NATION RELIGION KING, CAMBODIA FOREST COVER 2016, THE MINISTRY OF ENVIRONMENT, March 2018 |
| P Other wooded lands | 0          | In absence of the value, considering 0 is being a conservative approach.                                       |

#### Calculation of Mean Annual Increment<sup>25</sup>:

| Description                    | Value (tones/ha/year) | Source                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAI <sub>forest</sub>          | 1.72                  | Global Forest Resources Assessment 2000 by the FAO for "Distribution of total forest area by ecological zone"<br><br>Global Forest Resources Assessment 2020 Report Cambodia, FAO, 2020 for "forest area"<br><br>2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories for "Above-ground biomass growth rates for different ecological zones" (Chapter 4, Table 4.9) |
| MAI <sub>otherwoodedland</sub> | 1.72                  | Global Forest Resources Assessment 2000 by the FAO for "Distribution of total forest area by ecological zone"<br><br>Global Forest Resources Assessment 2020 Report Cambodia, FAO, 2020 for "forest area"<br><br>2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories for "Above-ground biomass growth"                                                             |

<sup>25</sup> Refer the fNRB calculation sheet for more details on MAI calculation.

|  |  |                                      |
|--|--|--------------------------------------|
|  |  | rates for different ecological zones |
|--|--|--------------------------------------|

Renewable Biomass (RB):

| Parameter | Value (Tonnes/year) | Data Source                 |
|-----------|---------------------|-----------------------------|
| RB        | 1,911,910           | Calculated based on eq. (1) |

The Global Forest Resources Assessment 2020 Report, Cambodia, by FAO for Extent of forest, other wooded land is being found the most reliable, accurate and updated data source. It is in line with the  $f_{NRB}$  CDM Tool 30 v.3.0.

As the woodfuel consumption data is available till 2020 vintage only (UN Stats), the forest area and other wooded land area data is also taken for 2020 vintage (by FAO) for consistency purpose.

The KINGDOM OF CAMBODIA, NATION RELIGION KING, CAMBODIA FOREST COVER 2016 Report, by THE MINISTRY OF ENVIRONMENT, published in March 2018, for protected forest area in Cambodia is being the most reliable, accurate and updated data source. It is also in line with the  $f_{NRB}$  calculation tool.

Further, the protected area under other wooded land data is not available hence, it is considered as 0, which is a conservative calculation approach.

The following data source has been used for calculating the mean annual increment (MAI):

- Global Forest Resources Assessment 2020, Report Cambodia, FAO, 2020 (data vintage 2020)
- Global Forest Resources Assessment 2000 by the FAO for "Distribution of total forest area by ecological zone" (Table 14)
- 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories for "Above-ground biomass growth rates for different ecological zones" (Chapter 4, Table 4.9).

**Step 2: Estimation of total consumption of woody biomass (H) in Cambodia as per equation (2) of the  $f_{NRB}$  tool**

The total consumption of woody biomass (H) is calculated using the following equation, accounting for all consumption within the applicable area:

$$H = T_{fw} \cdot D_{wood} + Ch \cdot CtW$$

**Where:**

Equation 2

- $T_{fw}$  = total energy consumption of wood fuel, in the applicable area in the relevant period (m<sup>3</sup>/year)
- $D_{wood}$  = wood density (t/m<sup>3</sup>)
- $Ch$  = Charcoal consumption by household in the applicable area in the relevant period (tonnes)
- $CtW$  = Charcoal to wood biomass factor

Data was drawn from the most recently available official reports/statistics, in this case from the UN's Statistics Division to demonstrate the recent trend in increase of woodfuel use at the household level, plus also the most recently available data (from 2020).

| Description                                                     | Value            | Data Source                                                                                                                                            |
|-----------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Fuelwood – Total energy supply (m<sup>3</sup>/yr) (2020)</b> | <b>7,117,000</b> | UN Data on fuel consumption<br><a href="http://data.un.org/Data.aspx?d=EDATA&amp;f=cmID%3aFW">http://data.un.org/Data.aspx?d=EDATA&amp;f=cmID%3aFW</a> |

The wood density and charcoal to wood biomass factor are given in below table:

| Description                      | Value | Source                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wood density (t/m <sup>3</sup> ) | 0.56  | IPCC 2006 IPCC Guidelines for National Greenhouse Gas Inventories, TABLE 4.13 BASIC WOOD DENSITY (D) OF TROPICAL TREE SPECIES (OVEN-DRY TONNES (MOIST M-3)) ( <a href="https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_04_Ch4_Forest_Land.pdf">https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_04_Ch4_Forest_Land.pdf</a> ) |

Hence, calculation leads to a total woody biomass consumption as below:

| Description                                      | Value            | Source     |
|--------------------------------------------------|------------------|------------|
| <b>H: Total woody biomass consumption (t/yr)</b> | <b>3,985,520</b> | Calculated |

**Step 3: Estimation of quantity of non-renewable biomass (t/year) as per equation (3') of the f<sub>NRB</sub> CDM Tool 30 V.3.0**

$$NRB = H - RB \quad \text{Equation (3)}$$

Where:

- NRB = Quantity of non-renewable biomass consumed in the applicable area in the relevant period (tonnes)
- RB = Quantity of renewable biomass that is available on a sustainable basis in the applicable area in the relevant period (tonnes).
- H = Total consumption of woody biomass in the applicable area in the relevant period (tonnes)

Non-renewable woody biomass (NRB) is the quantity of woody biomass consumed in the absence of the project activity in applicable area (H) minus the quantity of renewable biomass that can be sustainably harvested in the applicable area (RB).

| Parameter         | Value     |
|-------------------|-----------|
| H (Tonnes/year)   | 3,985,520 |
| RB (Tonnes/year)  | 1,911,910 |
| NRB (Tonnes/year) | 2,073,610 |

#### Step 4: Estimation of fraction of woody biomass that can be established as non-renewable ( $f_{NRB}$ ) as per equation (1) of the $f_{NRB}$ tool

$f_{NRB}$  is the fraction of woody biomass saved by the project activity that can be established as Non-renewable Biomass (NRB). It has a range from 0 to 1, where 1 equals to 100% of woody biomass saved by the project being non-renewable.

According to Tool 30, version 03.0 the following equation shall be used to calculate  $f_{NRB}$ :

$$f_{NRB} = NRB / (NRB + RB) \quad \text{Equation (4)}$$

$$f_{NRB} = 3,985,520 / (2,073,610 + 1,911,910)$$

$$f_{NRB} = 52.03\%$$

Through this detailed calculation in line with CDM tool 30 ver 3.0, we can conclude that  **$f_{NRB}$  for Cambodia is 52.03%** (or about 47.97% of the biomass is covered by the natural growth of the forest biomass).

#### 2) **Baseline survey (BS) of target population characteristics**

The objective of the Baseline Survey is to collect data necessary to provide evidence of the baseline situation, pre-project scenario and to be used as the reference scenario to measure the project progress.

As per the “Gold Standard Methodology for emission reductions from safe drinking water supply, Version 1.0 (03/05/2021)”, the following ex-ante parameters can be determined through the baseline surveys/study:

- SWDS 6: Stove technologies used in the project boundary
- SWDS 8: Percentage of Fuel f used in target population
- SWDS 11: Weighted average efficiency of the baseline water boiling devices
- SWDS12: Proportion of project end-users who in the baseline were already using safe water

### **Baseline Survey Sample Sizing & representativeness**

As required by the “Gold Standard Methodology for emission reductions from safe drinking water supply, Version 1.0 (03/05/2021)”, the baseline survey should be carried out for the baseline scenario defined previously using representative and random sampling, applying the following guidelines for minimum sample size as per the para 4.2.3 of the methodology:

- group size < 300: Minimum sample size 30 or population size, whichever is smaller
- group size 300 to 1,000: Minimum sample size 10% of group size
- group size > 1,000: Minimum sample size 100

As defined, the target households are located in rural areas across the whole country of Cambodia. Indeed, Cambodia counts 2,224,520 rural households in total<sup>26</sup>.

- Thus, the perimeter of this project includes 2,224,520 rural households.
- The sampling is done using generic population sampling rules to have a more robust representative sample, that is a confidence interval of 90% and a margin of error of 10% in line with para 4.2.2 of the applied methodology. There are 2,224,520 rural households across the country (geographic project zone).
- Applying this confidence interval and margin of error 90/10, we determined a sampling size of 69 households to survey<sup>27</sup>. However, as said above, the

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<sup>26</sup> Table 12.2.1, page 110, General Population Census of the Kingdom of Cambodia 2019, National Report on Final Census Results, National Institute of Statistics, Ministry of Planning, October 2020  
(<https://www.nis.gov.kh/nis/Census2019/Final%20Geneal%20Population%20Census%202019-English.pdf>)

<sup>27</sup> [Sample Size Calculator](#)



methodology is requesting a minimum sampling size of 100 when the population is superior to 1000 subjects.

- Finally, in order to comply with the rule 90/10, the methodology and to be the most conservative, a sample size of **297 rural households** was considered appropriate, which is **2.97 times more than the recommended Gold Standard applied methodology sample size**. This larger sample size will enable to account for possible outliers and unusable surveys.

### Sampling locations

3 provinces: Battambang, Kampong Speu and Takeo, that are representative of the rural households across the country have been chosen to do the Baseline Survey. They are representative of the whole country because of their different annual population growth, population density, sex-ratio, and type of land (Tonle sap, plateau & mountains, central plain) according to the final generation census of Cambodia 2019<sup>28</sup>. Indeed, the density of population is between 50 to 250 people/km<sup>2</sup> for the three provinces. Also, the annual growth is between -0.3 to 0.3 for Battambang, between 1.41 to 2.2 for Kampong Speu and between 0.31 to 0.8 for Takeo. These differences of annual growth allow to have representativity of the whole country in these 3 provinces. Regarding the population, the 3 provinces chosen, have between 770,000 to 1,200,000 population which is representative of rural areas in Cambodia.

### Preparation of survey

Within these 3 provinces, a sampling of locations where kiosks will be created/rehabilitated by 1001fontaines has been established in order to apply on this list, a random function to determine the location in order to realize the Baseline survey. The table below is gathering this list of potential kiosks.

Table 7: Partial list of future kiosks to rehabilitate/construct

| Province   | District     | Municipalities |
|------------|--------------|----------------|
| Battambang | Moung Russei | Chrey          |
| Battambang | Moung Russei | Taloas         |
| Battambang | Moung Russei | Prey Svay      |

<sup>28</sup> <https://www.nis.gov.kh/nis/Census2019/Final%20Geneal%20Population%20Census%202019-English.pdf>

|              |                 |                            |
|--------------|-----------------|----------------------------|
| Battambang   | Thma Koul       | Ou Ta Ki                   |
| Battambang   | Sangkae         | Kampong Preah              |
| Battambang   | Sangkae         | Ta Pon                     |
| Battambang   | Sangkae         | Ou Dambang Muoy            |
| Battambang   | Moung Ruessei   | Kakaoh                     |
| Battambang   | Banan           | Kantueu Pir                |
| Battambang   | Banan           | Phnom Sampov               |
| Battambang   | Rattanak Mondul | Reaksmei Sangha            |
| Battambang   | Kamrieng        | Ta Krey                    |
| Kampong Speu | Odongk          | Cheung Roas                |
| Kampong Speu | Odongk          | Prey Krasang               |
| Kampong Speu | Odongk          | Yutth Sameakki             |
| Kampong Speu | Odongk          | Damnak Reang               |
| Kampong Speu | Odongk          | Phnom Touch                |
| Takeo        | Bourei Cholsar  | Doung Khpos                |
| Takeo        | Bourei Cholsar  | Kouk Pou                   |
| Takeo        | Sam Raong       | Boeng Tranh Khang<br>Tboun |
| Takeo        | Prey kabbas     | Kdanh                      |
| Takeo        | Bati            | Pea Ream                   |
| Takeo        | Kirivong        | Ream andaeuk               |
| Takeo        | Kaoh Andaet     | Prey khla                  |

6 locations have been chosen after the application of a random function on this list,: 2 locations per province to be the more representative. In each of these 6 random locations, ~50 surveys done. The table below is highlighting the locations that emerged.

Table 8: Location for the Baseline survey

| Random function | Province     | District      | Commune       |
|-----------------|--------------|---------------|---------------|
| 0,951477612     | Battambang   | Moung Ruessei | Kakaoh        |
| 0,912527861     | Battambang   | Sangkae       | Kampong Preah |
| 0,956496521     | Kampong Speu | Odongk        | Prey Krasang  |
| 0,619705836     | Kampong Speu | Odongk        | Cheung Roas   |
| 0,963209196     | Takeo        | Kaoh Andaet   | Prey khla     |
| 0,452535095     | Takeo        | Bati          | Pea Ream      |

## Distribution of surveys across Sub-Counties and Wards

| Province     | District      | Commune       | Actual surveyed households |
|--------------|---------------|---------------|----------------------------|
| Battambang   | Moung Ruessei | Kakaoh        | 56                         |
| Battambang   | Sangkae       | Kampong Preah | 44                         |
| Kampong Speu | Odongk        | Prey Krasang  | 49                         |
| Kampong Speu | Odongk        | Cheung Roas   | 50                         |
| Takeo        | Kaoh Andaet   | Prey khla     | 51                         |
| Takeo        | Bati          | Pea Ream      | 47                         |
| <b>Total</b> |               |               | <b>297</b>                 |

### **Time of year for the baseline survey:**

The survey must be realized in the best conditions, which means when the weather is not rainy and during a dry period. Indeed, as long as the weather is not rainy, the baseline survey can be done without any data alteration.

This Baseline survey has been realized from 18<sup>th</sup> to 20<sup>th</sup> June 2021 when the weather conditions were favourable in Battambang<sup>29</sup>, in Kampong Speu<sup>30</sup>, and in Takeo<sup>31</sup>.

Indeed, as explained in the Kitchen Performance Test<sup>32</sup> from Clean Cooking Alliance page 4: "Temperature changes between rainy and dry seasons can change the demand for space-heating and pre-harvest and postharvest seasons can affect either household income or food consumption, which also affects fuel use". As it affects the fuel use and food consumption, it is not recommended to do the Baseline survey during this period otherwise the results would not be representative. Moreover, the access of rural households during this period can be difficult too.

### **Data collection**

The data collected is specific to the characteristics of the baseline scenario.

1. User follow up
  - a. Address or location

<sup>29</sup> <https://www.historique-meteo.net/asie-sud-est/cambodge/battambang/2021/06/> Battambang weather in June 2021. Only 2.39 mm of precipitations on average per day.

<sup>30</sup> <https://www.timeanddate.com/weather/@1831795/historic?month=6&year=2021> Odongk weather in June.

<sup>31</sup> [Météo Takeo en juin 2022 | Takeo 14 Jours | MeteoArt.com](https://www.meteoart.com/meteoart/taqueo/taqueo-14-jours-2022/) Takeo weather in June.

<sup>32</sup> <https://www.cleancookingalliance.org/binary-data/DOCUMENT/file/000/000/83-1.pdf>

- b. Mobile telephone number and/or landline telephone number (when possible)
2. End user characteristics
  - a. Total family member in household who drink water
  - b. Does household purify water before drinking
  - c. For what use household make water drinkable?
3. Baseline technology and fuels
  - a. How household purify their water?
  - b. What type of cooking stove household use to boil water?
  - c. What combustible household use for the stove?
  - d. How household get the combustible?
  - e. At what price household get this volume of combustible
4. Sustainable Development Indicators
  - a. In the last 4 weeks, have your children (include from 11 years old) been sick?
  - b. Have other adult members get sick with diarrhea?
  - c. What is the reason for this sickness?
  - d. How much would you pay to have safe bottled water delivered at home

## **Results and Analysis**

The result was a total number of **297** households surveyed

### **SWDS 6: Stove technologies used in the project boundary:**

| Stove Technology                                                          | No. of Users <sup>33</sup> | % of Technology Used in Baseline |
|---------------------------------------------------------------------------|----------------------------|----------------------------------|
| HH using three stone stoves/<br>traditional stoves / clay stove only      | 136.5                      | 89%                              |
| HH Using Improved traditional<br>method / Improved Woodfired Jiko<br>Only | 0                          | 0%                               |
| HH Using LPG Stove                                                        | 17                         | 11%                              |
| HH Using Electric Stove                                                   | 0.5                        | 0%                               |

### **SWDS 8: Percentage of Fuel f used in target population**

| Fuel Type   | No. of Households | % of Households |
|-------------|-------------------|-----------------|
| HH use Wood | 129               | 84%             |

<sup>33</sup> Some of the households may using more than one technology. In this case equal weightage is given to each of the technology. For example, if a household is using two technology, then 0.5 weightage is given to each technology.

|                    |            |             |
|--------------------|------------|-------------|
| HH use Charcoal    | 8          | 5%          |
| HH use Gas         | 16.5       | 11%         |
| HH use Electricity | 0.5        | 0%          |
| <b>Total</b>       | <b>154</b> | <b>100%</b> |

**SWDS 11: Weighted average efficiency of the baseline water boiling devices**

| Stove Technology                                                    | No. of Users | Default Efficiency <sup>34</sup> |
|---------------------------------------------------------------------|--------------|----------------------------------|
| HH using three stone stoves traditional stoves / clay stove only    | 136.5        | 10%                              |
| HH Using Improved traditional method / Improved Woodfired Jiko Only | 0            | 20%                              |
| HH Using LPG Stove                                                  | 17           | 55%                              |
| HH Using Electric Stove                                             | 0.5          | 50%                              |
| <b>Weighted Average Thermal Efficiency</b>                          | <b>15.1%</b> |                                  |

**SWDS12: Proportion of project end-users who in the baseline were already using safe water:**

| Parameter                                                                                                                           | Percentage of Users (%) |
|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Proportion of project end-users who in the baseline were already using safe water, from a water treatment method other than boiling | <b>13%</b>              |
| Proportion of project end-users who in the baseline were already using safe water, from purchasing it                               | <b>22%</b>              |
| % of HH who do not purify water but think the water they are getting from water source is safe for drinking                         | <b>4%</b>               |
| <b>Cb</b>                                                                                                                           | <b>39%</b>              |

## B.5. Demonstration of additionality

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).

According to the "Community services activity requirements" Version 1.2, page para 4.1.9 on 6, published in October 2019<sup>35</sup>, projects that are in the Least Developed Countries (LDC) according to the UN Office of the High Representative; are considered as deemed additional.

|                                                                                |                                                        |
|--------------------------------------------------------------------------------|--------------------------------------------------------|
| Describe how the proposed project meets the criteria for deemed additionality. | Cambodia is part of the list of the LDC. <sup>36</sup> |
|--------------------------------------------------------------------------------|--------------------------------------------------------|

<sup>34</sup> For traditional three stone fire (10%) and improved wood stove (20%), the default efficiency is as per the applied methodology. For LPG and electric stove, the default efficiency (50%) is considered as per the PP other project experience.

<sup>35</sup> [https://globalgoals.goldstandard.org/standards/201\\_V1.2\\_AR\\_Community-Services-Activity-Requirements.pdf](https://globalgoals.goldstandard.org/standards/201_V1.2_AR_Community-Services-Activity-Requirements.pdf)

<sup>36</sup> <https://unctadstat.unctad.org/CountryProfile/GeneralProfile/en-GB/116/index.html>

### B.5.1 Prior Consideration

>>

**Time of first submission is within one years from the project start date:** Time of first submission (21/12/2021) is more than one year from the project start date (09/03/2020).

The Project Developer (EcoAct SAS) has requested a deviation from Gold Standard to deviate from Para 4.1.49 (b) of Principles and Requirements (version 1.2, October 2019). GS has granted deviation from para 4.1.49.(b) of the Principles and Requirements considering the COVID situations<sup>37</sup>.

Further as per the approved deviation request

- *PD will successfully demonstrate, along with relevant evidence, to demonstrate prior consideration of carbon revenues.*
- *The validating VVB shall, through appropriate means at its disposal, evaluate the Project's compliance with the above-mentioned conditions and provides its opinion in the Validation Report. Certification body shall review both the PD's submission and the VVB's opinion of the same and take appropriate steps.*

The PD will demonstrate the prior consideration of carbon revenues to the Validating VVB. The chronology of events under project are as follows:

#### 1. Discussion of Project Certification & Carbon Finance in 2019

The certification of the kiosks that will be deployed from 2020 onwards has been discussed in the Board Committee Meeting held on 12/11/2019. The extract of the minutes of meeting is as follows:

*-We now plan to start a new certification process shortly to cover the kiosks that will be deployed from 2020 onwards. This could possibly be with a different partner.*

*It is reminded that the issuance of carbon credits is necessary to ensure the long-term sustainability of the Cambodian program. The revenues generated additionally to our other revenue streams will allow the network of kiosks to reach financial break-even in 2019, for the first time.*

*Without the additionality from carbon finance, the viability of the network is jeopardized as this would incur having less cashflow available to reinvest, do enough maintenance, etc.*

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<sup>37</sup> Refer DEV\_253 approved on 28/06/2022 from GS.

*Therefore, the management team wishes to pursue its efforts and find a way to have a carbon credit certificate for the kiosks that will be deployed in 2020. This has been discussed with the management team and the board members of Teuk Saat 1001, who share the same vision. Teuk Saat 1001 has asked 1001fontaines to continue to lead this effort.*

*The board members acknowledge and understand the financial additionality coming from carbon credits, and its need to ensure the sustainability of the network of kiosks. The board members approve the launch of a new certification process for the Cambodian program.*

## **2. Project Start in 2020:**

1001fontaines (PP) has been providing safe drinking water to households since March 2020 (first kiosks installed on 09/03/2020) due to the construction of water kiosks in several regions of Cambodia.

The Safe Water Kiosk O2 (Water Kiosk Code) has started distributing water from 09/03/2020. The water sale report of O2 Kiosk from commcare software (in pdf format) and extract in excel format has already been submitted to SustainCert during the Preliminary Review of the project and also submitted along with this deviation Request.

## **3. Document Submission to SC for Preliminary Review in 2021:**

Regarding the Carbon Certification of the project, the documents have been submitted to SustainCert on 21/12/2021 for Preliminary Review.

However, and for several reasons, project developer (EcoAct SAS) is requesting Gold Standard to have the project start date as 09/03/2020. This is the exact date of first sales of water happened. In order to be fully compliant with the GS4GG requirements, submission of the required documents for preliminary review should have been before 08/03/2021. While the documents effectively submitted on 21/12/2021. The different reasons for the delay in submission are detailed below.

**3.1 Covid Situation in 2020 & 2021:** First, project developer (EcoAct SAS) would like to underline the covid situation in Cambodia (project host country) and in France (the country of both 1001fontaines and EcoAct headquarters located, and project team situated) have been facing in 2020 and 2021.

The carbon certification of the kiosks that will be deployed from 2020 onwards have been discussed in the 1001fontaines Board Meeting held on 12<sup>th</sup> Nov 2019.

Then the design and implementation of "Anakot Thmei Safe Water Project" have been started in early 2020, but the covid situation in Europe hit badly as well as more importantly, in Cambodia, across the country where the project located. The covid 19 situation across these two countries affected the project development work and preparation of all the documentation required for preliminary review under Gold Standard (Carbon Certification).

EcoAct and 1001fontaines main offices are in Paris, France and experts dedicated to Anakot Thmei Safe Water Project as well located there. Starting from March 2020 in France, lockdown was decided for several months. During the two years, more than 4 months, activity was considerably reduced for project team and partly explain the delay



in preparing the draft project design document as well as other required documents mandatory for the preliminary review. Work from Home was not very familiar in the company before Covid 19 and the new working situation slowed down the new project development activities somehow. Also, project team was not able to travel from France to Cambodia to work closer with the field staff. Health consequences (being Covid positives) faced by some of the EcoAct & 1001fontaines team members and experts also delayed the document preparation.

The host country of the project, Cambodia, has also faced difficult sanitary conditions. As described in the PDD, Anakot Thmei Safe Water Project covers the entire country with kiosks to be implemented in different regions across the country. To prepare the project implementation and implement its activities on the ground requires a great mobility for local field workers to move between the different locations and the local main office in Phnom Penh. For several months between 2020 and 2021, field teams were not allowed to move from one region to another for the carbon certification purpose (LSC, baseline survey etc.) and keep the project moving forward, collect data, information and ensure all requirements could be fulfilled at every location. Some quarantine periods were even decided at the highest level in the Kingdom of Cambodia in 2021.

As an example, it has been impossible to conduct a baseline survey for months due to local restrictions. This essential work in designing the project requires field teams to travel from the main office to the different locations where kiosks are already implemented as well as in potential new areas where kiosks are expected to be implemented. Globally, project developer aims at focusing on rural and peri-urban areas. This means workers on the field were expected to move from the capital city Phnom Penh where the main office is, to several remote areas. This is exactly what authorities wanted to avoid during the covid period to stop spread of covid 19. Cambodia being less advanced country, hospital and health capacity of the country are limited and vast expansion of the virus was to be avoided.

Also, the covid situation came along with great uncertainty in planning. Indeed, project developer intended to complete preliminary review preparation in time, then only a few weeks late, then months, and it came to around one year. Delays were observed all along the year and could not be planned from day one. At first, project developer also expected to conduct the local stakeholder consultation before Preliminary review to submit the report along with other required documents. It was postponed several times until decision was made to submit the draft stakeholder consultation report with self assessment of Sustainable Development Goals (SDGs) and safeguards assessment to Gold Standard for preliminary review in line with para 2.3.2 of GS4GG COVID 19: Interim Measures v4-17.05.2021.

**3.2. Launch of GS New Safe Drinking Water Supply Methodology in 2021:** In addition to the worldwide covid situation that affected project and project team members, project developer also believes that some additional delay in project documentation preparation is due to the change/revision of GS water methodology, although the new water methodology published on 03/05/2021 and came into force from 02/08/2021. At first, the project was expected to be develop and registered using the Annex 3 of TPDDTEC ver 3.1. The Annex 3 included most of the water projects developed/manage by EcoAct and it made sense for EcoAct to follow a methodology it

has already worked with. Annex 3 of the TPDDTEC ver 3.1 was indeed expected to be the chosen Impact Quantification methodology for aforesaid project. However, project team learnt about the updates of annex 3 in Q2- Q3 of 2021. We were thus expecting changes and new requirements to be mandatory for the months/years coming. Then it's been brought up to our attention that the outcome could be a dedicated "standalone" methodology. So, it had been decided to include the changes/new requirements at the earliest stage of project design.

This was to avoid main changes in project activities to align with the new methodology. Several changes were very important compared to the first estimations project team made, changing noticeably the project design. Two main areas of change: the Emission Reduction calculations have changed between Annex 3 of TPDDTEC ver 3.1 and the new Safe Drinking Water Supply methodology ver 1.0, default values (for example, caps on claimable water volumes). This lowered the outcome: the quantity of carbon credits that could be expected from the project in the 5 years of crediting period.

The other main area of change was monitoring requirements. The number of households to be included in the project, changed to only include ones within a 1km/30min radius around project water source. In the specific case of Anakot Thmei Safe Water Project, this changed the heart of the project. Indeed, water is produced at the kiosk installation location and then transported via trucks or motorbikes and sold to beneficiaries a few km around the kiosks. Strictly, this meant only dozens of households could be included in the project only. Hence, it has impact on the GS VERs estimated to be generate from the project earlier and with this change, the financial additionality (whole project financing planning) of the project has also been impacted negatively. At some point of time PP were not sure that they will able to recover the kiosk fixed and variable costs.

It took quite some time to find an acceptable solution after having few email exchanges and calls with the Gold Standards, explaining the delay as well in document submission for preliminary review. New Ex-ante estimations had to be found, ex-ante calculations updated, ex-post monitoring process had to be updated. This took further time to the project team to finalize the documents and submit for review.

While the project has missed the one-year deadline to be submit to Gold Standard for the Preliminary Review from the Project Start Date, there are no impact on the quality of the water supplied by the kiosks, kiosks life etc. Further the Project Developer is now speeding up Third-Party Validation with VVB and then submit the project for Design Review to GS in this year only.

**1. System Maintenance & Life:** Also, the fact that the PP has submitted documents for preliminary review after one year from the project starts date, does not impact the quality of the water produced and supplied by the kiosks and kiosks life span too. The water kiosks installed in 2020 are well maintained in a proactive way (refer section B.6.2 of the submitted PDD for more details) so that the water filtration process keeps running correctly. Even though the technology was installed in early 2020, the maintenance management runs since this start date of project to keep the equipment (water kiosks) up to date, working and efficient and to provide safe drinking water to households in Cambodia who do not have access to safe drinking water and need to boil water to make it safe for drinking.

For example, the UV lamps use in water kiosks have an 8,000 hour of usage life expectancy and which can be replaced after the 8,000 hours of operation and the covid

delay does not affect the maintenance planning. Further the kiosks use sand filters and other kind of filters of water filtration that can be easily replaced and with replacement provide safe water quality and new life to the system. Finally, only 29 kiosks were installed in 2020, out of 300 kiosks PP is planning to install under the project.

**2. Validation of Project with VVB:** Further, EcoAct along with the 1001fontaines has selected the Validation and Verification Body (VVB) for project validation and now in process of signing the validation contract with VVB once the deviation request has been approved.

B.5.2 Ongoing Financial Need

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N/A

## B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

| Sustainable Development Goals Targeted | Most relevant SDG Target                                                                                                                                                                    | SDG Impact<br><br>Indicator (Proposed or SDG Indicator)                       |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 13 Climate Action (mandatory)          | N/A                                                                                                                                                                                         | Average Emission Reductions per year                                          |
| 6 Clean Water and Sanitation           | 6.1: By 2030, achieve universal and equitable access to safe and affordable drinking water for all                                                                                          | Average Quantity of clean water distributed to project beneficiaries per year |
| 8 Decent Work and Economic Growth      | 8.5: By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value | Average Number of jobs created per year                                       |

### B.6.1 Explanation of methodological choices/approaches for estimating the SDG Impact

>>

**13 Climate Action (mandatory):** The outcome for SDG 13 will be quantified as CO<sub>2</sub> emission reductions by applying the GS Emission reductions from Safe Drinking Water Supply v.1.0 methodology.

**The emission reductions are calculated as follows:**

$$ER_y = BE_y - PE_y - LE_y \quad (\text{equation 11 of meth})$$

Where:

- $ER_y$  = Emission reductions in year y (t CO<sub>2e</sub>/yr)
- $BE_y$  = Baseline emissions in year y (t CO<sub>2e</sub>/yr) (calculated as explained below)
- $PE_y$  = Project emissions in year y (t CO<sub>2e</sub>/yr) (calculated as explained below)
- $LE_y$  = Leakage emissions in year y (t CO<sub>2e</sub>/yr) (calculated as explained below)

**Baseline emissions calculation:**

$$BE_y = EF_b \times (1 - C_b - X_{cleanboil,y}) \times Q_y \times M_{q,y} \quad (\text{equation 3 of meth})$$

Where:

- $BE_y$  = Baseline emissions from the use of fuel to obtain safe water in the baseline (tCO<sub>2e</sub>)
- $EF_b$  = Emission factor for the use of fuel to obtain safe water in the baseline (tCO<sub>2e</sub>/L)
- $C_b$  = Proportion of project end-users who in the baseline were already using a safe water supply that did not require boiling (%)
- $X_{cleanboil,y}$  = Proportion of project end-users that boil safe water in the project year y (%)
- $Q_y$  = Quantity of safe drinking water provided by the project in year y (L)
- $M_{q,y}$  = Modifier for the water quality in year y

**Emission factor calculation:**

$$EF_b = SE_{w,b,y} \times \sum (x_f \times (EF_{b,f,CO2} \times f_{NRB,f,y} + EF_{b,f,nonCO2})) \div 10^9 \quad (\text{equation 1 of meth})$$

Where:

- $EF_b$  = Emission factor for the use of fuel to obtain safe water in the baseline (tCO<sub>2e</sub>/L)
- $SE_{w,b,y}$  = Specific energy required to boil water (kJ/L), to be calculated as per the paragraph below
- $x_f$  = Proportion of fuel f used in the baseline (fraction determined based on an energy basis)
- $EF_{b,f,CO2}$  = CO<sub>2</sub> emission factor from use of fuel f (tCO<sub>2</sub>/TJ)
- $EF_{b,,O2}$  = Non-CO<sub>2</sub> emission factor arising from use of fuel f, when the baseline fuel f is biomass or charcoal (tCO<sub>2e</sub>/TJ). This parameter is omitted when f is a fossil fuel.

- $f_{NRB,y}$ =Fractional non-renewability status of woody biomass fuel during year y (fraction). For biomass, it is the fraction of woody biomass that can be established as non-renewable. This parameter is omitted when f is a fossil fuel.
- f=Index for baseline fuel types

## Method 1 – CWT technology

Quantity of safe drinking water provided

$$Q_y = \min(Q_{m,y}, Q_{pop,y}) \quad (\text{equation 4 of meth})$$

Where:

- $Q_{m,y}$ =Monitored quantity of safe water provided by the project in year y(L).
- $Q_{pop,y}$ =Quantity of safe drinking water that could be consumed by project end-users in year y(L)

$$Q_{pop,y} = \sum HH_{p,y} \times HN_{p,y} \times QPW_p \times DO_{p,y} \quad (\text{equation 5 of meth})$$

Where:

- $HH_{p,y}$ = Number of premises type p served by the project in year y
- $HN_{p,y}$ = Number of individuals per premises type p (e.g. household, school) in year y
- $QPW_p$ = Volume of drinking water per person per day for premises type p(L). Apply the default value or monitored value through water consumption field tests in the project scenario, capped at 5.5 L per person per day.
- $DO_{p,y}$ =Days the project technology is operational for end-users in premises pin year y

## Project emissions calculation:

$$PE_y = PE_{ff,y} + PE_{ec,p,y} \quad (\text{equation 8 of meth})$$

Where:

- $PE_y$ =Project emissions in year y (tCO<sub>2</sub>)
- $PE_{ff,p,y}$ =Project emissions from fossil fuel use in year y (tCO<sub>2</sub>)
- $PE_{ec,p,y}$ =Project emissions from electricity use in year y (tCO<sub>2</sub>)

>>

## Project emissions from fossil fuel use are determined as follows.

$$PE_{ff,y} = \sum P_{p,f,y} \times NCV_f \times EF_f \quad (\text{equation 9 of meth})$$

Where:

- $P_{p,f,y}$ =Quantity of fossil fuel f that is consumed in the project during year y (mass or volume units)

- $NCV_f$  = Net calorific value of fossil fuel f (TJ/fuel units)
- $EF_f$  = Emission factor of fossil fuel f (tCO<sub>2e</sub>/TJ)

>>

Project emissions from electricity use are estimated as follows.

$$PE_{ec,i} = \sum EC_{p,y} \times EF_{ec} \times (1 + TDL_{ec}) \text{ (equation 10 of meth)}$$

Where:

- $EC_{p,y}$  = Quantity of electricity that is used by the project during year y (kWh)
- $EF_{ec}$  = Emission factor associated with the electricity use (tCO<sub>2</sub>/kWh)
- $TDL_{ec}$  = Transmission and distribution losses associated with the electricity use (%)

#### Leakage calculations:

Leakage relating to non-renewable woody biomass shall be assessed, other types are excluded for simplification:

- a. Population who does not participate in the project and used lower emitting energy sources instead use the non-renewable biomass saved under the project activity
- b. The project significantly reduces the NRB fraction within an area where other GHG mitigation project activities account for NRB fraction in their baseline scenario
- c. The project population compensates for loss of the space heating effect of water boiling by adopting some other form of space heating or by retaining some baseline woodfuel-burning practices

**SDG 6 Clean Water and Sanitation:** Section B.7.1. of the PDD outlines how SDG 6 will be monitored. The quantification of clean water distributed will be monitored continuously throughout the project lifetime. The indicator is the cumulative quantity of safe water provided.

**Baseline:** baseline is the purification of water by households using firewood to boil water. There is no safe water distributed.

#### **Project outcome:**

$$Q_y = \sum_{year}^y Q_{m,year} \text{ (equation 12)}$$

Where:

- $Q_y$  = cumulative quantity of clean water distributed to project beneficiaries from beginning of project until end of year y (L). It is a cumulative indicator.
- $Q_{m,year}$  = Monitored quantity of safe water provided by the project to beneficiaries in year y (L).

**8 Decent Work and Economic Growth:** Section B.7.1. of the PDD outlines how SDG 8 will be monitored. The construction and operation of the water kiosks requires agents. These workers will be selected, trained and accompanied by Teuk Saat in order to operate the water kiosks. The number of jobs related to each kiosk is known as a contract is signed with each person.

**Baseline:** baseline is the purification of water by households using firewood to boil water. There is no job created.

Project outcome:  $N_{jc}$  is monitored as being the number of jobs created in year  $y$

$$N_{jy} = \sum_{year}^y N_{Jyear} \quad (\text{equation 13})$$

Where:

- $N_{jy}$  = cumulative number of jobs created between beginning of project and end of year  $y$
- $N_{Jyear}$  = number of jobs created in year “year”

## B.6.2 Data and parameters fixed ex ante

### SDG 13

| Parameter ID                                         | SDWS 1                                                                                                                                                                                                             |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data/parameter                                       | Number of households per CWT                                                                                                                                                                                       |
| Unit                                                 | Number of end-users per CWT                                                                                                                                                                                        |
| Description                                          | End user's premises (e.g. households, institutions) within 5-15 km distance of project water source.                                                                                                               |
| Source of data                                       | 1001fontaines water kiosks data base                                                                                                                                                                               |
| Value(s) applied                                     | Number of households per CWT: 2,000                                                                                                                                                                                |
| Choice of data or Measurement methods and procedures | As per the applied methodology                                                                                                                                                                                     |
| Purpose of data                                      | Calculation of baseline scenario                                                                                                                                                                                   |
| Additional comment                                   | As per the CWT definition on page 3 of the methodology the distribution's point (distribution vehicle's) can be considered as a source of safe water. Thus, the project end-users (households, schools, commercial |



establishments etc.) will be within 1 km or less from these distribution points (motorbike, trucks etc.). Water is distributed at the kiosk's location but also via an efficient distribution network (trucks, motorcycles etc.) to households. Each kiosk is indeed equipped with 1-2 vehicles or more that distribute water to households within a radius of at least 5 km. The number of households potentially served by the distribution system will be therefore calculated. There is a possibility that some of the households can also visit to the kiosk to purchase water hence these households will also remain within 1km radius.

Each kiosk has one or two distribution vehicles that travel a route with at least 5 distribution points at distances of more than 2km from each other.

On average one distribution point serves 400 households through the home delivery system (1km rule was apply), and one route is about 5 distribution points, hence  $5 \times 400 = 2,000$  households served by one kiosk every day.

This estimate was compared to 1001fontaines' estimates which consider that all the population within a 5km radius of the kiosk or more gain access to safe water and the home delivery service, and even part of the population located farther than 5km (15 km) depending on local specific context. For each kiosk the demographic information is based on national census and commune council's information.

The number of houses has been counted for these distribution points which are considered as the water sources of this project.

On average one distribution point can serves 400 households, through the home delivery system (1km rule was apply). One route is about 5 distribution points, so  $5 \times 400 = 2,000$  households can be served by one kiosk every day.



| Parameter ID                                         | SDWS 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--|---------|-------------------------------------------------------------|------------|------------------------------------------------------|-------------|-----------------------------------------------------|-------------------|--------------------------|
| Data/parameter                                       | Project technology description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Unit                                                 | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Description                                          | Water kiosks are composed of several parts: mainly motor pump, UV lamp, solar panel etc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Source of data                                       | Manufacturer specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Value(s) applied                                     | <p>Manufacture Name: 1001fonatines<br/>           Product Name: Water Kiosk<br/>           Technology Type: UV Filtration System<br/>           System Capacity:</p> <table border="1"> <thead> <tr> <th colspan="2">Hardware specification</th></tr> </thead> <tbody> <tr> <td>UV lamp</td><td>Lamp power: 36W, 6A<br/>Life time: about 8000 hours of using</td></tr> <tr> <td>Motor pump</td><td>Maximum flow rate: 1500 l/min<br/>Life time : 3 years</td></tr> <tr> <td>Solar panel</td><td>12 VDC supply with regulator<br/>Life time: 20 years</td></tr> <tr> <td>Designed capacity</td><td>4,800 l/day<sup>2</sup></td></tr> </tbody> </table> | Hardware specification |  | UV lamp | Lamp power: 36W, 6A<br>Life time: about 8000 hours of using | Motor pump | Maximum flow rate: 1500 l/min<br>Life time : 3 years | Solar panel | 12 VDC supply with regulator<br>Life time: 20 years | Designed capacity | 4,800 l/day <sup>2</sup> |
| Hardware specification                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| UV lamp                                              | Lamp power: 36W, 6A<br>Life time: about 8000 hours of using                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Motor pump                                           | Maximum flow rate: 1500 l/min<br>Life time : 3 years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Solar panel                                          | 12 VDC supply with regulator<br>Life time: 20 years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Designed capacity                                    | 4,800 l/day <sup>2</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Choice of data or Measurement methods and procedures | Manufacturer specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Purpose of data                                      | Calculation of baseline & project emissions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Additional comment                                   | <p>Please note that the UV lamps use in water kiosks have an 8,000 hour of usage life expectancy and which can be replaced after the 8,000 hours of operation at multiple times. Further the water kiosks use, sand filters and other kind of filters for water filtration that can be easily replaced and with such replacement, kiosk provide safe water quality and new life to the system. Hence the water kiosks have life more than 15 years as it renewed every time on maintenance and replacement.</p>                                                                                                                                       |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |

|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter ID                                         | SDWS 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Data/parameter                                       | Project technology performance level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Unit                                                 | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Description                                          | <p>TS1001 Water quality standards is matching WHO Guidelines for drinking-water quality, 4th edition (Table 7.10, WHO, 2017). Each month, two samples for each water kiosk are collected and analyzed in Teuk Saat laboratories. The first sample is taken at the outlet of the treatment system, the second from a sealed bottle to ensure that the hygiene procedures guaranteeing water quality till the point of use are respected. Water quality results are recorded in the quality app and corrective actions taken if quality alerts are detected. The list of parameters covers microbiological (E-coli, total coliform) as well as physical and chemical parameters. List can be provided to VVB.</p> |
| Source of data                                       | Water quality test report                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Value(s) applied                                     | 95%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Choice of data or Measurement methods and procedures | Water quality tests in line with the national standard to WHO guidelines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Purpose of data                                      | Calculation of SDG 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Additional comment                                   | <p><i>Laboratories used for water quality testing must be approved by local health authorities and/or have quality accreditation; and The laboratory used shall have evidence to demonstrate that it has an adequate quality management plan in place which addresses both quality assurance and quality control test procedures. Table 4.6 Checklist for effective analytical quality assurance of WHO Guidelines, 1997 may be used as a guideline for laboratory compliance with quality assurance practices.</i></p> <p>Teuk Saat has 4 provincial laboratories which can perform water analyses on microbiological and physio-chemical parameters for the water kiosks located in 19 provinces.</p>         |

|  |                                                                                                                                                                                                                                                                               |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Each structure is equipped and operated by 2 full-time technicians, able to test water each open day and deliver results 1 to 2 days after testing. In case of poor water quality results, their regional technicians are available on site to solve the water quality issue. |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                      |                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Parameter ID</b>                                  | <b>SDWS 4</b>                                                                                                                                                                                                                                                                                             |
| Data/parameter                                       | Regulatory framework for safe water supply                                                                                                                                                                                                                                                                |
| Unit                                                 | NA                                                                                                                                                                                                                                                                                                        |
| Description                                          | The project is in line with the 2019-2023 Nation Action Plan on water access, coordinated by the Ministry of Rural Development. Regarding water quality standards, the project follows rigorously WHO requirements while the Cambodian Ministry of Rural Development is editing Water Quality Guidelines. |
| Source of data                                       | See table 2 – section B2 Applicability of methodology                                                                                                                                                                                                                                                     |
| Value(s) applied                                     | -                                                                                                                                                                                                                                                                                                         |
| Choice of data or Measurement methods and procedures | -                                                                                                                                                                                                                                                                                                         |
| Purpose of data                                      | -                                                                                                                                                                                                                                                                                                         |
| Additional comment                                   | -                                                                                                                                                                                                                                                                                                         |

|                     |                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Parameter ID</b> | <b>SDWS 5</b>                                                                                               |
| Data/parameter      | Water sources in the project boundary                                                                       |
| Unit                | NA                                                                                                          |
| Description         | Identify the improved water sources in the project boundary.                                                |
| Source of data      | Table 9, Page 30, KINGDOM OF CAMBODIA Nation Religion King Report of Cambodia Socio-Economic Survey 2019/20 |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                   |       |  |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------|--|
| Value(s) applied   | Improved Water Source (rural): 74.7<br>Unimproved Water Source (rural): 25.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                   |       |  |
|                    | Water sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Cambodia                          | Rural |  |
|                    | Improved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 79.7                              | 74.7  |  |
|                    | Piped into dwelling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 25.8                              | 10.1  |  |
|                    | Piped into compound, yard or plot                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3.7                               | 3     |  |
|                    | Public tap/standpipe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1                                 | 0.8   |  |
|                    | Tubed/piped well or borehole                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 30.1                              | 38.4  |  |
|                    | Protected well                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4.1                               | 5.4   |  |
|                    | Rainwater collection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 5.8                               | 7.7   |  |
|                    | Bottled water                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 9.1                               | 9.4   |  |
|                    | Unimproved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 20.3                              | 25.3  |  |
|                    | Unprotected well                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4.4                               | 6.1   |  |
|                    | Unprotected spring                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0.1                               | 0.1   |  |
|                    | Tanker-truck                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3.9                               | 3.9   |  |
|                    | Cart with small tank/drum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2.9                               | 3.5   |  |
|                    | Surface water (Pond, river, stream,...)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 8.6                               | 11.5  |  |
|                    | Other                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0.3                               | 0.3   |  |
|                    | Total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 100                               | 100   |  |
|                    | Choice of data or Measurement methods and procedures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Official government publications  |       |  |
|                    | Purpose of data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Calculation of baseline emissions |       |  |
| Additional comment | <p>It is important to note that “an improved water source is not necessarily safe, but an improved source is more likely to provide safe water” according to the Report of Cambodia Socio-Economic Survey, 2019/20.</p> <p>Although this is necessary information for use in the methodology calculations, the improved water distribution of 74.7% in rural Cambodia used in the standards does not give an accurate indicator of water quality. Thus, most of these improved water sources in Cambodia are not safe, as indicated for example by the Teuk Saat and WaterAid water quality study results - Kampong Chhnang, 2020 and the Water quality for young children in Cambodia study – Poirot</p> |                                   |       |  |

E. and all, 2018. According to this study, “as approximately 59% of water samples were found to be contaminated with coliforms at the point of collection at a level above 11 CFU/100 ml in our study, including 36% of samples from improved sources having presence of E. coli, these findings call into question the quality of drinking water obtained from what are generally considered “improved sources.”

This puts consumers at risk of believing bottled water is safe to drink, according to the 2018 WASH Data Pilot Survey Report in Kampong Chhnang Province conducted by WaterAid, MRD and NIA.

| Parameter ID                                                        | SDWS 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                  |  |                  |                            |                                  |                                                                  |       |     |                                                                     |   |    |                    |    |     |                         |     |       |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|--|------------------|----------------------------|----------------------------------|------------------------------------------------------------------|-------|-----|---------------------------------------------------------------------|---|----|--------------------|----|-----|-------------------------|-----|-------|
| Data/parameter                                                      | Stove technologies used in the project boundary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                  |  |                  |                            |                                  |                                                                  |       |     |                                                                     |   |    |                    |    |     |                         |     |       |
| Unit                                                                | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                  |  |                  |                            |                                  |                                                                  |       |     |                                                                     |   |    |                    |    |     |                         |     |       |
| Description                                                         | The proportion of different stove types used in premises in the geographical area of the project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                  |  |                  |                            |                                  |                                                                  |       |     |                                                                     |   |    |                    |    |     |                         |     |       |
| Source of data                                                      | Baseline survey                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                  |  |                  |                            |                                  |                                                                  |       |     |                                                                     |   |    |                    |    |     |                         |     |       |
| Value(s) applied                                                    | <table><tr><th>Stove Technology</th><th>No. of Users<sup>38</sup></th><th>% of Technology Used in Baseline</th></tr><tr><td>HH using three stone stoves traditional stoves / clay stove only</td><td>136.5</td><td>89%</td></tr><tr><td>HH Using Improved traditional method / Improved Woodfired Jiko Only</td><td>0</td><td>0%</td></tr><tr><td>HH Using LPG Stove</td><td>17</td><td>11%</td></tr><tr><td>HH Using Electric Stove</td><td>0.5</td><td>0.32%</td></tr></table> <p>Traditional three stone firewood stove: 89%</p> <p>LPG Stove: 11%</p> <p>Electric Stove: 0.32%<sup>39</sup></p> |                                  |  | Stove Technology | No. of Users <sup>38</sup> | % of Technology Used in Baseline | HH using three stone stoves traditional stoves / clay stove only | 136.5 | 89% | HH Using Improved traditional method / Improved Woodfired Jiko Only | 0 | 0% | HH Using LPG Stove | 17 | 11% | HH Using Electric Stove | 0.5 | 0.32% |
| Stove Technology                                                    | No. of Users <sup>38</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | % of Technology Used in Baseline |  |                  |                            |                                  |                                                                  |       |     |                                                                     |   |    |                    |    |     |                         |     |       |
| HH using three stone stoves traditional stoves / clay stove only    | 136.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 89%                              |  |                  |                            |                                  |                                                                  |       |     |                                                                     |   |    |                    |    |     |                         |     |       |
| HH Using Improved traditional method / Improved Woodfired Jiko Only | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0%                               |  |                  |                            |                                  |                                                                  |       |     |                                                                     |   |    |                    |    |     |                         |     |       |
| HH Using LPG Stove                                                  | 17                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 11%                              |  |                  |                            |                                  |                                                                  |       |     |                                                                     |   |    |                    |    |     |                         |     |       |
| HH Using Electric Stove                                             | 0.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0.32%                            |  |                  |                            |                                  |                                                                  |       |     |                                                                     |   |    |                    |    |     |                         |     |       |

<sup>38</sup> Some of the households may using more than one technology. In this case equal weightage is given to each of the technology. For example, if a household is using two technology, then 0.5 weightage is given to each technology.

<sup>39</sup> As less than 0.5% population use electric stove for water boiling, it is not considered in calculation for convenient purpose.

|                                                      |                                                                                                                                           |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Choice of data or Measurement methods and procedures | Following the methodology “ Gold Standard Methodology for emission reductions from safe drinking water supply, Version 1.0 (03/05/2021)”. |
| Purpose of data                                      | Calculation of baseline scenario                                                                                                          |
| Additional comment                                   | N/A                                                                                                                                       |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter ID     | SDWS 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Data/parameter   | Expected technical life of project technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Unit             | Month/Years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Description      | <p>The different components of the kiosk do not have the same lifespan, so when a component reaches the end of its life, it is replaced so that the kiosk can continue to operate. A maintenance framework is in place, covering roles and responsibilities of the entrepreneur and Teuk Saat.</p> <p>Maintenance tasks are shared between entrepreneur and Social Franchise support team.</p> <p>The water entrepreneur is responsible for the proper functioning of the whole water kiosk including for basic maintenance:</p> <ul style="list-style-type: none"> <li>- Backwashing for pre-filtration tanks;</li> <li>- Cleaning RO membrane;</li> <li>- Checking UV quartz sleeve;</li> <li>- Checking micro-filters;</li> <li>- Disinfecting drinking output pipe;</li> </ul> |
| Source of data   | Technical specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Value(s) applied | <p>UV lamp: 8000 hours of using</p> <p>Motor pump: 3 years</p> <p>Solar panel: 20 years</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                                                      | <table border="1"> <thead> <tr> <th colspan="2">Hardware specification</th></tr> </thead> <tbody> <tr> <td>UV lamp</td><td>Lamp power: 36W, 6A<br/>Life time: about 8000 hours of using</td></tr> <tr> <td>Motor pump</td><td>Maximum flow rate: 1500 l/min<br/>Life time : 3 years</td></tr> <tr> <td>Solar panel</td><td>12 VDC supply with regulator<br/>Life time: 20 years</td></tr> <tr> <td>Designed capacity</td><td>4,800 l/day<sup>2</sup></td></tr> </tbody> </table>                        | Hardware specification |  | UV lamp | Lamp power: 36W, 6A<br>Life time: about 8000 hours of using | Motor pump | Maximum flow rate: 1500 l/min<br>Life time : 3 years | Solar panel | 12 VDC supply with regulator<br>Life time: 20 years | Designed capacity | 4,800 l/day <sup>2</sup> |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--|---------|-------------------------------------------------------------|------------|------------------------------------------------------|-------------|-----------------------------------------------------|-------------------|--------------------------|
| Hardware specification                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| UV lamp                                              | Lamp power: 36W, 6A<br>Life time: about 8000 hours of using                                                                                                                                                                                                                                                                                                                                                                                                                                             |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Motor pump                                           | Maximum flow rate: 1500 l/min<br>Life time : 3 years                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Solar panel                                          | 12 VDC supply with regulator<br>Life time: 20 years                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Designed capacity                                    | 4,800 l/day <sup>2</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Choice of data or Measurement methods and procedures | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Purpose of data                                      | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |
| Additional comment                                   | Please note that the UV lamps use in water kiosks have an 8,000 hour of usage life expectancy and which can be replaced after the 8,000 hours of operation at multiple times. Further the water kiosks use, sand filters and other kind of filters for water filtration that can be easily replaced and with such replacement, kiosk provide safe water quality and new life to the system. Hence the water kiosks has life more than 15 years as it renewed every time on maintenance and replacement. |                        |  |         |                                                             |            |                                                      |             |                                                     |                   |                          |

|                                                      |                                                                                                                                          |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| ID Parameter                                         | SDWS 8                                                                                                                                   |
| Data/parameter                                       | $xf$                                                                                                                                     |
| Unit                                                 | Percentage of wood use in target population                                                                                              |
| Description                                          | % of target population using wood for water boiling                                                                                      |
| Source of data                                       | Baseline survey                                                                                                                          |
| Value(s) applied                                     | 84%                                                                                                                                      |
| Choice of data or Measurement methods and procedures | Following the methodology "Gold Standard Methodology for emission reductions from safe drinking water supply, Version 1.0 (03/05/2021)". |
| Purpose of data                                      | Calculation of baseline scenario                                                                                                         |
| Additional comment                                   | N/A                                                                                                                                      |

|                                                      |                                                                                                                                           |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| ID Parameter                                         | SDWS 8 charcoal                                                                                                                           |
| Data/parameter                                       | $xf$                                                                                                                                      |
| Unit                                                 | Percentage of charcoal use in target population                                                                                           |
| Description                                          | % of target population using charcoal for water boiling                                                                                   |
| Source of data                                       | Baseline survey                                                                                                                           |
| Value(s) applied                                     | 5%                                                                                                                                        |
| Choice of data or Measurement methods and procedures | Following the methodology " Gold Standard Methodology for emission reductions from safe drinking water supply, Version 1.0 (03/05/2021)". |
| Purpose of data                                      | Calculation of baseline scenario                                                                                                          |
| Additional comment                                   | N/A                                                                                                                                       |

|                                                      |                                                    |
|------------------------------------------------------|----------------------------------------------------|
| ID Parameter                                         | SDWS 8 gas                                         |
| Data/parameter                                       | $xf$                                               |
| Unit                                                 | Percentage of gas use in target population         |
| Description                                          | % of target population using gas for water boiling |
| Source of data                                       | Baseline survey                                    |
| Value(s) applied                                     | 11%                                                |
| Choice of data or Measurement methods and procedures | Households using gas for the stove                 |
| Purpose of data                                      | Calculation of baseline scenario                   |

|                |                  |
|----------------|------------------|
| ID Parameter   | SDWS 9           |
| Data/parameter | $F_{b,wood,C02}$ |
| Unit           | tCO2/TJ          |



|                                                      |                                                              |
|------------------------------------------------------|--------------------------------------------------------------|
| Description                                          | CO2 emission factor from use wood that is reduced            |
| Source of data                                       | IPCC 2006 Guidelines for National Greenhouse Gas Inventories |
| Value(s) applied                                     | 112                                                          |
| Choice of data or Measurement methods and procedures | Default value for firewood/firewood waste is applied         |
| Purpose of data                                      | Calculation of baseline and project scenario                 |
| Additional comment                                   |                                                              |

|                                                      |                                                              |
|------------------------------------------------------|--------------------------------------------------------------|
| ID Parameter                                         | SDWS 9                                                       |
| Data/parameter                                       | <i>EF<sub>b, charcoal, CO2</sub></i>                         |
| Unit                                                 | tCO <sub>2</sub> /TJ                                         |
| Description                                          | CO2 emission factor from use charcoal that is reduced        |
| Source of data                                       | IPCC 2006 Guidelines for National Greenhouse Gas Inventories |
| Value(s) applied                                     | 165.22                                                       |
| Choice of data or Measurement methods and procedures | Default value for charcoal is applied                        |
| Purpose of data                                      | Calculation of baseline and project scenario                 |
| Additional comment                                   |                                                              |

|                |                                                  |
|----------------|--------------------------------------------------|
| ID Parameter   | SDWS 9                                           |
| Data/parameter | <i>EF<sub>b, gas, CO2</sub></i>                  |
| Unit           | tCO <sub>2</sub> /TJ                             |
| Description    | CO2 emission factor from use gas that is reduced |

|                                                      |                                                              |
|------------------------------------------------------|--------------------------------------------------------------|
| Source of data                                       | IPCC 2006 Guidelines for National Greenhouse Gas Inventories |
| Value(s) applied                                     | 63.1                                                         |
| Choice of data or Measurement methods and procedures | Default value for gas is applied                             |
| Purpose of data                                      | Calculation of baseline and project scenarii                 |
| Additional comment                                   |                                                              |

|                                                      |                                                                         |
|------------------------------------------------------|-------------------------------------------------------------------------|
| ID Parameter                                         | SDWS 10                                                                 |
| Data/parameter                                       | <i>EFb,wood,nonCO2</i>                                                  |
| Unit                                                 | tCO2/TJ                                                                 |
| Description                                          | Non-CO2 emission factor arising from the use of biomass that is reduced |
| Source of data                                       | IPCC default value AR5 GWP                                              |
| Value(s) applied                                     | 9.46 tCO2e/TJ                                                           |
| Choice of data or Measurement methods and procedures | Default IPCC value                                                      |
| Purpose of data                                      | Calculation of baseline and project scenarii                            |
| Additional comment                                   |                                                                         |

|                |                                                                          |
|----------------|--------------------------------------------------------------------------|
| ID Parameter   | SDWS 10                                                                  |
| Data/parameter | <i>EFb,charcoal,nonCO2</i>                                               |
| Unit           | tCO2/TJ                                                                  |
| Description    | Non-CO2 emission factor arising from the use of charcoal that is reduced |

|                                                      |                                             |
|------------------------------------------------------|---------------------------------------------|
| Source of data                                       | IPCC default value AR5 GWP                  |
| Value(s) applied                                     | 44.83 tCO <sub>2</sub> e/TJ                 |
| Choice of data or Measurement methods and procedures | Default IPCC value                          |
| Purpose of data                                      | Calculation of baseline and project scenari |
| Additional comment                                   |                                             |

| ID Parameter                                                        | SDWS 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                  |  |                  |              |                                  |                                                                  |       |     |                                                                     |   |     |                    |    |     |                         |     |                   |                                     |        |  |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|--|------------------|--------------|----------------------------------|------------------------------------------------------------------|-------|-----|---------------------------------------------------------------------|---|-----|--------------------|----|-----|-------------------------|-----|-------------------|-------------------------------------|--------|--|
| Data/parameter                                                      | $\eta_{wb}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                  |  |                  |              |                                  |                                                                  |       |     |                                                                     |   |     |                    |    |     |                         |     |                   |                                     |        |  |
| Unit                                                                | Percentage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                  |  |                  |              |                                  |                                                                  |       |     |                                                                     |   |     |                    |    |     |                         |     |                   |                                     |        |  |
| Description                                                         | Efficiency of baseline water boiling devices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                  |  |                  |              |                                  |                                                                  |       |     |                                                                     |   |     |                    |    |     |                         |     |                   |                                     |        |  |
| Source of data                                                      | Baseline survey                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                  |  |                  |              |                                  |                                                                  |       |     |                                                                     |   |     |                    |    |     |                         |     |                   |                                     |        |  |
| Value(s) applied                                                    | <div>15.10%</div> <table><thead><tr><th>Stove Technology</th><th>No. of Users</th><th>Default Efficiency<sup>40</sup></th></tr></thead><tbody><tr><td>HH using three stone stoves traditional stoves / clay stove only</td><td>136.5</td><td>10%</td></tr><tr><td>HH Using Improved traditional method / Improved Woodfired Jiko Only</td><td>0</td><td>20%</td></tr><tr><td>HH Using LPG Stove</td><td>17</td><td>55%</td></tr><tr><td>HH Using Electric Stove</td><td>0.5</td><td>50%<sup>41</sup></td></tr><tr><td>Weighted Average Thermal Efficiency</td><td colspan="2">15.10%</td></tr></tbody></table> |                                  |  | Stove Technology | No. of Users | Default Efficiency <sup>40</sup> | HH using three stone stoves traditional stoves / clay stove only | 136.5 | 10% | HH Using Improved traditional method / Improved Woodfired Jiko Only | 0 | 20% | HH Using LPG Stove | 17 | 55% | HH Using Electric Stove | 0.5 | 50% <sup>41</sup> | Weighted Average Thermal Efficiency | 15.10% |  |
| Stove Technology                                                    | No. of Users                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Default Efficiency <sup>40</sup> |  |                  |              |                                  |                                                                  |       |     |                                                                     |   |     |                    |    |     |                         |     |                   |                                     |        |  |
| HH using three stone stoves traditional stoves / clay stove only    | 136.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 10%                              |  |                  |              |                                  |                                                                  |       |     |                                                                     |   |     |                    |    |     |                         |     |                   |                                     |        |  |
| HH Using Improved traditional method / Improved Woodfired Jiko Only | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 20%                              |  |                  |              |                                  |                                                                  |       |     |                                                                     |   |     |                    |    |     |                         |     |                   |                                     |        |  |
| HH Using LPG Stove                                                  | 17                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 55%                              |  |                  |              |                                  |                                                                  |       |     |                                                                     |   |     |                    |    |     |                         |     |                   |                                     |        |  |
| HH Using Electric Stove                                             | 0.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 50% <sup>41</sup>                |  |                  |              |                                  |                                                                  |       |     |                                                                     |   |     |                    |    |     |                         |     |                   |                                     |        |  |
| Weighted Average Thermal Efficiency                                 | 15.10%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                  |  |                  |              |                                  |                                                                  |       |     |                                                                     |   |     |                    |    |     |                         |     |                   |                                     |        |  |

<sup>40</sup> For traditional three stone fire (10%) and improved wood stove (20%), the default efficiency is as per the applied methodology. For LPG stoves studies show that LPG stoves have an efficiency of 50-60%. Since the project is not promoting improved fossil fuel stoves, and avoiding use of fossil fuels, a conservative value of 50% is applied for reporting purpose.

<sup>41</sup> The average efficiency of electric stove is 49.1%. As a conservative approach PP has considered thermal efficiency of electric stove as 50%.

[https://www.researchgate.net/publication/341875219\\_Quantifying\\_Energy\\_Losses\\_on\\_Electric\\_Cooking\\_Stove#:~:text=average%20efficiency%20of%20electric%20stove%20is%2049.1%25](https://www.researchgate.net/publication/341875219_Quantifying_Energy_Losses_on_Electric_Cooking_Stove#:~:text=average%20efficiency%20of%20electric%20stove%20is%2049.1%25).

|                                                      |                                                                                                                                           |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Choice of data or Measurement methods and procedures | Following the methodology " Gold Standard Methodology for emission reductions from safe drinking water supply, Version 1.0 (03/05/2021)". |
| Purpose of data                                      | Calculation of baseline scenario                                                                                                          |
| Additional comment                                   | N/A                                                                                                                                       |

|                                                      |                                                                                                                                                                               |                         |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| ID Parameter                                         | SDWS 12                                                                                                                                                                       |                         |
| Data/parameter                                       | C <sub>b</sub>                                                                                                                                                                |                         |
| Unit                                                 | Percentage                                                                                                                                                                    |                         |
| Description                                          | Proportion of project end-users who in the baseline were already using safe water, either from an improved water source, or from a water treatment method other than boiling. |                         |
| Source of data                                       | Baseline Survey                                                                                                                                                               |                         |
| Value(s) applied                                     | 39%                                                                                                                                                                           |                         |
|                                                      | Parameter                                                                                                                                                                     | Percentage of Users (%) |
|                                                      | Proportion of project end-users who in the baseline were already using safe water, from a water treatment method other than boiling                                           | 13%                     |
|                                                      | Proportion of project end-users who in the baseline were already using safe water, from purchasing it                                                                         | 22%                     |
|                                                      | % of HH who do not purify water but think the water they are getting from water source is safe for drinking                                                                   | 4%                      |
|                                                      | <b>C<sub>b</sub></b>                                                                                                                                                          | <b>39%</b>              |
| Choice of data or Measurement methods and procedures | Value is calculated based on baseline survey conducted for the target users                                                                                                   |                         |
| Purpose of data                                      | Determination of baseline emissions                                                                                                                                           |                         |
| Additional comment                                   | -                                                                                                                                                                             |                         |

|                                                      |                                        |
|------------------------------------------------------|----------------------------------------|
| ID Parameter                                         | SDWS 13                                |
| Data/parameter                                       | qi                                     |
| Unit                                                 | Liters per day                         |
| Description                                          | Capacity of the water kiosk technology |
| Source of data                                       | Manufacturer specifications            |
| Value(s) applied                                     | 4,800                                  |
| Choice of data or Measurement methods and procedures | -                                      |
| Purpose of data                                      | Calculation of project scenario        |
| Additional comment                                   | -                                      |

|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data/parameter                                       | $SE_{w,b,y}$                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Unit                                                 | kJ/L                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Description                                          | Specific energy required to boil water using the baseline technology                                                                                                                                                                                                                                                                                                                                                                            |
| Source of data                                       | Safe drinking water supply methodology default value                                                                                                                                                                                                                                                                                                                                                                                            |
| Value(s) applied                                     | 23.90                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Choice of data or Measurement methods and procedures | Calculated as follow: $SE_{w,b,y}=360.83/\eta_{wb}$ with 360.83 kJ/L being the default amount of energy required to obtain 1 L of water after 5 minutes of boiling from a first principles approach and $\eta_{wb}$ the efficiency of the stoves for baseline water boiling. As beneficiaries used traditional three-stone stoves in baseline scenario, $\eta_{wb}$ is calculated as 15.10% from baseline survey as per the methodology (3.6.2) |
| Purpose of data                                      | Calculation of baseline scenario                                                                                                                                                                                                                                                                                                                                                                                                                |
| Additional comment                                   | -                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Parameter ID                                         | SDWS 14                                                   |
|------------------------------------------------------|-----------------------------------------------------------|
| Data/parameter                                       | NCV <sub>f</sub>                                          |
| Unit                                                 | MJ/kg                                                     |
| Description                                          | Net calorific value of diesel                             |
| Source of data                                       | Table 1.2 of Chapter1 of Vol. 2 (Energy) of the 2006 IPCC |
| Value(s) applied                                     | 43                                                        |
| Choice of data or Measurement methods and procedures | -                                                         |
| Purpose of data                                      | Project emissions                                         |
| Additional comment                                   | -                                                         |

|                                                      |                                                          |
|------------------------------------------------------|----------------------------------------------------------|
| Data/parameter                                       | ρ <sub>i,y</sub>                                         |
| Unit                                                 | kg/L                                                     |
| Description                                          | Density of diesel                                        |
| Source of data                                       | Table A3.8 of Energy Statistics Manual by OECD/IEA, 2005 |
| Value(s) applied                                     | 0.8439                                                   |
| Choice of data or Measurement methods and procedures | -                                                        |
| Purpose of data                                      | Project emissions                                        |
| Additional comment                                   | -                                                        |

| Parameter ID   | SDWS 15         |
|----------------|-----------------|
| Data/parameter | EF <sub>f</sub> |

|                                                      |                                                           |
|------------------------------------------------------|-----------------------------------------------------------|
| Unit                                                 | TCO <sub>2</sub> /TJ                                      |
| Description                                          | Emission factor for diesel                                |
| Source of data                                       | Table 1.4 of Chapter1 of Vol. 2 (Energy) of the 2006 IPCC |
| Value(s) applied                                     | 74.1                                                      |
| Choice of data or Measurement methods and procedures | -                                                         |
| Purpose of data                                      | Project emissions                                         |
| Additional comment                                   |                                                           |

|                                                      |                                                                                      |
|------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Parameter ID</b>                                  | <b>SDWS 16</b>                                                                       |
| Data/parameter                                       | EF <sub>ec</sub>                                                                     |
| Unit                                                 | tCO <sub>2</sub> /kWh                                                                |
| Description                                          | Emission factor for electricity use                                                  |
| Source of data                                       | Default value of methodology for emission reductions from safe drinking water supply |
| Value(s) applied                                     | 0,001                                                                                |
| Choice of data or Measurement methods and procedures | -                                                                                    |
| Purpose of data                                      | Project emissions                                                                    |
| Additional comment                                   |                                                                                      |

|                     |                   |
|---------------------|-------------------|
| <b>Parameter ID</b> | <b>SDWS 17</b>    |
| Data/parameter      | TDL <sub>ec</sub> |
| Unit                | %                 |

|                                                      |                                                                                      |
|------------------------------------------------------|--------------------------------------------------------------------------------------|
| Description                                          | Transmission and distribution losses associated with the electricity use             |
| Source of data                                       | Default value of methodology for emission reductions from safe drinking water supply |
| Value(s) applied                                     | 20%                                                                                  |
| Choice of data or Measurement methods and procedures | -                                                                                    |
| Purpose of data                                      | Project emissions                                                                    |
| Additional comment                                   | -                                                                                    |

|                                                      |                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Parameter ID</b>                                  | <b>SDWS 21</b>                                                                                                                                                                                                                                                                         |
| Data/parameter                                       | $f_{NRB,f,y}$                                                                                                                                                                                                                                                                          |
| Unit                                                 | Percentage                                                                                                                                                                                                                                                                             |
| Description                                          | Fractional non-renewability status of woody biomass fuel during year y.<br><br>Determined ex-ante and fixed for a given crediting period                                                                                                                                               |
| Source of data                                       | Literature review.<br>See section B4 Establishment and description of baseline scenario                                                                                                                                                                                                |
| Value(s) applied                                     | 52.03%                                                                                                                                                                                                                                                                                 |
| Choice of data or Measurement methods and procedures | CDM Tool 30: Calculation of the fraction of non-renewable biomass version 3.0<br><br><a href="https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v1.pdf/history_view">https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v1.pdf/history_view</a> |
| Purpose of data                                      | SDG 13 claims: Emission reductions calculations                                                                                                                                                                                                                                        |
| Additional comment                                   | Determined ex-ante and fixed for a given crediting period (as it is fixed ex-ante, hence included $f_{NRB,f,y}$ in the "data and parameters fixed ex ante" section of the PDD),                                                                                                        |



| Parameter ID                       | SDWS 24                                                              |
|------------------------------------|----------------------------------------------------------------------|
| Data/parameter                     | QPW <sub>p</sub>                                                     |
| Unit                               | Liters/person/day                                                    |
| Description                        | Volume of drinking water per person per day for premises type p      |
| Source of data                     | Default value of the methodology - Full-day premises: 4 L/person/day |
| Value(s) applied                   | 4 L/person/day                                                       |
| Measurement methods and procedures | Meth Default Value                                                   |
| Monitoring frequency               | NA                                                                   |
| QA/QC procedures                   | -                                                                    |
| Purpose of data                    | SDG 13 claims : Emission reductions calculations                     |
| Additional comment                 | -                                                                    |

| Parameter ID                       | SDWS 35                                                                                       |
|------------------------------------|-----------------------------------------------------------------------------------------------|
| Data/parameter                     | LE <sub>y</sub>                                                                               |
| Unit                               | tCO <sub>2</sub> e per year                                                                   |
| Description                        | Leakage emissions during year y                                                               |
| Source of data                     | Sources established by following Leakage emissions section                                    |
| Value(s) applied                   | 5%<br>Ex ante : default value                                                                 |
| Measurement methods and procedures | As the default value has been applied, hence no measurement methods and procedures applicable |
| Monitoring frequency               | At crediting period renewal                                                                   |
| QA/QC procedures                   | -                                                                                             |

|                    |                                                  |
|--------------------|--------------------------------------------------|
| Purpose of data    | SDG 13 claims : Emission reductions calculations |
| Additional comment | -                                                |

### B.6.3 Ex ante estimation of SDG Impact

#### >> (SDG 13)

The emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y - LE_y$$

Where:

- $ER_y$  = Emission reductions in year y (tCO<sub>2</sub>e/yr)
- $BE_y$  = Baseline emissions in year y (t CO<sub>2</sub>e/yr)
- $PE_y$  = Project emissions in year y (t CO<sub>2</sub>e/yr)
- $LE_y$  = Leakage emissions in year y (t CO<sub>2</sub>e/yr)

#### Baseline emissions:

$$BE_y = EF_b \times (1 - C_b - X_{cleanboil,y}) \times Q_y \times M_{q,y}$$

Where:

|                   | Definition                                                                                                                   | Numeric value determined ex-ante                                               |
|-------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| $BE_y$            | Baseline emissions from the use of fuel to obtain safe water in the baseline (tCO <sub>2</sub> e)                            |                                                                                |
| $C_b$             | Proportion of project end-users who in the baseline were already using a safe water supply that did not require boiling (%). | With 39% estimated ex-ante as per Baseline Survey.                             |
| $X_{cleanboil,y}$ | Proportion of project end-users that boil safe water in the project year y (%)                                               | With 0% estimated from the Baseline survey                                     |
| $Q_y$             | Quantity of safe drinking water provided by the project in year y (L)                                                        | With 658,825 L estimated. 1900L/day with 347 days of operations (365*0.95=347) |
| $M_{q,y}$         | Modifier for the water quality in year y                                                                                     | With a modifier estimated at 95% ex-ante.                                      |

With  $EF_b = SE_{w,b,y} \times \sum (x_f \times (EF_{b,f,CO2} \times f_{NRB,f,y} + EF_{b,f,nonCO2}))_f$

Where:

|                    | Definition                                                                                                                                                                                                                                                          | Numeric value determined ex-ante                                                                                                                                |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $EF_b$             | Emission factor for the use of fuel to obtain safe water in the baseline (tCO <sub>2e</sub> /L)                                                                                                                                                                     | -                                                                                                                                                               |
| $SE_{w,b,y}$       | Specific energy required to boil water (kJ/L),                                                                                                                                                                                                                      | With $SE_{w,b,y} = 360.83 / \eta_{wb} = 360.83 / 15.1\% = 2389.6$ kJ/L                                                                                          |
| $x_f$              | Proportion of fuel f used in the baseline (fraction determined based on an energy basis).                                                                                                                                                                           | With $x_{wood} = 84\%$ and $x_{charcoal} = 5\%$ and $x_{gas} = 11\%$                                                                                            |
| $EF_{b,f,CO_2}$    | CO <sub>2</sub> emission factor from use of fuel f (tCO <sub>2</sub> /TJ)                                                                                                                                                                                           | With $EF_{b,wood,CO_2} = 112.0$ tCO <sub>2</sub> /TJ and $EF_{b,charcoal,CO_2} = 165.22$ tCO <sub>2</sub> /TJ and $EF_{b,gas,CO_2} = 63.1$ tCO <sub>2</sub> /TJ |
| $EF_{b,f,nonCO_2}$ | Non-CO <sub>2</sub> emission factor arising from use of fuel f, when the baseline fuel f is biomass or charcoal (tCO <sub>2e</sub> /TJ). This parameter is omitted when f is a fossil fuel.                                                                         | With, $EF_{b,wood,nonCO_2} = 9.46$ tCO <sub>2</sub> /TJ and $EF_{b,charcoal,nonCO_2} = 44.83$ tCO <sub>2</sub> /TJ                                              |
| $f_{NRB,f,y}$      | Fractional non-renewability status of woody biomass fuel during year y (fraction). For biomass, it is the fraction of woody biomass that can be established as non-renewable. This parameter is omitted when f is a fossil fuel. f = Index for baseline fuel types. | With $f_{NRB,f,y} = 52.03\%$                                                                                                                                    |

Which makes:

$$EF_b = 2389.6 * [(0.84 * (112 * 52.03\% + 9.46) + (0.05 * (165.22 * 52.03\% + 44.83)) + (0.11 * 63.1)] = 0.000168 \text{ tCO}_2\text{e/L.}$$

And where  $Q_y = \min(Q_{m,y}, Q_{pop,y}) = 658,825$  L

Where:

- $Q_m$ , = Monitored quantity of safe water provided by the project in year y (L). With an ex-ante estimation of 658,825 L.yr<sup>-1</sup> for each Kiosk.
- $Q_{pop}$ , = Quantity of safe drinking water that could be consumed by project end-users in year y (L), with an ex-ante estimation 11,928,200 L. yr<sup>-1</sup> for each Kiosk.

And where:

$$Qpop,y = \sum p HH_{p,y} \times HN_{p,y} \times QPW_p \times DO_{p,y}$$

Where:

- $HH_{p,y}$  = Number of premises type p served by the project in year y, an average of 8600 households per kiosk
- $U_{p,y}$  = Proportion of premises in the 1km radius that actually use water at least every-two-days, 100%.
- $HN_{p,y}$  = Number of individuals per premises type p (e.g. household, school) in year y, an average of 4,46 people per household according to the baseline survey
- $QPW_p$  = Volume of drinking water per person per day for premises type p (L). The default value of 4 L per person was considered.
- $DO_p$  = Days the project technology is operational for end-users in premises p in year y. An operating rate of 95% has been applied.

We find:

$$Qpop_r = 8,600 \times 1.0 \times 4 \times 365 \times 0.95 = 11,928,200 \text{ L}$$

Hence:

$$BE_y = 0.000252 \times (1 - 39\% - 0\%) \times 658,825 \times 0.95$$

|                                                 |
|-------------------------------------------------|
| $BE_y = 64.21 \text{ tCO}_2\text{e/year/kiosk}$ |
|-------------------------------------------------|

## Project Emissions

$$PE_y = PE_{ff,y} + PE_{ec,p,y}$$

Where :

- $PE_y$  = Project emissions in year y ( $\text{tCO}_2$ )
- $PE_{ff,p,y}$  = Project emissions from fossil fuel use in year y ( $\text{tCO}_2$ )
- $PE_{ec,p,y}$  = Project emissions from electricity use in year y ( $\text{tCO}_2$ )

Where:

$$PE_{ff,y} = \sum Pp,f,y \times NCV_f \times pf,y \times EF_f$$

- $Pp,f,y$  = Quantity of diesel that is consumed in the project during year y = 900 L per kiosk

- $NCV_f$  = Net calorific value of diesel = 43 MJ/kg
- $\rho_{f,y}$  = Density of diesel = 0.8439 kg/L
- $EF_f$  = Emission factor of diesel = 74.10 tCO<sub>2</sub>e/TJ

The project activity involves transportation by road for delivery of water jugs but transportation is not the main project activity.

Therefore:

$$PE_{ff,,} = 2.42 \text{ tCO}_2\text{e}$$

Where

$$PE_{ec,,} = \sum EC_{p,y} \times EF_{ec} \times (1 + TD_{Lec})$$

- $EC_{p,y}$  = Quantity of electricity that is consumed in the project during year y = 552 kWh per kiosk
- $EF_{ec}$  = Emission factor associated with electricity used = 0.001 tCO<sub>2</sub>e/kWh (default value given by the methodology)
- $TD_{Lec}$  = Transmission and distribution losses associated with the electricity use = 20% (default value given by the methodology)

Therefore:

$$PE_{ec,,} = 0.66 \text{ tCO}_2\text{e}$$

This value is negligible as the kiosks will be solar-powered.

### Emissions Reductions:

As demonstrated above:

$$ER_y = 58.07 \text{ tCO}_2\text{e}$$

$$BE_y = 64.21 \text{ tCO}_2\text{e}$$

$$PE_y = 3.08 \text{ tCO}_2\text{e}$$

$LE_y = 3.06$  as it represents a net to gross adjustment factor of 0.95 of ( $BE_y - PE_y$ ) to account for leakages.

Leakage has been assessed as per 3.8.2.1 with an ex-ante evaluation for each of the following sources:

*"a. Members of the population who do not participate in the project, and previously used lower emitting energy sources, instead use the non-renewable biomass saved under the project activity"*

The population that does not participate in the project is the population that already has water treatment systems (13% use techniques other than boiling) or that buys bottled water (22% buy water) and 4% consider the water they use to be potable. There is no reason for these households to begin to use non-renewable biomass, especially if they have access to the 1001fontaines project.

*"b. The project significantly reduces the NRB fraction within an area where other GHG mitigation project activities account for NRB fraction in their baseline scenario".*

The project is a small-scale project and thus will not substantially affect the NRB fraction calculation at the national scale of similar projects currently assessing their baseline scenario.

*c. The project population compensates for loss of the space heating effect of water boiling by adopting some other form of space heating or by retaining some baseline wood fuel-burning practices. Leakage risks deemed very low can be ignored as long as the case for their insignificance is substantiated.*

Cambodia is a tropical country where mean temperature is warm (more than 28°). Hence space heating leakage is considered as non-existent for this project.

#### **(SDG 6)**

Baseline: no safe water distributed.

Project outcome:

$$Qy = \sum_{year}^y Qm, year$$

Where:

$Qy$  = cumulative quantity of clean water distributed to project beneficiaries from beginning of project until end of year  $y$  (L). It is a cumulative indicator.

$Qm, ear$  = Monitored quantity of safe water provided by the project to beneficiaries in year  $y$ (L).

Ex-ante estimation is calculated as follow, with  $Q_{th}$  being the quantity sold on average<sup>42</sup> and  $Nb_{days}$  being the number of days kiosks are active per year:

$$Qm, = Q_{th} \times Nb_{days} = 1900 \times 365 \times 95\% = 658,825 \text{ L.yr}^{-1} \text{ for each Kiosk.}$$

Ex-ante estimation is a distribution of **658,825 L/year/kiosk**.

---

<sup>42</sup> Data from 1001fontaines expertise and experience from other projects

## (SDG 8)

Baseline: no job created.

Project outcome:  $N_{jc}$  is monitored as being the number of jobs created in year  $y$

$$N_{jy} = \sum_{year}^y N_{jyear}$$

Where:

$N_{jy}$  = cumulative number of jobs created between beginning of project and end of year  $y$

$N_{jyear}$  = number of jobs created in year "year". Ex-ante estimation is the creation of 4 **jobs/kiosk**.

### B.6.4 Summary of ex ante estimates of each SDG Impact

SDG 13: (unit: VERs generated)

| Year                                            | Baseline estimate | Project estimate | Leakage estimate | Net benefit |
|-------------------------------------------------|-------------------|------------------|------------------|-------------|
| Year 1                                          | 3,756             | 180              | 179              | 3,397       |
| Year 2                                          | 5,329             | 256              | 254              | 4,820       |
| Year 3                                          | 6,677             | 321              | 318              | 6,039       |
| Year 4                                          | 8,764             | 421              | 417              | 7,926       |
| Year 5                                          | 9,888             | 475              | 471              | 8,942       |
| Total                                           | 34,415            | 1,652            | 1,638            | 31,124      |
| <b>Total number of crediting years</b>          | <b>5</b>          |                  |                  |             |
| <b>Annual average over the crediting period</b> | 6,883             | 330              | 328              | 6,225       |

SDG 6: (unit: quantity of safe water distributed)

| Year   | Baseline estimate | Project estimate | Net benefit |
|--------|-------------------|------------------|-------------|
| Year 1 | 0                 | 38,541,263       | 38,541,263  |
| Year 2 | 0                 | 54,682,475       | 54,682,475  |
| Year 3 | 0                 | 68,517,800       | 68,517,800  |
| Year 4 | 0                 | 89,929,613       | 89,929,613  |

|                                                 |          |             |             |
|-------------------------------------------------|----------|-------------|-------------|
| Year 5                                          | 0        | 101,459,050 | 101,459,050 |
| Total                                           | 0        | 353,130,200 | 353,130,200 |
| <b>Total number of crediting years</b>          | <b>5</b> |             |             |
| <b>Annual average over the crediting period</b> | 0        | 70,626,040  | 70,626,040  |

SDG 8: (unit: number of jobs created)

| Year                                            | Baseline estimate | Project estimate | Net benefit |
|-------------------------------------------------|-------------------|------------------|-------------|
| Year 1                                          | 0                 | 148              | 148         |
| Year 2                                          | 0                 | 48               | 48          |
| Year 3                                          | 0                 | 120              | 120         |
| Year 4                                          | 0                 | 140              | 140         |
| Year 5                                          | 0                 | 0                | 0           |
| Total                                           | <b>0</b>          | 456              | 456         |
| <b>Total number of crediting years</b>          | <b>5</b>          |                  |             |
| <b>Annual average over the crediting period</b> | <b>0</b>          | <b>91</b>        | <b>91</b>   |

## B.7. Monitoring plan

B.7.1 Data and parameters to be monitored

### A. Related to water quality

| Parameter ID   | SDWS 18                                                                                                                                                                   |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data/parameter | <i>M<sub>q,y</sub></i>                                                                                                                                                    |
| Unit           | Fraction                                                                                                                                                                  |
| Description    | Ongoing water quality indicated as the fraction of the samples that pass microbial quality standard requirements specified in Kingdom of Cambodia: Drinking Water Quality |



|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Standards as well as in WHO Guideline values for verification of microbial quality i.e., all water directly intended for drinking must not have detectable E.Coli in any 100 ml sample i.e., less than 1 Colony Forming Unit (CFU) of E.Coli /100 ml.</p> <p>Water is transported in sealed bottle or jugs from water kiosk to end user (households, schools, commercial establishments etc.) location water quality at the point of, protecting drinking water from contamination during transport as well as storage in the home.</p> <p>1001fontaines will test the quality of water from a sealed jugs or bottle when it reaches the end-user premises (e.g. household, institution) to ensure that the hygiene procedures guaranteeing water quality until the point of use are respected.</p>       |
| Source of data                     | <p>Water Quality Test Reports</p> <p>1001fontaines test the quality of water from a sealed jugs or bottles right before the delivery of water to the end users (households, schools, commercial establishments etc.)</p> <p>Water is transported in sealed bottle or jugs from water kiosk to the end user (households, schools, commercial establishments etc.) location.</p> <p>Once the water bottle/jug are sealed, they are delivered to the end users (households, schools, commercial establishments etc.) within 24 hours or less. So, the safe water remains at kiosks for only few hours and then delivered to the end users (households, schools, commercial establishments etc.) hence there is no change in the water quality or no possibility of contamination during the transportation.</p> |
| Value(s) applied                   | 95%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Measurement methods and procedures | <p>The purified water was collected and tested by Teuk Saat laboratory on a monthly basis. At least biannual test was sent out and tested by accredited laboratory.</p> <p>The applied technology complies with the Kingdom of Cambodia: Drinking Water Quality Standards as well as</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>"WHO Guideline values for verification of microbial quality i.e., all water directly intended for drinking must not have detectable E.Coli in any 100 ml sample i.e., less than 1 Colony Forming Unit (CFU) of E.Coli /100 ml. A presence of less than 1 E. coli CFU/100 ml in the treated water is acceptable.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Monitoring frequency | Monthly                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| QA/QC procedures     | <p>4 provincial laboratories can perform water analyses microbiological and physio-chemical parameters for the water kiosks following the rules below:</p> <ol style="list-style-type: none"> <li>1. Laboratories used for water quality testing are approved by local health authorities and/or have quality accreditation; and</li> <li>2. The laboratory used must demonstrate that it has an adequate quality management plan in place which addresses both quality assurance and quality control test procedures.</li> <li>3. Field testing kits also are eligible, e.g. based on Colony Forming Unit method or Most Probable Number method. To use the field testing kits the project shall meet the following requirements: <ol style="list-style-type: none"> <li>a. Testing kits must be approved by national agency or meet standards set by relevant international organization e.g. US-EPA, and</li> <li>b. Testing kits shall be tested for its accuracy and robustness prior to application for project level monitoring, whereby local or accredited laboratory shall conduct water quality tests using testing kits and a relevant ISO standard or an equivalent standard, in parallel with field testing kits.</li> </ol> </li> </ol> |
| Purpose of data      | To calculate SDG 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Additional comment   | <p>Applies in all cases If the proportion of samples not meeting Safe Drinking Water Quality Standards exceeds a threshold, no emission reductions can be claimed for the corresponding monitoring period. Thresholds: - Project or VPA year 1: 20% - Project or VPA year 2: 15% - Project or VPA year 3 or above: 10% When the crediting period is renewed, the year number count continues, i.e. the second crediting period would encompass year 6, year 7, year 8, etc. Additionally, when the threshold is exceeded, the project shall provide an explanation for why this occurred and provide a remediation plan.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## SDG 8

| Parameter ID                       | SDWS 19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data/parameter                     | <i>N/year</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Unit                               | Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Description                        | Number of fulltime jobs created in year y                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Source of data                     | Employment Records (Contract Agreement / Salary Payment Records)                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Value(s) applied                   | 4 per kiosk<br><br>One: Entrepreneur<br><br>Three: Operators                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Measurement methods and procedures | Employment Records (Contract Agreement / Salary Payment Records)                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Monitoring frequency               | Annually                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| QA/QC procedures                   | Transparent data analysis and reporting                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Purpose of data                    | SDG 8 Claims                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Additional comment                 | 1001f Teuk Saat signed one employment contract with each entrepreneur who manage the kiosk.<br><br>An entrepreneur hires around three operators for operation, maintenance, and management of a kiosk. The operators got hired based on their availability and can associate with kiosks operation on short- or long-term basis based on their availability. Hence there is no contract between entrepreneur and operators while the entrepreneur pays to the operators on monthly basis and keep it recorded. |

| Parameter ID     | SDWS 20                           |
|------------------|-----------------------------------|
| Data / Parameter | Water hygiene education campaigns |
| Unit             | NA                                |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                        | <p>Hygiene campaigns carried out among project safe water end users ( households, schools, commercial establishments etc.)</p> <p>The following guidelines apply for conducting these campaigns.</p> <ul style="list-style-type: none"> <li>- Hygiene refers to access to sanitation amenities, equipment and infrastructure, as well as to the behavior in respect to regular and correct use of such amenities. It also refers to behavior that prevents infections from water-related diseases.</li> <li>- The project developer will report the activities conducted each year in a detailed "Report of annual hygiene campaigns results" and summarize the results in the project monitoring reports.</li> <li>- Any major changes in the health status of the water users as a result of contaminated water (e.g. an outbreak of water-related disease) must be reported and, if relevant, a strategy put in place to address it through the subsequent hygiene campaign.</li> <li>- The detailed method used to assess hygienic handling of clean water must be provided with the PDD and verified by the VVB.</li> <li>- The details of the method should be adjusted to suit the circumstances of each project and also to suit learning year on year.</li> </ul> <p>The impacts of the hygiene campaign shall be assessed using the WHO/UNICEF Joint Monitoring Programme Core questions for drinking water and hygiene to determine the fraction of the households and institutions where Safe water and Hygiene practices are found to fulfill "safely managed" or "basic" requirements. In-person or telephone or by messaging (e.g. text, app) based survey shall be conduct covering all the JMP core questions for drinking water and core questions for hygiene. This hygiene campaign will be carried on the project survey.</p> |
| Source of data                     | Project Survey                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Value(s) applied                   | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Measurement methods and procedures | <p>Project Survey (Door to door or in-person or telephone or by messaging (e.g. text, app))</p> <ul style="list-style-type: none"> <li>- Introduction of the benefits of safe water to the Key Opinion Leaders and households living in the</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>commune using dedicated awareness collateral (posters, explanation)</p> <ul style="list-style-type: none"> <li>- The survey will cover the JMP core questions<sup>43</sup> for drinking water and core questions for hygiene.</li> </ul> <p>The impacts of the hygiene campaign shall be assessed using the WHO/UNICEF Joint Monitoring Programme Core questions for drinking water and hygiene to determine the fraction of the households and institutions where Safe water and Hygiene practices are found to fulfill “safely managed” or “basic” requirements.</p> <p>Stratified sampling approach (vintage wise) have been used for selection of kiosk from installation database for project survey. Once the kiosks samples have been selected, PP will use random sampling approach for selection of households/beneficiary (who purchased water from kiosk) on regular basis and conduct the project survey.</p> <p>As the project is a community level project and PP could not maintain the households/beneficiary data, random sampling approach best suited for selection of households for project survey once the kiosks have been selected.</p> <p>The no. of kiosks and households/beneficiaries selected for the project survey (sample size) will be in line with the meth requirement.</p> |
| Monitoring frequency | Annually                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| QA/QC procedures     | The fraction of the households where Safe water and Hygiene practices are found to fulfill “safely managed” or “basic” requirements is expected to increase over time because of the hygiene campaigns.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Purpose of data      | To calculate SDG                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Additional comment   | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## B. Related to emissions reduction

|                     |                     |
|---------------------|---------------------|
| <b>Parameter ID</b> | <b>SDWS 22</b>      |
| Data/parameter      | <i>Xcleanboil,y</i> |

<sup>43</sup> <https://washdata.org/monitoring/methods/core-questions>

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unit                               | Percentage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Description                        | Proportion of project end-users that boil safe water in year y                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Source of data                     | Project Survey                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Value(s) applied                   | 0%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Measurement methods and procedures | <p>Project survey</p> <p>The project survey will collect the information regarding the households boil water in project scenario. The survey questionnaire will have question related to water boiling in project scenario.</p> <p>This survey may be performed in person, by telephone, by messaging (e.g. text, app), appropriate to the context. For sampling, 1001fontaines will follow the section 4.2  General requirements for sampling, below of the methodology</p> <p>Stratified sampling approach (vintage wise) have been used for selection of kiosk from installation database for project survey. Once the kiosks samples have been selected, PP will use random sampling approach for selection of households/beneficiary (who purchased water from kiosk) on regular basis and conduct the project survey.</p> <p>As the project is a community level project and PP could not maintain the households/beneficiary data, random sampling approach best suited for selection of households for project survey once the kiosks have been selected.</p> <p>The no. of kiosks and households/beneficiaries selected for the project survey (sample size) will be in line with the meth requirement.</p> |
| Monitoring frequency               | Annually                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Purpose of data                    | SDG 13 claims: Emission reductions calculations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Additional comment                 | Some of the project households collect surface water for cooking purpose. As the surface water is not safe, there may be possibility that some end users may boil this water for cooking purpose but not for drinking.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Parameter ID                       | SDWS 23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data/parameter                     | $Q_{m,y}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Unit                               | Liters/year                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Description                        | Monitored quantity of safe water provided by the kiosks of the project in year y                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Source of data                     | Site Information Management System (SIMS) database (sales / distribution database)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Value(s) applied                   | <p>658,825 L</p> <p>Ex-ante: capacity's kiosks which shall be 1900L/kiosks/day</p> <p>95% of 365 days as per technical specifications provided by 1001fontaines</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Measurement methods and procedures | <p>The number of purified water bottles/jugs (20L or other) delivered/sold will be measured. 1001f accounting only the actual quantity of water sold and not claiming credit based on the quantity produced (avoiding leakage of water if any), most conservative approach.</p> <p>Entrepreneurs record operation data in a notebook on a daily basis (number of functional days, no. of bottles/jugs sold/distributed, expenses). Advisors collect Water Kiosk's operation data on a monthly basis and record it on their tablet, through an online application called commcare. Data is consolidated at national level, enabling ongoing monitoring of the network and standardized monthly reporting.</p> |
| Monitoring frequency               | Continuously                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| QA/QC procedures                   | The quantity of purified water was verified by the platform team (Teuk Saat 1001), reported to the Teuk Saat office and kept in the SIMS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Purpose of data                    | SDG 13 Claims                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Additional comment                 | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Parameter ID | SDWS 25 |
|--------------|---------|
|--------------|---------|

|                                    |                                                                                                                                                                                  |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data/parameter                     | $HN_{p,y}$                                                                                                                                                                       |
| Unit                               | Number                                                                                                                                                                           |
| Description                        | Number of individuals per premises type p (e.g. household, school) in the project boundary in year y                                                                             |
| Source of data                     | General population census of the Kingdom of Cambodia 2019 national report on final census results National Institute of Statistics, Ministry of Planning, Oct-2020 <sup>44</sup> |
| Value(s) applied                   | 4.3                                                                                                                                                                              |
| Measurement methods and procedures | Not applicable                                                                                                                                                                   |
| Monitoring frequency               | Annually                                                                                                                                                                         |
| Purpose of data                    | SDG 13 claims: Emission reductions calculations                                                                                                                                  |
| Additional comment                 | -                                                                                                                                                                                |

| Parameter ID                       | SDWS 26                                                                                                                                                                              |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data/parameter                     | $HH_{p,y}$                                                                                                                                                                           |
| Unit                               | Number                                                                                                                                                                               |
| Description                        | Number of premises type p served by the project in year y                                                                                                                            |
| Source of data                     | Project Survey<br><br>Survey of the sampled premises within to check how often the premises used the project water source during the year.                                           |
| Value(s) applied                   | 2,000                                                                                                                                                                                |
| Measurement methods and procedures | Project survey may be performed in person, by telephone, by messaging (e.g. text, app), appropriate to the context. Premises that report at least every-two-days use may be counted. |

<sup>44</sup> [https://data.opendatacambodia.net/en/library\\_record/general-population-census-of-the-kingdom-of-cambodia-2019-national-report-on-final-census-results/resource/dd6f15e0-0d7a-4ca7-b660-cbe42213f4ff](https://data.opendatacambodia.net/en/library_record/general-population-census-of-the-kingdom-of-cambodia-2019-national-report-on-final-census-results/resource/dd6f15e0-0d7a-4ca7-b660-cbe42213f4ff)



|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>Stratified sampling approach (vintage wise) have been used for selection of kiosk from installation database for project survey. Once the kiosks samples have been selected, PP will use random sampling approach for selection of households/beneficiary (who purchased water from kiosk) on regular basis and conduct the project survey.</p> <p>As the project is a community level project and PP could not maintain the households/beneficiary data, random sampling approach best suited for selection of households for project survey once the kiosks have been selected.</p> <p>The no. of kiosks and households/beneficiaries selected for the project survey (sample size) will be in line with the meth requirement.</p> |
| Monitoring frequency | Annually                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| QA/QC procedures     | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Purpose of data      | SDG 13 claims: Emission reductions calculations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Additional comment   | On average one distribution point serves 400 households, and one route is about 5 distribution points, hence $5 \times 400 = 2,000$ households served by one kiosk every day.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Parameter ID                       | SDWS 27                                                                                                                                                              |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data/parameter                     | $DO_{p,y}$                                                                                                                                                           |
| Unit                               | Days                                                                                                                                                                 |
| Description                        | Days the project technology is operational for end-users in premises p in year y                                                                                     |
| Source of data                     | Demonstrate from Site Information Management System (SIMS) database (log of operation and maintenance system)                                                        |
| Value(s) applied                   | 347<br><br>Ex-ante: Hypothesis kiosks are functional 95% of the year                                                                                                 |
| Measurement methods and procedures | Entrepreneurs record operation data in a notebook on a daily basis (number of functional days). Advisors collect Water Kiosk's operation data on a monthly basis and |

|                      |                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>record it on their tablet, through an online application called commcare. Data is consolidated at national level, enabling ongoing monitoring of the network and standardized monthly reporting.</p> <p>The technicians are alerted by the entrepreneurs when a technical issue (UV, filtration, solar etc) happens. Data and intervention reports are recorded in the Technical app.</p> |
| Monitoring frequency | Annually                                                                                                                                                                                                                                                                                                                                                                                     |
| QA/QC procedures     | -                                                                                                                                                                                                                                                                                                                                                                                            |
| Purpose of data      | SDG 13 claims : Emission reductions calculations                                                                                                                                                                                                                                                                                                                                             |
| Additional comment   | -                                                                                                                                                                                                                                                                                                                                                                                            |

| Parameter ID                       | SDWS 33                                                                                                                                                                                                         |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data/parameter                     | $Pp,f,y$                                                                                                                                                                                                        |
| Unit                               | mass or volume units (e.g. kg, Litres, standard m3)                                                                                                                                                             |
| Description                        | Quantity of fossil fuel f that is consumed in the project during year y for the home delivery system                                                                                                            |
| Source of data                     | Site Information Management System (SIMS) database (Data based on fuel invoice or purchase receipt)                                                                                                             |
| Value(s) applied                   | <p>75 L/month</p> <p>Ex-ante: Estimation based on kiosk invoices</p>                                                                                                                                            |
| Measurement methods and procedures | <p>To be calculated based on fuel invoice or purchase receipt.</p> <p>The actual expense of the fuel consumption was recorded and converted to the equivalent of consumption in litres based on fuel price.</p> |
| Monitoring frequency               | Continually                                                                                                                                                                                                     |
| QA/QC procedures                   | -                                                                                                                                                                                                               |
| Purpose of data                    | To calculate project emissions                                                                                                                                                                                  |
| Additional comment                 | -                                                                                                                                                                                                               |

|                                    |                                                                                                                                                                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -                                  | <b>SDWS 34</b>                                                                                                                                                                                                                                                                 |
| Data/parameter                     | $ECp,y$                                                                                                                                                                                                                                                                        |
| Unit                               | kwh                                                                                                                                                                                                                                                                            |
| Description                        | <p>Quantity of electricity that is used by the project during year y (kWh)</p> <p>Usually the pumping system is using solar panel but some kiosks are using electricity from the grid. This consumption from the grid is usually very low, carbon emission are negligible.</p> |
| Source of data                     | Site Information Management System (SIMS) database (Electricity invoices)                                                                                                                                                                                                      |
| Value(s) applied                   | <p>46 kWh per month per kiosk</p> <p>Ex ante : Sampling of electricity invoices</p>                                                                                                                                                                                            |
| Measurement methods and procedures | To be calculated based based on Electricity invoices                                                                                                                                                                                                                           |
| Monitoring frequency               | Continually                                                                                                                                                                                                                                                                    |
| QA/QC procedures                   | -                                                                                                                                                                                                                                                                              |
| Purpose of data                    | To calculate project emissions                                                                                                                                                                                                                                                 |
| Additional comment                 | -                                                                                                                                                                                                                                                                              |

|                |                                                                                                                                                                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter ID   | <b>SDWS 2</b>                                                                                                                                                                                                                                                           |
| Data/parameter | Project technology description                                                                                                                                                                                                                                          |
| Unit           | NA                                                                                                                                                                                                                                                                      |
| Description    | <p>The detailed description of the planned project technology shall include as a minimum:</p> <p><b>Rehabilitated technologies:</b><br/>In case the project technology (CWT) is a rehabilitation, the following is also required as part of technology description:</p> |

|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      | <ul style="list-style-type: none"> <li>- Evidence of Non-operational time prior to proposed rehabilitation (at minimum with evidence letter from local representative or government, etc.);</li> <li>- Evidence of lack of an existing maintenance or repair plan (at minimum with evidence letter from local representative or government, etc.);</li> <li>- Original installation date/month (approximate month/year);and</li> <li>- Information/evidence to confirm the details of rehabilitation activity (e.g. parts replaced, specifications followed, personnel conducting the repairs and date of retrofitting)</li> </ul> |
| Source of data                                       | Rehabilitated technologies: <ul style="list-style-type: none"> <li>- Manufacturer specifications and</li> <li>- Technical reports from a qualified entity that undertakes the rehabilitation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Value(s) applied                                     | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Choice of data or Measurement methods and procedures | As per the applied methodology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Purpose of data                                      | Calculation of baseline & project emissions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Additional comment                                   | Any information not available at validation, the validating VVB shall raise a FAR for verification. The verifying VVB shall verify the information at the first verification and in case of progressive at first verification for the technology.                                                                                                                                                                                                                                                                                                                                                                                  |

### B.7.2 Sampling plan

>>

The following parameter will be monitored through annually through project survey :

- SDWS 20: Water hygiene education campaigns
- SDWS 22:  $X_{cleanboil,y}$
- SDWS 26:  $HHp,y$

The following parameters will be recorded on monthly basis.

- SDWS 18:  $Mq,y$
- SDWS 19: SDG Claims
- SDWS 23:  $Qm,y$

- SDWS 27:  $DO_{p,y}$
- SDWS 33:  $PP_{f,y}$
- SDWS 34:  $EC_{p,y}$

The sampling plan for the monitored parameters follows the General requirements for sampling from METHODOLOGY FOR EMISSION REDUCTIONS FROM SAFE DRINKING WATER SUPPLY along with "Standard for sampling and surveys for CDM projects activities and programme of activities" (ver 9.0) and can be detailed as follows.

- **Objectives and reliability requirements**

The objective of the monitoring is to determine a reliable estimate of the mean (average) values of the parameters needed to calculate emission reductions, mentioned in the previous section. As the project will be using the estimation of the means for calculations, the confidence/precision levels must comply with the 90/10 rule for the tests conducted, as mentioned in the section 4.2 of the methodology used.

- **Sampling method**

Stratified sampling approach (vintage wise) will be used for selection of kiosk from installation database for project survey. Once the kiosks samples have been selected, PP will use random sampling approach for selection of households/schools/beneficiary (who purchased water from kiosk) on regular basis and conduct the project survey.

As the project is a community level project and PP could not maintain the households/beneficiary data, random sampling approach best suited for selection of households/schools/beneficiary for project survey once the kiosks have been selected.

The no. of kiosks and households/beneficiaries selected for the project survey (sample size) will be in line with the meth requirement.

- **Sample size**

When a baseline and project survey is used the following sample size guidelines should be applied, unless otherwise stated for specific parameters:

| Group size  | Minimum sample size                         |
|-------------|---------------------------------------------|
| <300        | 30 or population size, whichever is smaller |
| 300 to 1000 | 10% of group size                           |
| > 1000      | 100                                         |

In sampling, a kiosk will be considered as a unit of the Group. The total no. of kiosk installed by end of the monitoring period will be considered as total group for sampling.

Hence, the sample size will be dependent on the no. of kiosks installed. Once the sample size is being determined using the above table, PP will select 4-5 households/users/schools per kiosk for project survey. Hence, the sample size will satisfies the minimum sample size in terms of samples as well as in terms of end users.

### B.7.3 Other elements of monitoring plan

>> Entrepreneurs operating the kiosks record operation data in a notebook on a daily basis. Indicators such as number of functional days, sales, etc. are recorded. 1001fontaines advisors collect such data monthly and record it on tablets through a specific online app. Data is then consolidated at national level, enabling ongoing monitoring of the network and standardized monthly reporting. For example, daily sales are compared and cross-checked with water meters from kiosks.

Also, technicians are alerted by the entrepreneurs when technical issue appears (regarding UV treatment, filtration, solar panels...). Such information and data are described in reports that are also recorded in the technical app.

## SECTION C. DURATION AND CREDITING PERIOD

### C.1. Duration of project

#### C.1.1 Start date of project

>>

09/03/2020

The start date of water distribution/start of sale of water from O2 kiosk under project<sup>45</sup>.

The water sale report of O2 Kiosk from commcare software (in pdf format)

and extract in excel format has already been submitted to SustainCert during the Preliminary Review of the project and also submitted to Validating VVB.

#### C.1.2 Expected operational lifetime of project

>>

15 years

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<sup>45</sup> Kindly refer to the approved deviation request (DEV\_253) for more details.

## C.2. Crediting period of project

C.2.1 Start date of crediting period

>>

06/06/2021

C.2.2 Total length of crediting period

>>

5 years, twice renewable, total 15 years

## SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

### D.1 Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarized below.

| Principles | Mitigation Measures added to the Monitoring Plan |
|------------|--------------------------------------------------|
| N/A        | N/A                                              |

### D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

|                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy? | <p>The project meets the mandatory gender-sensitive requirements by complying with the related Safeguarding Principles and Requirements through both project activities and project implementation (consultations).</p> <p>Project activities (kiosks operation and water distribution) will affect both women and men ensuring equality:</p> <ul style="list-style-type: none"> <li>- project developer ensures jobs created are equally accessible to women and men</li> <li>- safe water will be distributed to all potential beneficiaries regardless of their gender.</li> </ul> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                           | <p>In fact, the project will mainly benefit to women. In the baseline scenario, women and children (especially girls) are usually responsible for collecting woods for water purification. This time is usually time they do not spend for agricultural work or any income-generating activity, household work, school/children education... the distribution of safe water will enable them to save this time of wood collection but also caring time and helps to reduced water-related diseases.</p> <p>Project design is presented to all stakeholders and the project owner will ensure the equal participation of women to consultation meetings and will encourage them to provide feedback on the project design. In the same way, both women and men will have the opportunity to provide with feedbacks through the continuous input mechanism.</p> |
| Question 2 - Explain how the project aligns with existing country policies, strategies and best practices | <p>The project is fully aligned with local policies, strategies and best practices. It is perfectly aligned with the five-year strategic plan for strengthening gender mainstreaming and women's empowerment Neay Rattanak V as it contributes directly to nearly all 6 strategies developed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?                 | <p>Based on the answers above and Appendix 1 (in particular Principle 2), no expert is required.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



|                                                                                                  |                                                                                                                                       |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation? | Based on the answers above and Appendix 1 (in particular Principle 2), no expert is required to assist with gender issues at the LSC. |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|

## SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

### E.1 Summary of stakeholder mitigation measures

>>

| Gender of Stakeholder | Stakeholder comment                                                                                        | Was comment taken into account (Yes/No)? | Explanation/ Justification (Why? How?)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                                                                                                            |                                          | <ul style="list-style-type: none"> <li><b>Affordability</b></li> </ul> <p>A factual analysis of the level of poverty of the population shows that the recommended price of O-we water (1.800 riels for 20 litres) is actually affordable for most of the population:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Ma Sovanna Gender: F  | is the water really affordable?<br>Why not doing pipe network?<br>How to reach the most vulnerable people? | Not Applicable                           | <p>The Cambodian Ministry of Planning (MoP), supported by the World Bank, has announced in November 2021 that it had updated the country's poverty line. It is now 8908 riels (\$2.2) per capita per day in rural Cambodia, vs. 3503 riels (\$0.84) in 2014 (study by MoP and ADB, Asian Development Bank). Despite this 154% surge, the percentage of the rural population living under the poverty line has decreased, from 24.6% to 22.8%.</p> <p>For an O-we customer at the poverty line level, drinking everyday 1.5 litres of safe water represents a cost of 135 riels per day, i.e. no more than 1.5 % of his/her income. This same calculation amounted to 3.3% of the income in 2014, meaning that O-we water has become more affordable.</p> <p>The United Nations Development Programme (UNDP) suggests that water costs should not exceed 3% of the</p> |

income to be considered as fully affordable. We can therefore assert that, for at least 77 % of the villages population (those living over the poverty line), drinking O-we water every day is in fact affordable.

The lack of reliable data prevents us from fine-tuning the analysis on the population living under the poverty and be more assertive, but, given the gap between our 1.5% and the 3% threshold, it is likely that it is also affordable for part of the remaining 23% of the rural population.

- **Bottled water and piped water complementarity**

Pipe water network requires important investments (pipe network construction, water treatment units with more or less capacity and technicality) of which the cost increases as population density reduces. While in urban and peri-urban areas pipe water solutions can be paid off which results a relatively high number of “clients”, the situation is much different in rural areas where population density is much lower. This explains why piped networks still reach only 16% of rural households (General population census 2019).

The water kiosk offers a complementary solution, focusing on drinking water needs by providing higher quality guarantee up to the point of use, while remaining affordable because of reduced investment costs. Community-owned water kiosks are recognized as being part of the Rural Water Supply strategy to achieve universal access to safe water and proved to be particularly resilient to climate change. Teuk Saat 1001 has been building a national based expertise and local capacities to ensure deployment and sustainable management of such infrastructures, also working in cooperation with piped network stakeholders to foster efficient coverage of rural areas (read our position paper on complementarity of water supply solutions to know more about our approach).

- **The way forward reaching the most vulnerable**

To ensure that children, highly vulnerable to waterborne diseases, had access to safe drinking water Teuk Saat developed the Water in School program in 2008. Relying on our network of entrepreneurs, we deliver a 20L bottle of water every day to every class of

the nearby primary schools. The service is free for the schools and the entrepreneurs are paid by subsidies. This also generates a decrease in school absenteeism by up to 75%. Today, we count 310,000 children getting water for free at school.

Given the severity of the COVID-19 wave with its direct consequences for rural livelihoods, we designed a specific response program in partnership with communes' authorities targeting 13,000 ID poor households living below the poverty line. This program enabled local entrepreneurs to provide free 20L containers, key hygiene items, and free refills of safe drinking water for 3 months to 57 330 people.

Such targeted programs have the potential to be scaled provided integrated into public policies and budget planning. Teuk Saat 1001 is active into sectoral discussion and work with national and local administrations in order to implement effective strategy to left no one behind.

|                                     |                                                                                                                                                                                   |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| Minh<br>Lim,<br>CWA<br>Gender:<br>M | Is the price determined by TS1001 or which institution? Is it sold the same or different price in different areas? Laboratories and suggestion on kiosk connected to pipe network | No | The price (1.800 riels for 20 litres) is recommended by Teuk Saat 1001 based on the socio-economic analysis detailed above. The average price amounted to 1514 riels in 2021, showing that local entrepreneurs keep a certain level of autonomy. In case the price practiced by an entrepreneur is not matching the community situation, O-We committee can report to Teuk Saat 1001 and Teuk Saat 1001 will intervene, supporting the entrepreneur in implementing the good practices. |
| Sokchen<br>Unicef<br>Gender:<br>F   | How to calculate carbon? Now water kiosks use solar                                                                                                                               | No | The project is being developed in line with the Gold Standard Methodology"Gold Standard Methodology for emission reductions from safe drinking water supply,                                                                                                                                                                                                                                                                                                                            |

|                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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|                                                                                                                                                | <p>system, how about delivery vehicles, is there a way for solar system for delivery vehicles?</p> <p>Version 1.0<sup>46</sup> . The methodology provides formulas to calculate the Emission Reduction (CO2 saving). Kindly refer to the methodology for more details.</p> <p>As of now the delivery vehicles run on fossil fuel (Petrol/Diesel). The emissions due to combustion of fossil fuel in delivery vehicle is accounted under Project Emission (PEy) and subtracted from the Baseline Emission achieved.</p> <p>Running of the delivery vehicle with solar panel is not very common as of now. We will investigate it when the technology will be more developed and easily available also cost friendly.</p> |
| <p>Timeframe of the certification process</p> <p>Maxence Souvet Gender: M</p> <p>Carbon finance enough to fund kiosk deployment.</p> <p>No</p> | <p>The project certification with Gold Standard is in progress. The Preliminary Review of the Project has already been completed (project received Green Signal from GS to go ahead). Now we are planning for the Validation of the Project and then will do Project Design Certification with GS. We hope that the Certification of the project will complete before end of this year (i.e. 2022).</p> <p>Carbon Finance will help in repair and maintenance of the kiosks, reinvest in development of new kiosks and help in ensuring the sustainability of the network of kiosks.</p>                                                                                                                                |

## E.2 Final continuous input / grievance mechanism

| Method                                                                  | Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.                                                                               |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Continuous Input / Grievance Expression Process Book (mandatory)</p> | <p>Continuous input / Grievance Expression process book is available at 1001 fontaines' office at <b>#100, Street 95, Sangkat Beoung Trobek, Khan Chamkar Mon, Phnom Penh, Cambodia</b></p> |

<sup>46</sup>[https://globalgoals.goldstandard.org/standards/429\\_V1.0\\_EE\\_SWS\\_Emission-reductions-from-Safe-Drinking-Water-Supply.pdf](https://globalgoals.goldstandard.org/standards/429_V1.0_EE_SWS_Emission-reductions-from-Safe-Drinking-Water-Supply.pdf)

|                           |                                                                                                                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GS Contact<br>(mandatory) | <a href="mailto:help@goldstandard.org">help@goldstandard.org</a> provided on all documentation and<br>signs of the project                                                                   |
| Email / internet          | The continuous contribution will also be made possible via following<br>email addresses:<br><a href="mailto:feedback.anakotthmei@teuksaat1001.com">feedback.anakotthmei@teuksaat1001.com</a> |
| Phone                     | A project phone also dedicated to feedbacks from stakeholders<br>will be set-up and communicated to stakeholders.                                                                            |

## APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form.

| Assessment Questions/<br>Requirements                                                                                                                                                                                                                                                                                                                                         | Justification of Relevance<br>(Yes/potentially/no) | How Project will achieve Requirements<br>through design, management or risk<br>mitigation.                                                                                                                                                                                                                                    | Mitigation<br>Measures<br>added to<br>the<br>Monitoring<br>Plan (if<br>required) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Principle 1. Human Rights</b>                                                                                                                                                                                                                                                                                                                                              |                                                    |                                                                                                                                                                                                                                                                                                                               |                                                                                  |
| <ol style="list-style-type: none"> <li>1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights</li> <li>2. The Project shall not discriminate with regards to participation and inclusion</li> </ol> | No                                                 | The project respects human rights requirements especially through the respect of internationally proclaimed human rights and Declaration of Human Rights. Also, the project will not discriminate in any way: safe water is distributed to all potential beneficiaries without any requirement or discrimination of any kind. | N/A                                                                              |

| Principle 2. Gender Equality                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <p>1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women</p> <p>2. Projects shall apply the principles of non-discrimination, equal treatment, and equal pay for equal work</p> <p>3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks</p> <p>4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)</p> | No | <p>The Project will increase access of women to safe water. In particular, women and children may save time in wood collection that, in the baseline scenario, is required to boil water for purification. It lowers women's workload and provide them with time to engage in other activities. Also, the improved access to safe water is foreseen to lead to decreased water born diseases, therefore reducing the time women spend for caring activities.</p> <p>Safe water is distributed to all local beneficiaries, both women and men.</p> <p>During the local stakeholder consultation and throughout the project's lifetime, women are welcome and encouraged to provide feedback about the project through various means of communication. All events (LSC meetings, hygiene campaigns) are planned and organized to avoid any discrimination of women or other marginalized groups.</p> <p>For jobs created, the equal pay for equal work principle is respected by the project implementer (1001fontaines) as both women</p> | N/A |

|                                                                                                                                                    |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |
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|                                                                                                                                                    |    | <p>and men operate water kiosks and associated activities.</p> <p>The Project is perfectly aligned with the five year strategic plan for strengthening gender mainstreaming and women's empowerment Neay Rattanak V as it contributes directly to nearly all 6 strategies developed.</p> <p>In addition, Teuk Saat has an advanced employment policy. Despite the physical activities of operating water kiosks, 33% of jobs created are occupied by woman. The project also develops a new position mainly for women that aims at increasing awareness on safe water issue. Important training is provided to such ambassadors.</p> |     |
| <b>Principle 3. Community Health, Safety and Working Conditions</b>                                                                                |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |
| 1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community | No | <p>The project activities don't include exposing the community to increased health risks and is not adversely affecting the health of workers and the community. Both the construction of water kiosks and their operation will respect the country's regulation and will not include any hazardous chemicals or material.</p> <p>In regard to the construction of the kiosks, criteria used to select providers include safety rules and work conditions.</p> <p>In regard to running the kiosks, the franchise system enables the project implementer</p>                                                                          | N/A |



|                                                                                                                                                                  |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
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|                                                                                                                                                                  |       | <p>(1001fontaines) to ensure safety rules, workers protection processes are followed in line with local regulations.</p> <p>In addition, development of kiosks 2.0 aims at automatization of manutention tasks in order to improve work conditions.</p>                                                                                                                                                                                                                                                     |     |
| <b>Principle 4.1 Sites of Cultural and Historical Heritage</b>                                                                                                   |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture? | >> No | The project activity doesn't include sites, structures or objects with historical, cultural, artistic, traditional, or religious values or intangible forms of culture. Places where water kiosks are implemented are chosen in relation with local municipalities therefore it does not require alternation, damage or removal of any historical, artistic, traditional, religion or cultural heritage issues. In particular, the presence of grave, funeral site, pagoda is a criteria in site selection. | N/A |
| <b>Principle 4.2 Forced Eviction and Displacement</b>                                                                                                            |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |

|                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                       |     |
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| Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?                                                                                                                                                                                                  | No | Water kiosks are constructed on municipal land that are available and owned by commune in agreement with the municipalities and local communities. Thus, the implementation does not cause relocation of any people.                                                                                                                                                                                  | N/A |
| <b>Principle 4.3 Land Tenure and Other Rights</b>                                                                                                                                                                                                                                                                            |    |                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| <p>a. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?</p> <p>b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?</p> | No | Water kiosks are constructed on land completely owned by the municipality it is implemented in. It is a requirement in site selection. An agreement is signed between the project implementer (1001fontaines), the kiosk operator and the land owner (commune) stating the location of construction of the kiosk as well as the water source that will be used. There is no uncertainty at any point. | N/A |
| <b>Principle 4.4 - Indigenous people</b>                                                                                                                                                                                                                                                                                     |    |                                                                                                                                                                                                                                                                                                                                                                                                       |     |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?                                                       | No | There are no indigenous peoples present in or within the area of influence of the project (ie surrounding water kiosks).                                                                                                                                                                                                                                    | N/A |
| <b>Principle 5. Corruption</b>                                                                                                                                                                                        |    |                                                                                                                                                                                                                                                                                                                                                             |     |
| 1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects                                                                                          | No | <p>The Project doesn't involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects.</p> <p>Cambodia has ratified the UN Convention against Corruption.</p>                                                                                                                                                          | N/A |
| <b>Principle 6.1 Labour Rights</b>                                                                                                                                                                                    |    |                                                                                                                                                                                                                                                                                                                                                             |     |
| 1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions | No | <p>The project implementer (1001fontaines) is fully aligned with Cambodia's national labour law and with the principles and standards embodied in the ILO fundamental conventions.</p> <p>Kiosks' operators are microentrepreneurs with franchise contracts with project implementer (1001fontaines) and municipality that are documented and archived.</p> | N/A |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |
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| <p>2. Workers shall be able to establish and join labour organisations</p> <p>3. Working agreements with all individual workers shall be documented and implemented and include:</p> <ul style="list-style-type: none"> <li>a) Working hours (must not exceed 48 hours per week on a regular basis), AND</li> <li>b) Duties and tasks, AND</li> <li>c) Remuneration (must include provision for payment of overtime), AND</li> <li>d) Modalities on health insurance, AND</li> <li>e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND</li> <li>f) Provision for annual leave of not less than</li> </ul> |  | <p>The PD does not restrict workers to be able to establish and join Labour organisations.</p> <p>No child labour is used at all.</p> <p>The project implementer (1001fontaines) ensures the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures. Workers operating water kiosks are properly trained and regularly accompanied by the project developer especially in water quality analysis in order to respect the WHO standards. Further than that, kiosk operators shall be trained well enough to ensure post-project sustainability.</p> |  |
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| <p>10 days per year, not including sick and casual leave.</p> <p>4. No child labour is allowed (Exceptions for children working on their families' property requires an <a href="#">Expert Stakeholder</a> opinion)</p> <p>5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures</p> |  |  |  |
| <b>Principle 6.2 Negative Economic Consequences</b>                                                                                                                                                                                                                                                                                                                                                                                  |  |  |  |

|                                                                                                          |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |
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| <p>1. Does the project cause negative economic consequences during and after project implementation?</p> | <p>No</p> | <p>First, the construction of water kiosks is financed by the project implementer (1001fontaines) (through carbon finance), without any cost for local beneficiaries nor municipalities. No negative economic consequence is expected here.</p> <p>Second, operational costs of the water kiosks shall be covered by the sale of water. The optimization of kiosks activities enables the project implementer (1001fontaines) to sell the water at an affordable price for every potential beneficiary (less than 3% of income) so that it has no negative economic consequence for beneficiaries. In addition, local inhabitants and women and children in particular will save time as wood collection time will be reduced. Women might thus engage in income-generating activity. Also, each kiosk enables the creation of several jobs for its operation, improving the economic activity of the workers.</p> <p>Third and last, kiosks shall continue to be operated after the project as microentrepreneurs shall be trained well enough to pursue purification and distribution activities.</p> | <p>N/A</p> |
|----------------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|

| Principle 7.1 Emissions                                                                                                                                                                          |    |                                                                                                                                                                                                                                                                                                                                                                                 |     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Will the Project increase greenhouse gas emissions over the Baseline Scenario?                                                                                                                   | No | GHG emissions will be reduced due to implementation the project. Project activity is to replace water purification using firewood by distributing safe water purified in water kiosks. The project will reduce consumption of biomass through the provision of access to clean water. The GHG emissions will be monitored and verified in line with Gold Standard requirements. | N/A |
| Principle 7.2 Energy Supply                                                                                                                                                                      |    |                                                                                                                                                                                                                                                                                                                                                                                 |     |
| Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users? | No | Energy used by the water kiosk for purification will be provided by solar panels. Fuel used by trucks for water distribution will be accounted in GHG emissions.                                                                                                                                                                                                                | N/A |
| Principle 8.1 Impact on Natural Water Patterns/Flows                                                                                                                                             |    |                                                                                                                                                                                                                                                                                                                                                                                 |     |

|                                                                                                                                                                                                                                                        |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?                          | No | The project water systems will use ground or surface water but will not affect negative natural or pre-existing pattern of watercourses, ground-water and/or watersheds. The agreement signed between project implementer (1001fontaines), kiosk entrepreneur and municipality states precisely the source that is used by the kiosk. In case of scarcity, the agreement also states a "backup" source that can be used to keep providing safe water. | N/A |
| <b>Principle 8.2 Erosion and/or Water Body Instability</b>                                                                                                                                                                                             |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| a. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?<br>b. Is the Project's area of influence susceptible to excessive erosion and/or water body instability? | No | The project will not cause additional erosion and/or water body instability or disrupt the natural pattern of erosion as water pumped is directly purified within the water kiosks and bottles in 20L jugs. The project area of influence is not susceptible to or will lead to excessive erosion or water body instability.                                                                                                                          | N/A |
| <b>Principle 9.1 Landscape Modification and Soil</b>                                                                                                                                                                                                   |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |



|                                                                                                                                                                                                                                                                            |    |                                                                                                                                                                    |     |
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| Does the Project involve the use of land and soil for production of crops or other products?<br>>>                                                                                                                                                                         | No | Water kiosks are built on municipal land and don't involve the use of land and soil for production of crops or other products.                                     | N/A |
| <b>Principle 9.2 Vulnerability to Natural Disaster</b>                                                                                                                                                                                                                     |    |                                                                                                                                                                    |     |
| Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?                                                                                       | No | The Project provides supply of safe and clean drinking water and is not susceptible to or will lead to increased vulnerability to any extreme climatic conditions. | N/A |
| <b>Principle 9.3 Genetic Resources</b>                                                                                                                                                                                                                                     |    |                                                                                                                                                                    |     |
| Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?<br>>> | No | The project uses ground water sources that are not impacted or do involve GMOs.                                                                                    | N/A |

| Principle 9.4 Release of pollutants                                                                                              |    |                                                                                                                                                                         |     |
|----------------------------------------------------------------------------------------------------------------------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Could the Project potentially result in the release of pollutants to the environment?                                            | No | As the purification process consists in filtration and UV treatment, there is no potential for the release of pollutants to the environment.                            | N/A |
| Principle 9.5 Hazardous and Non-hazardous Waste                                                                                  |    |                                                                                                                                                                         |     |
| Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials? | No | Project doesn't involve hazardous and non-hazardous chemicals and/or materials                                                                                          | N/A |
| Principle 9.6 Pesticides & Fertilisers                                                                                           |    |                                                                                                                                                                         |     |
| Will the Project involve the application of pesticides and/or fertilisers?                                                       | No | The project does not involve the use of pesticides and/or fertilizers.                                                                                                  | N/A |
| Principle 9.7 Harvesting of Forests                                                                                              |    |                                                                                                                                                                         |     |
| Will the Project involve the harvesting of forests?                                                                              | No | The project does not involve the harvesting of forest. Indeed, beneficiaries will be less likely to collect wood to purify water as safe drinking water is distributed. | N/A |
| Principle 9.8 Food                                                                                                               |    |                                                                                                                                                                         |     |

|                                                                                                                                                                                    |    |                                                                                                                                                                                                                                                                                                                                                  |     |
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| Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?                             | No | The Project doesn't modify the quantity or nutritional quality of food available as it consists only in purification and distribution of safe water.                                                                                                                                                                                             | N/A |
| <b>Principle 9.9 Animal husbandry</b>                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                  |     |
| Will the Project involve animal husbandry?                                                                                                                                         | No | The Project doesn't involve animal husbandry.                                                                                                                                                                                                                                                                                                    | N/A |
| <b>Principle 9.10 High Conservation Value Areas and Critical Habitats</b>                                                                                                          |    |                                                                                                                                                                                                                                                                                                                                                  |     |
| Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified? | No | As the project reduces wood collection, it is more likely to affect forest areas in a positive way than negative. In addition, kiosks are constructed on municipal land in agreement with local politics so that they do not affect largely intact or HCV ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified. | N/A |
| <b>Principle 9.11 Endangered Species</b>                                                                                                                                           |    |                                                                                                                                                                                                                                                                                                                                                  |     |

|                                                                                                                                                                                                                                                                                                    |           |                                                                                                       |            |
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| <p>a. Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)?</p> <p>b. Does the Project potentially impact other areas where endangered species may be present through transboundary affects?</p> | <p>No</p> | <p>The project activity does not have any foreseeable negative impacts on any endangered species.</p> | <p>N/A</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------|------------|

## APPENDIX 2- CONTACT INFORMATION OF PROJECT PARTICIPANTS

|                                             |                                                               |
|---------------------------------------------|---------------------------------------------------------------|
| <b>Organization name</b>                    | <b>ECOACT SAS</b>                                             |
| Registration number with relevant authority | 492029475                                                     |
| Street/P.O. Box                             | 35 rue de Miromesnil                                          |
| Building                                    | -                                                             |
| City                                        | Paris                                                         |
| State/Region                                | Ile de France                                                 |
| Postcode                                    | 75 008                                                        |
| Country                                     | France                                                        |
| Telephone                                   | tel: +33 (0) 7 89 09 51 78                                    |
| E-mail                                      | contact@eco-act.com                                           |
| Website                                     | <a href="https://eco-act.com/fr/">https://eco-act.com/fr/</a> |
| Contact person                              | Mathilde Mignot                                               |
| Title                                       | Director – Nature Based Solution                              |
| Salutation                                  | Ms.                                                           |
| Last name                                   | Mignot                                                        |
| Middle name                                 | -                                                             |
| First name                                  | Mathilde                                                      |
| Department                                  | Nature Based Solution                                         |
| Mobile                                      | -                                                             |
| Direct tel.                                 | -                                                             |
| Personal e-mail                             | mathilde.mignot@atos.net                                      |

|                                             |                        |
|---------------------------------------------|------------------------|
| <b>Organization name</b>                    | <b>1001fontaines</b>   |
| Registration number with relevant authority | (SIREN) 490 348 745    |
| Street/P.O. Box                             | 17-21 rue Saint Fiacre |
| Building                                    |                        |
| City                                        | Paris                  |

|                 |                                              |
|-----------------|----------------------------------------------|
| State/Region    | Ile de France                                |
| Postcode        | 75002                                        |
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## APPENDIX 3- LUF ADDITIONAL INFORMATION

|                                                                                                                                                                              |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Risk of change to the Project Area during Project Certification Period:                                                                                                      |  |
| Risk of change to the Project activities during Project Certification Period:                                                                                                |  |
| Land-use history and current status of Project Area:                                                                                                                         |  |
| Socio-Economic history:                                                                                                                                                      |  |
| Forest management applied (past and future)                                                                                                                                  |  |
| Forest characteristics (including main tree species planted)                                                                                                                 |  |
| Main social impacts (risks and benefits)                                                                                                                                     |  |
| Main environmental impacts (risks and benefits)                                                                                                                              |  |
| Financial structure                                                                                                                                                          |  |
| Infrastructure (roads/houses etc):                                                                                                                                           |  |
| Water bodies:                                                                                                                                                                |  |
| Sites with special significance for indigenous people and local communities - resulting from the Stakeholder Consultation:                                                   |  |
| Where indigenous people and local communities are situated:                                                                                                                  |  |
| Where indigenous people and local communities have legal rights, customary rights or sites with special cultural, ecological, economic, religious or spiritual significance: |  |

## APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to Design Change [Requirements](#) for more information on procedures governing Design Changes

### Revision History

| Version | Date            | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.2     | 14 October 2020 | Hyperlinked section summary to enable quick access to key sections<br>Improved clarity on Key Project Information<br>Inclusion criteria table added<br>Gender sensitive requirements added<br>Prior consideration (1 yr rule) and Ongoing Financial Need added<br>Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity<br>Improved Clarity on SDG contribution/SDG Impact term used throughout<br>Clarity on Stakeholder Consultation information required<br>Provision of an <a href="#">accompanying Guide</a> to help the user understand detailed rules and requirements |
| 1.1     | 24 August 2017  | Updated to include section A.8 on 'gender sensitive' requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1.0     | 10 July 2017    | Initial adoption                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

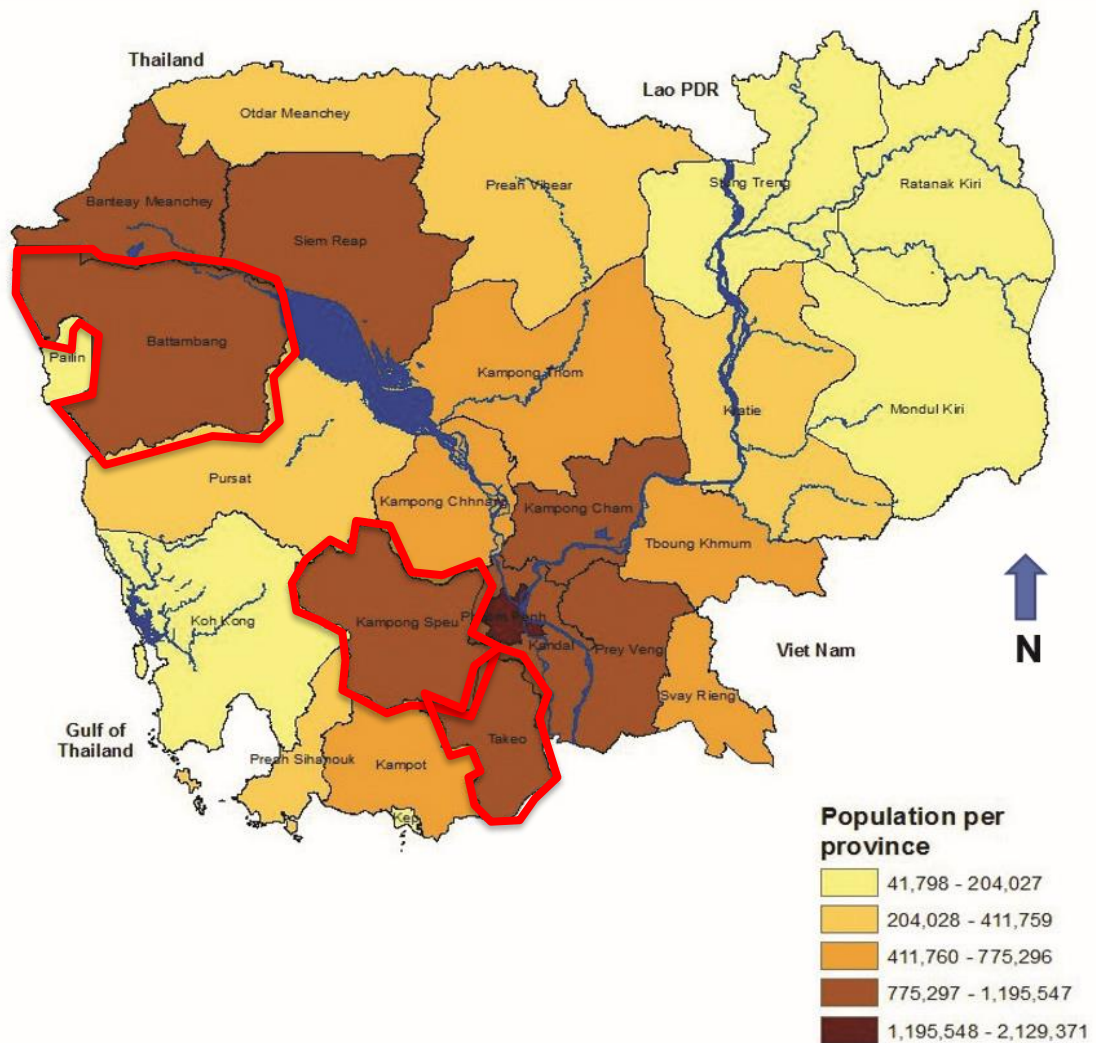


## ANNEX 1

Population by Province and sex 2019 in Cambodia. Highlighted in red the provinces concerned during the Baseline survey.

Source<sup>47</sup>

**Map 2.1. Population by province and sex, 2019**



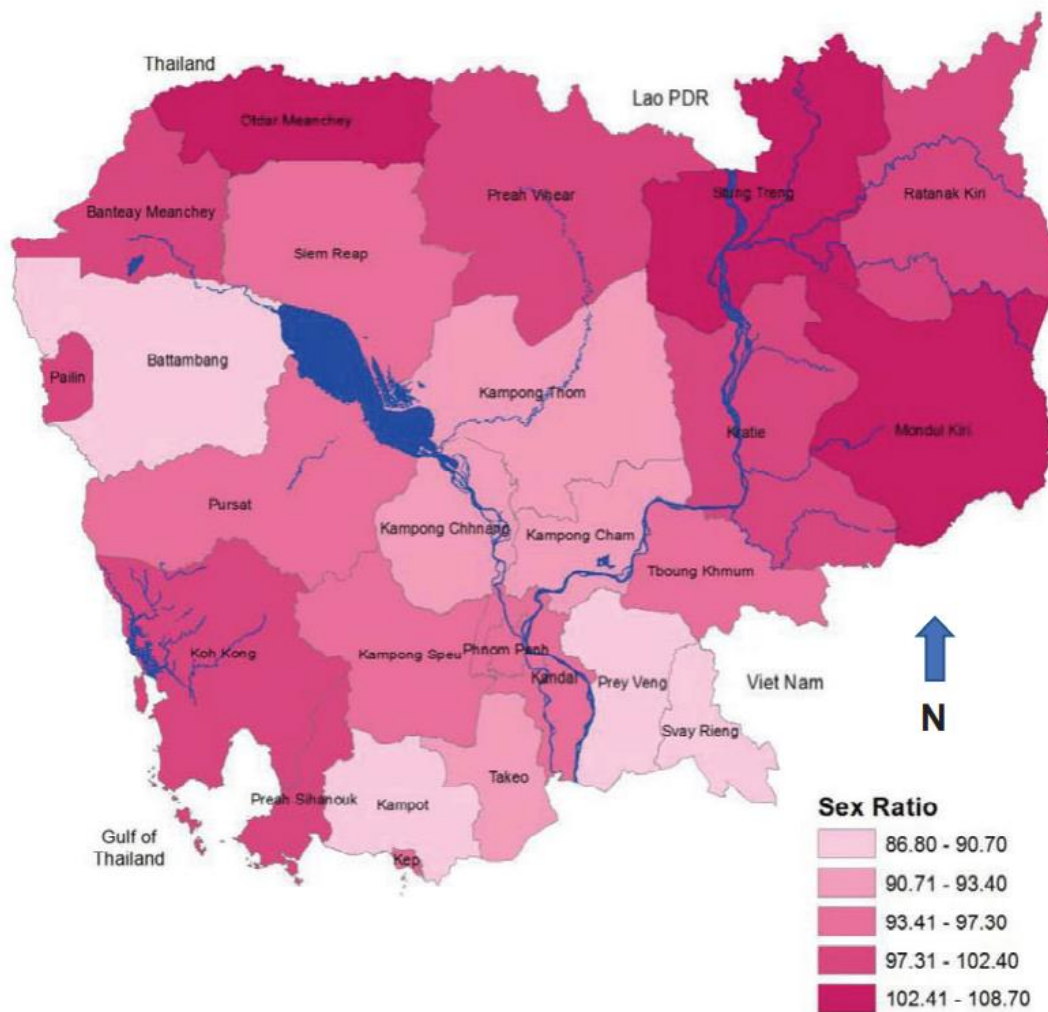
<sup>47</sup> <https://www.nis.gov.kh/nis/Census2019/Final%20Geneal%20Population%20Census%202019-English.pdf>

## ANNEX 2

Population by sex ratio in provinces in 2019 in Cambodia. Highlighted in red the provinces concerned during the Baseline survey.

Source<sup>48</sup>

**Map 2.5. Sex ratio by province, Cambodia 2019**



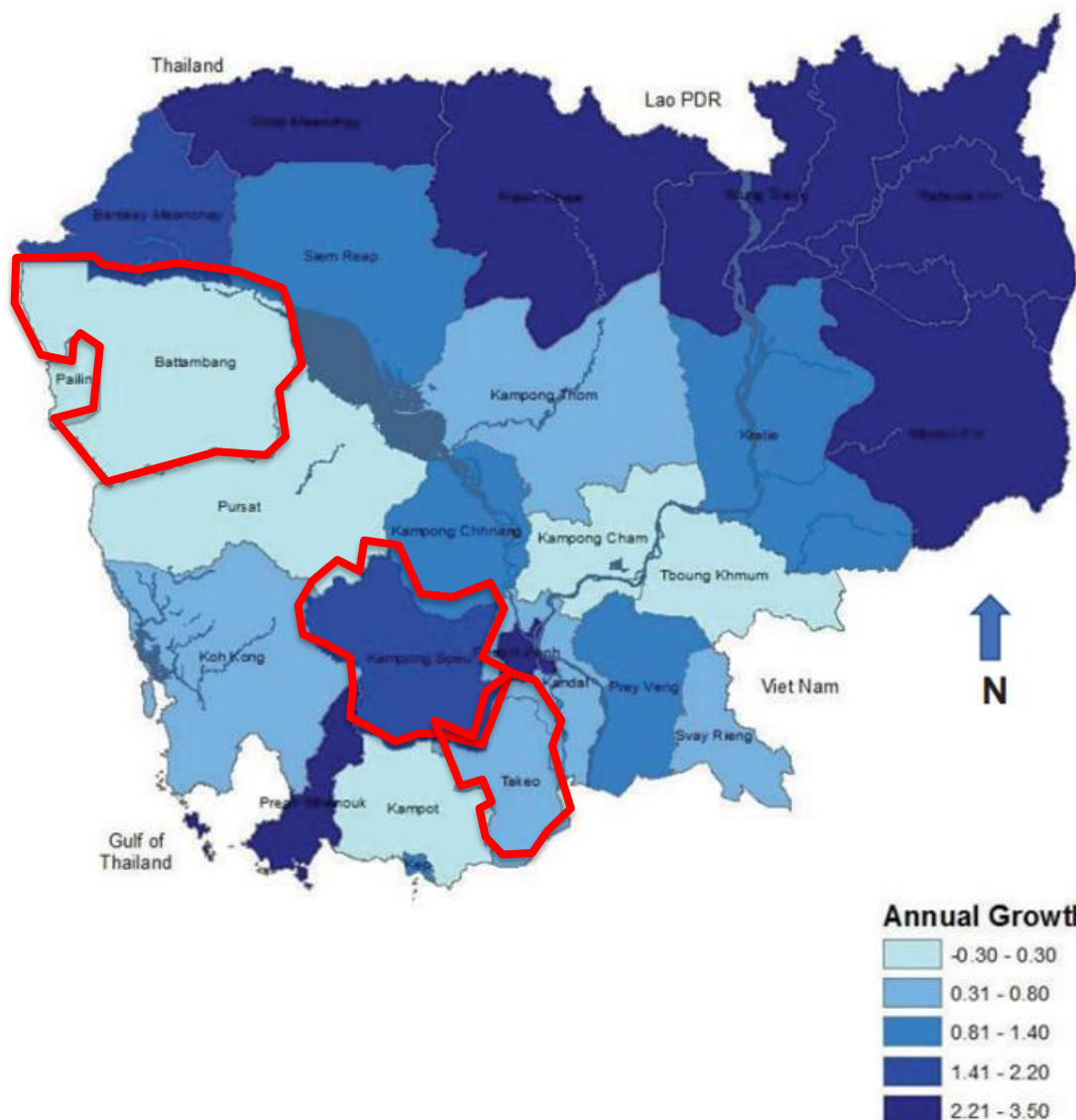
<sup>48</sup> <https://www.nis.gov.kh/nis/Census2019/Final%20Geneal%20Population%20Census%202019-English.pdf>

## ANNEX 3

Annual population growth in 2019 in Cambodia. Highlighted in red the provinces concerned during the Baseline survey.

Source<sup>49</sup>

**Map 2.2. Annual population growth, 2008-2019**

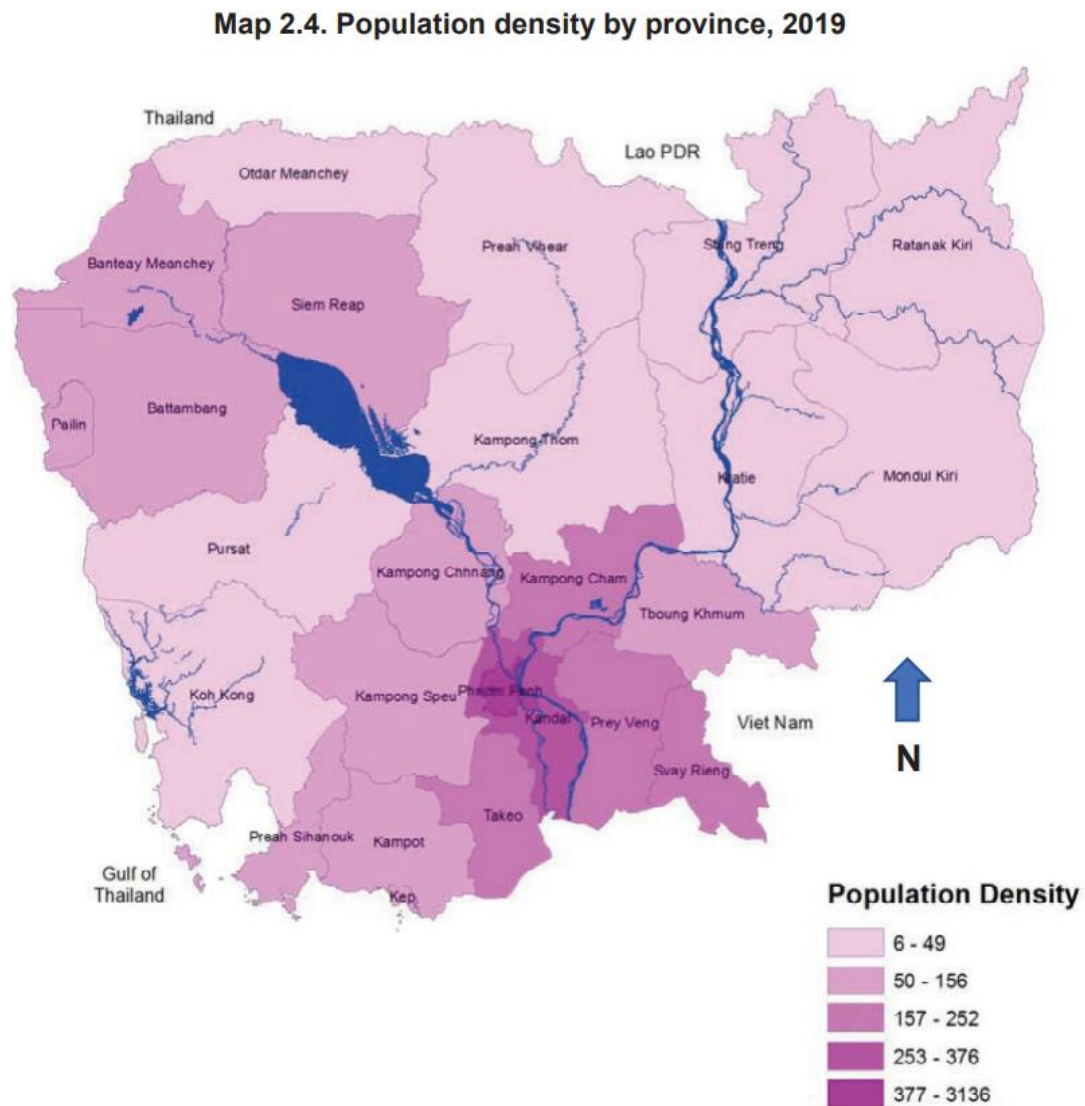


<sup>49</sup> <https://www.nis.gov.kh/nis/Census2019/Final%20Geneal%20Population%20Census%202019-English.pdf>

## ANNEX 4

Population density by province in 2019 in Cambodia. Highlighted in red the provinces concerned during the Baseline survey.

Source<sup>50</sup>



<sup>50</sup> <https://www.nis.gov.kh/nis/Census2019/Final%20Geneal%20Population%20Census%202019-English.pdf>